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Manzato

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(54) **CONFIGURATIONS FOR FOOTWEAR
EMPLOYING REEL BASED CLOSURE
SYSTEMS**

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(58) **Field of Classification Search**

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See application file for complete search history.

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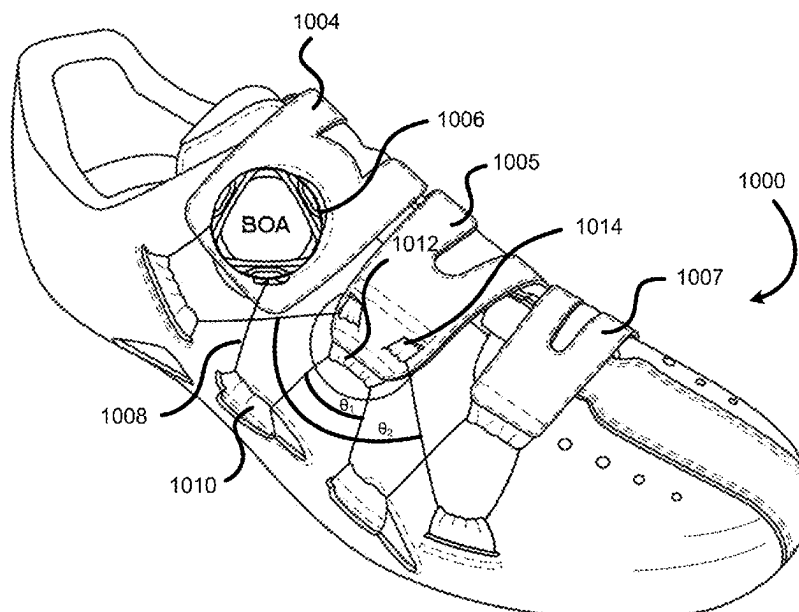
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(57)

ABSTRACT

A shoe includes a sole and an upper that is attached to the sole. The upper has a first side and a second side that fit around a wearer's foot when the shoe is fit about the wearer's foot. The shoe also includes a strap or panel having a proximal end that is attached to one side of the shoe and a distal end that extends toward an opposite side of the shoe. A tension member is guided or directed about a path along the shoe and a tightening mechanism is operable with the tension member to effect tensioning of the tension member. The strap or panel includes a first guide and a second guide and the tension member is routed through both the first guide and the second guide of the strap or panel.

15 Claims, 58 Drawing Sheets



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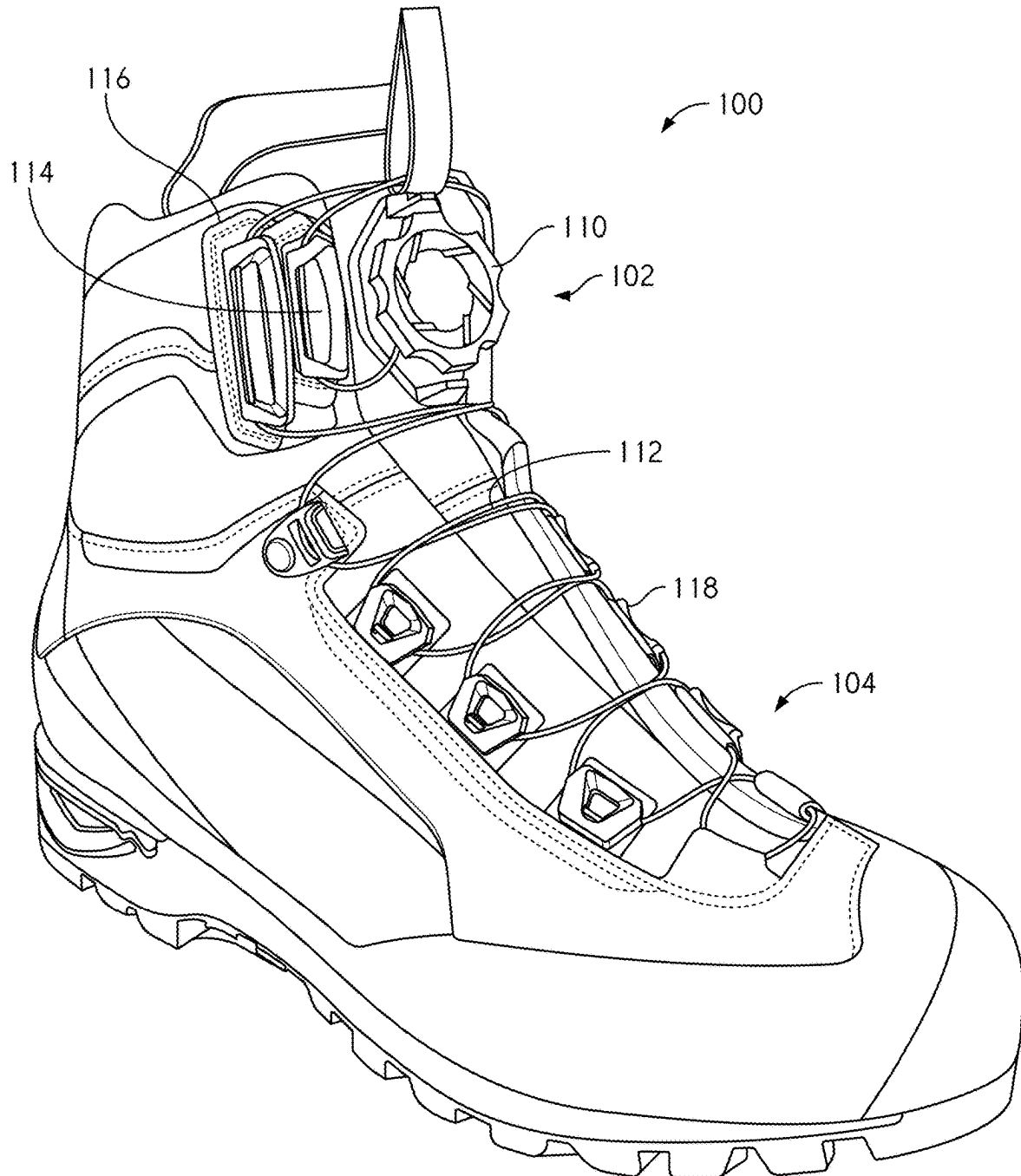


FIG. 1A

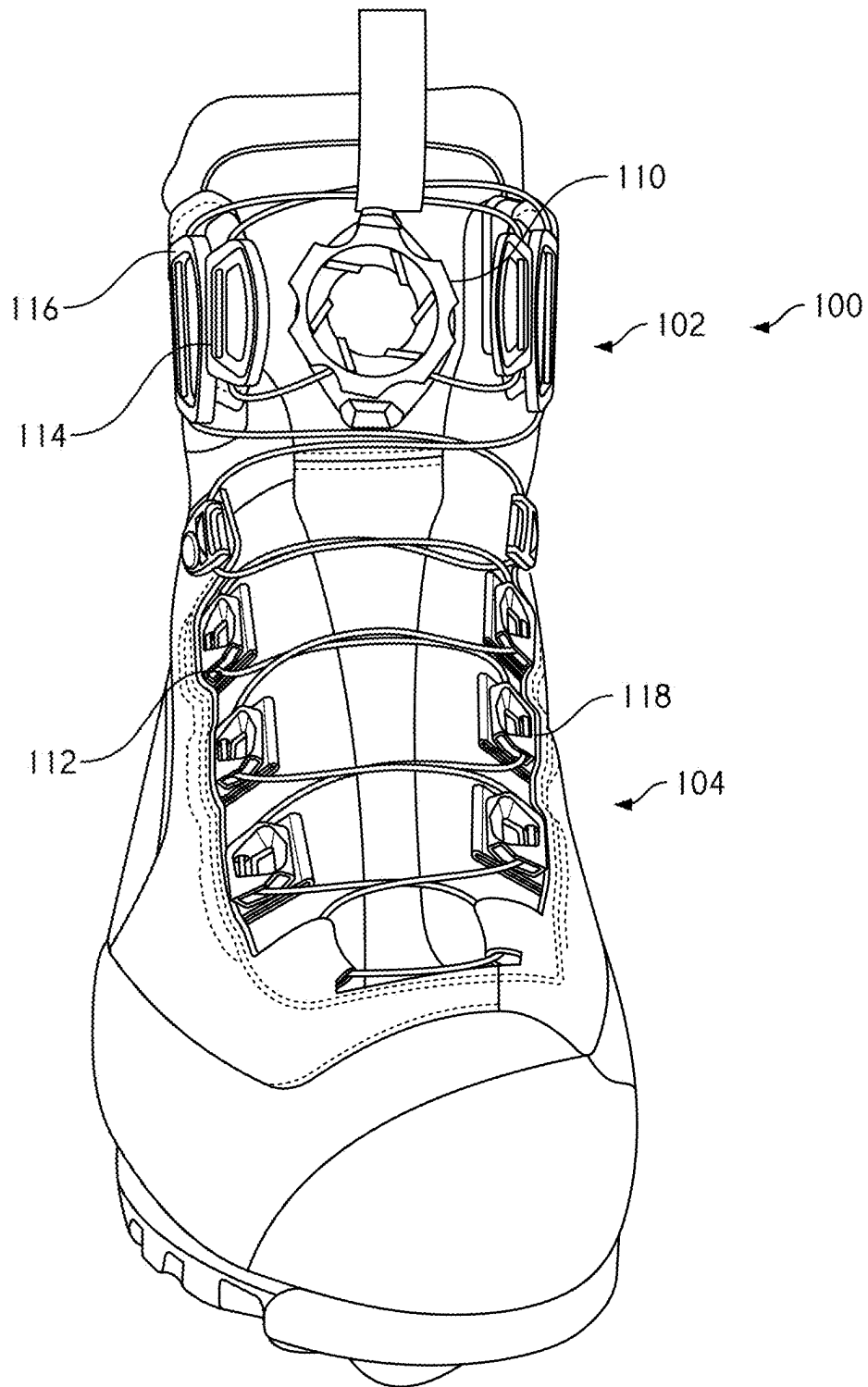


FIG. 1B

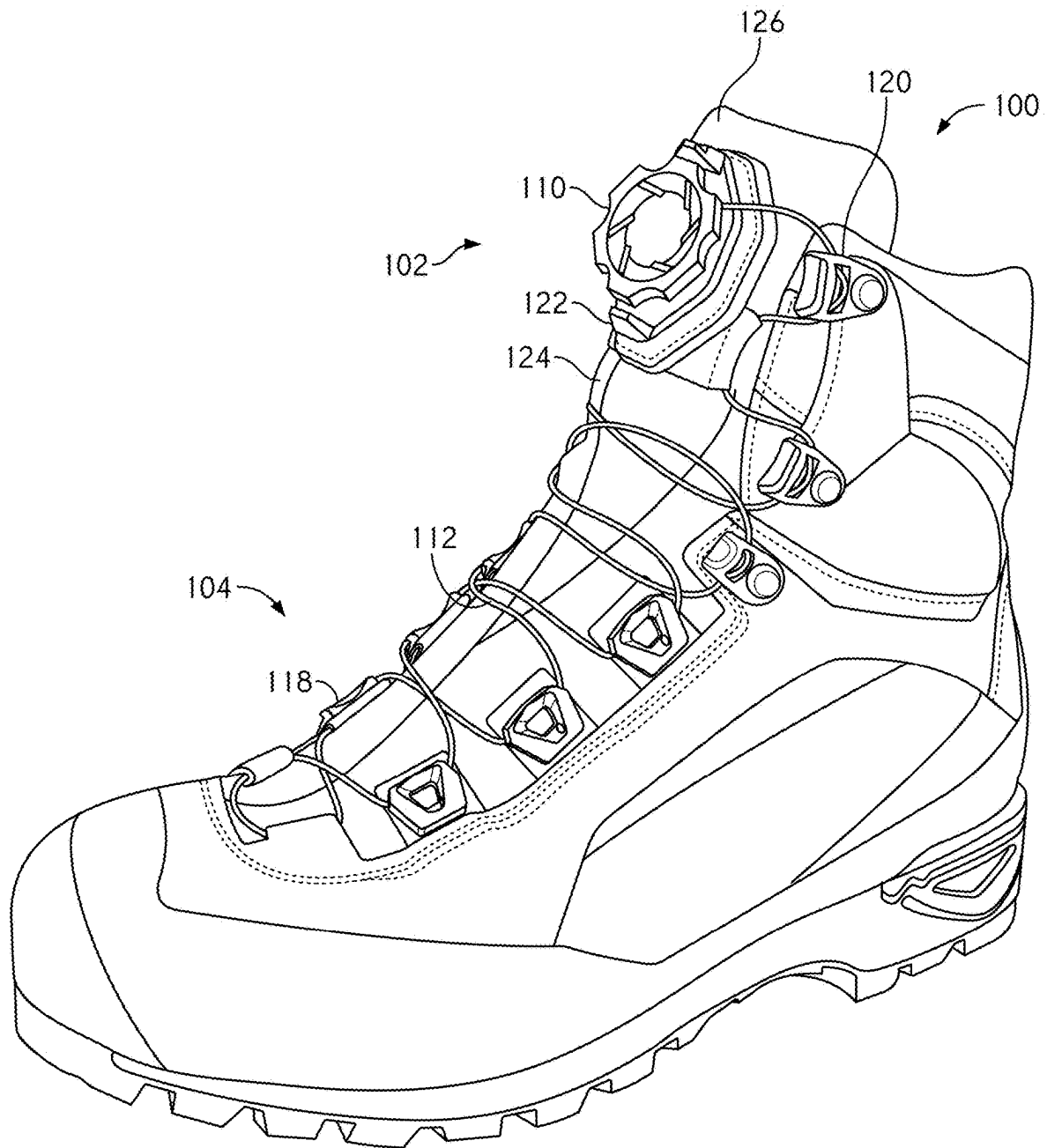


FIG. 2A

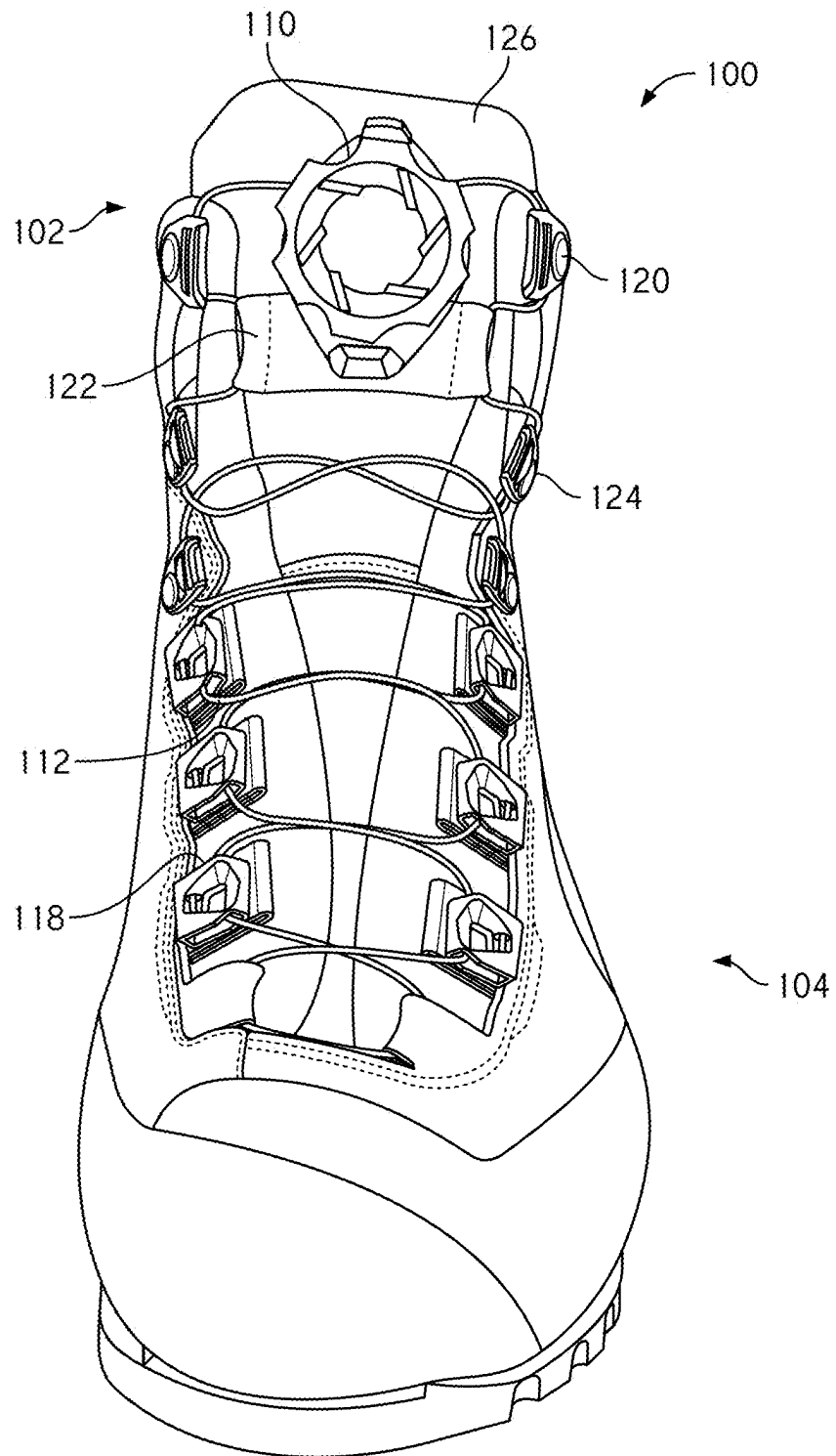


FIG. 2B

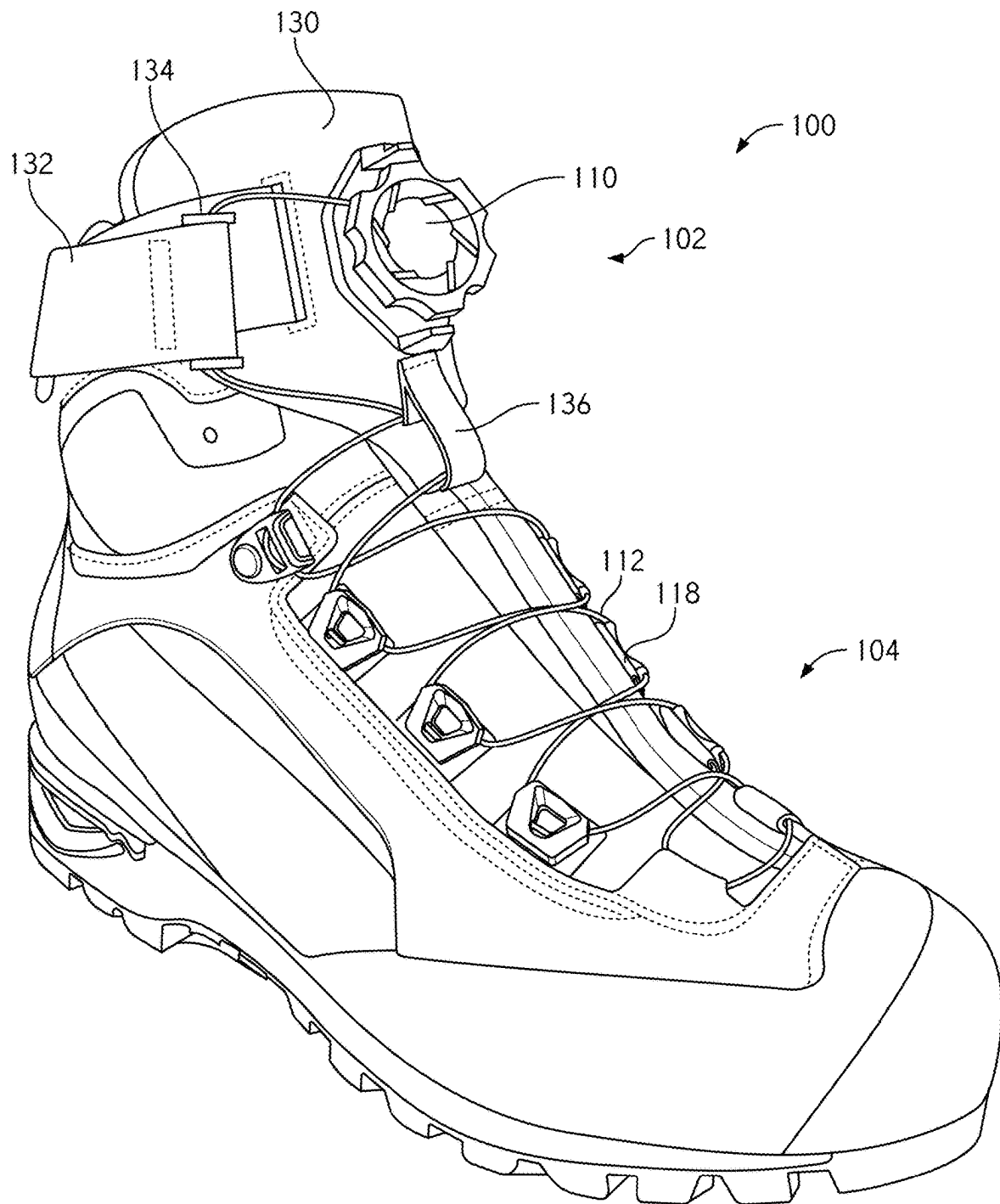


FIG. 3A

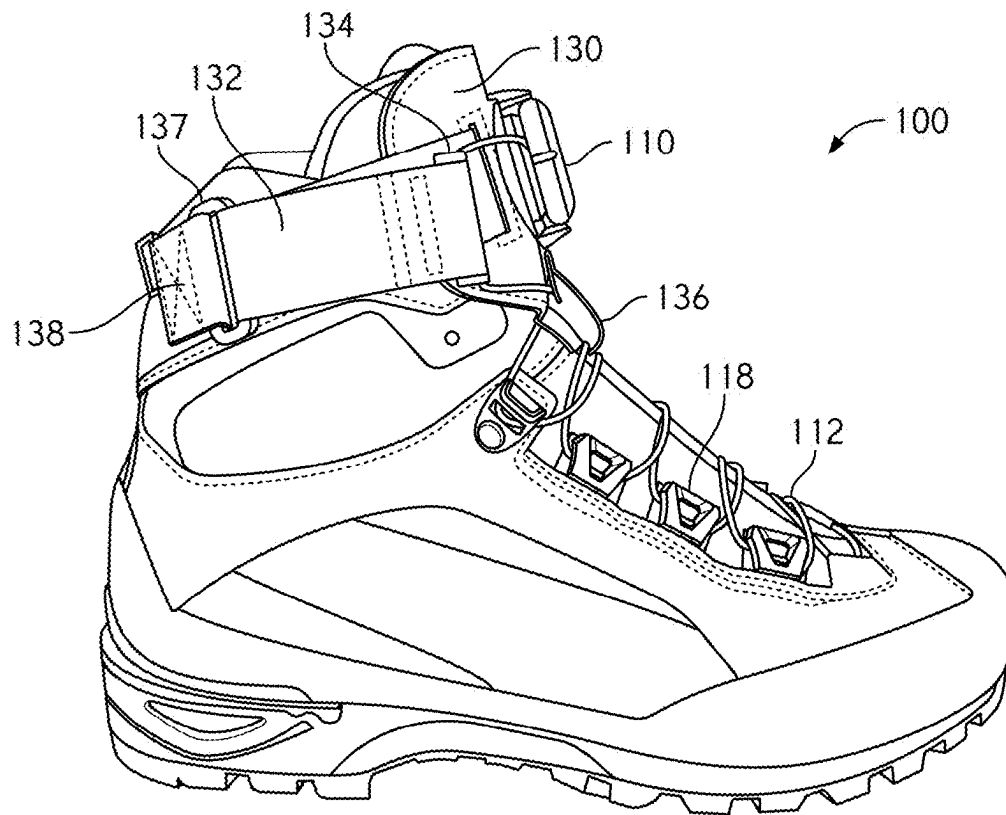


FIG. 3B

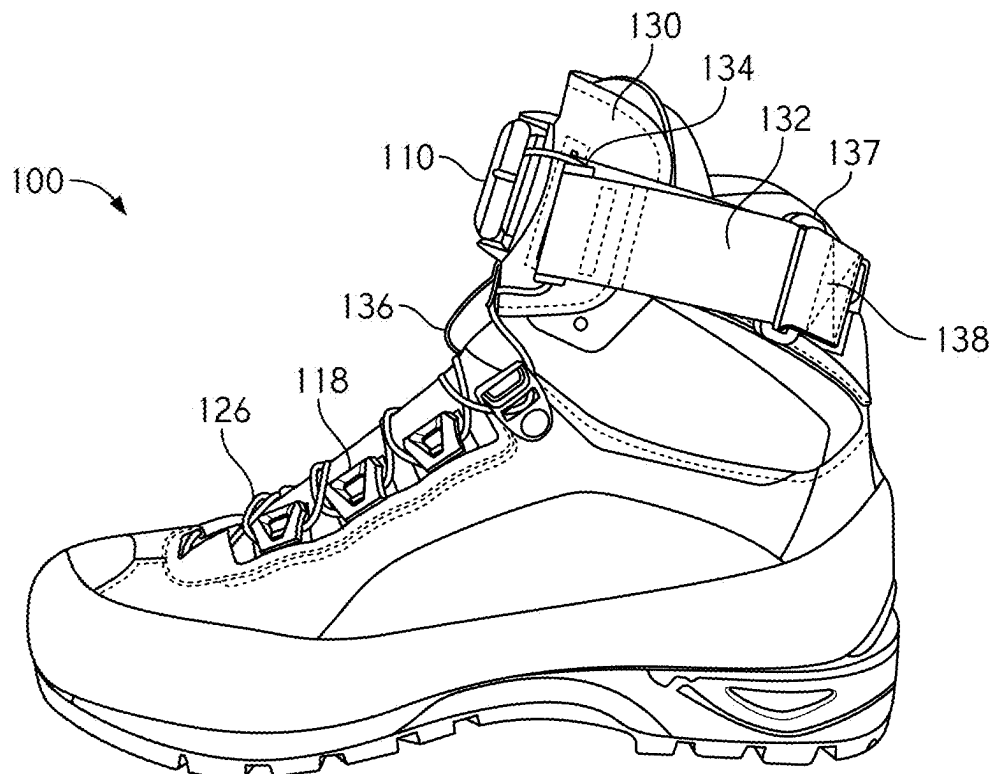


FIG. 3C

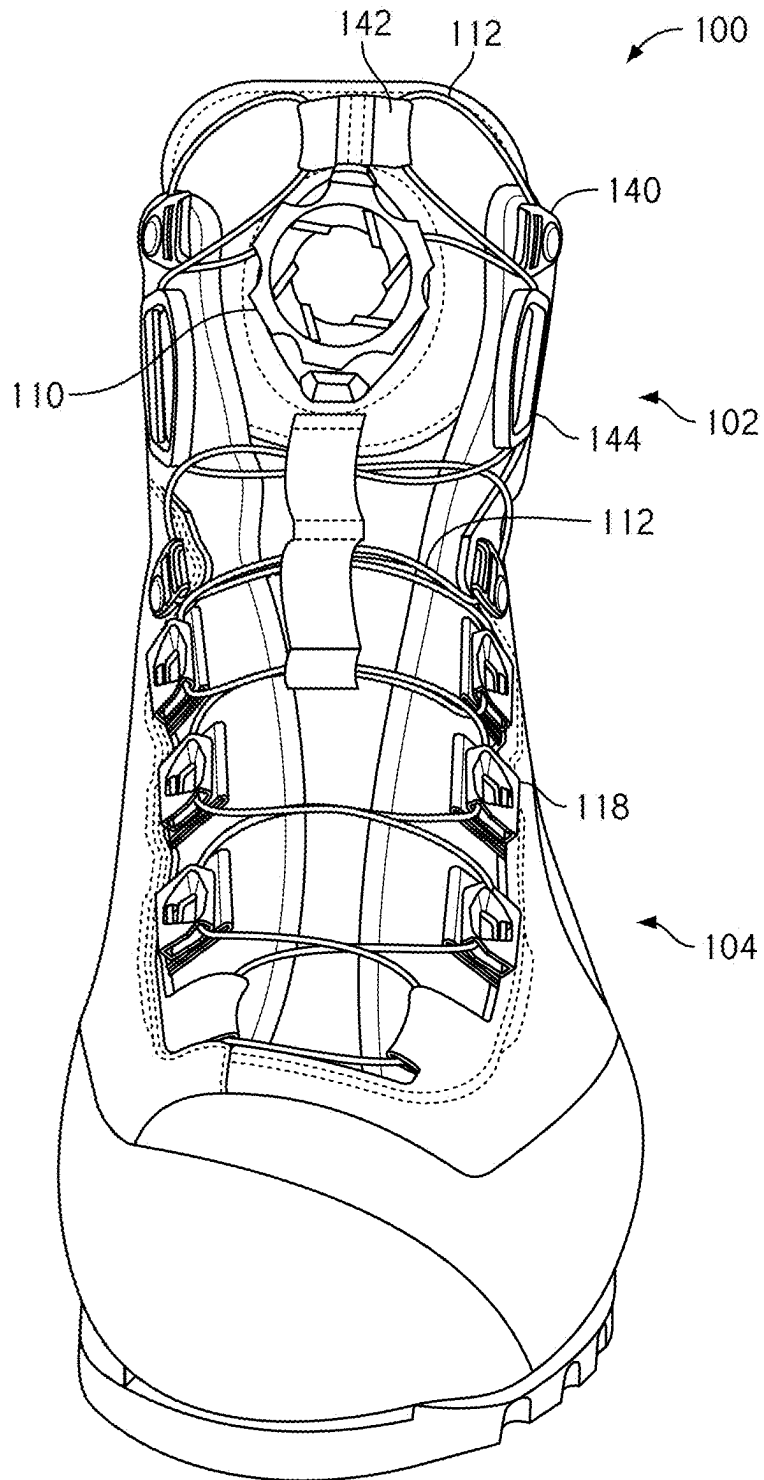


FIG. 4A

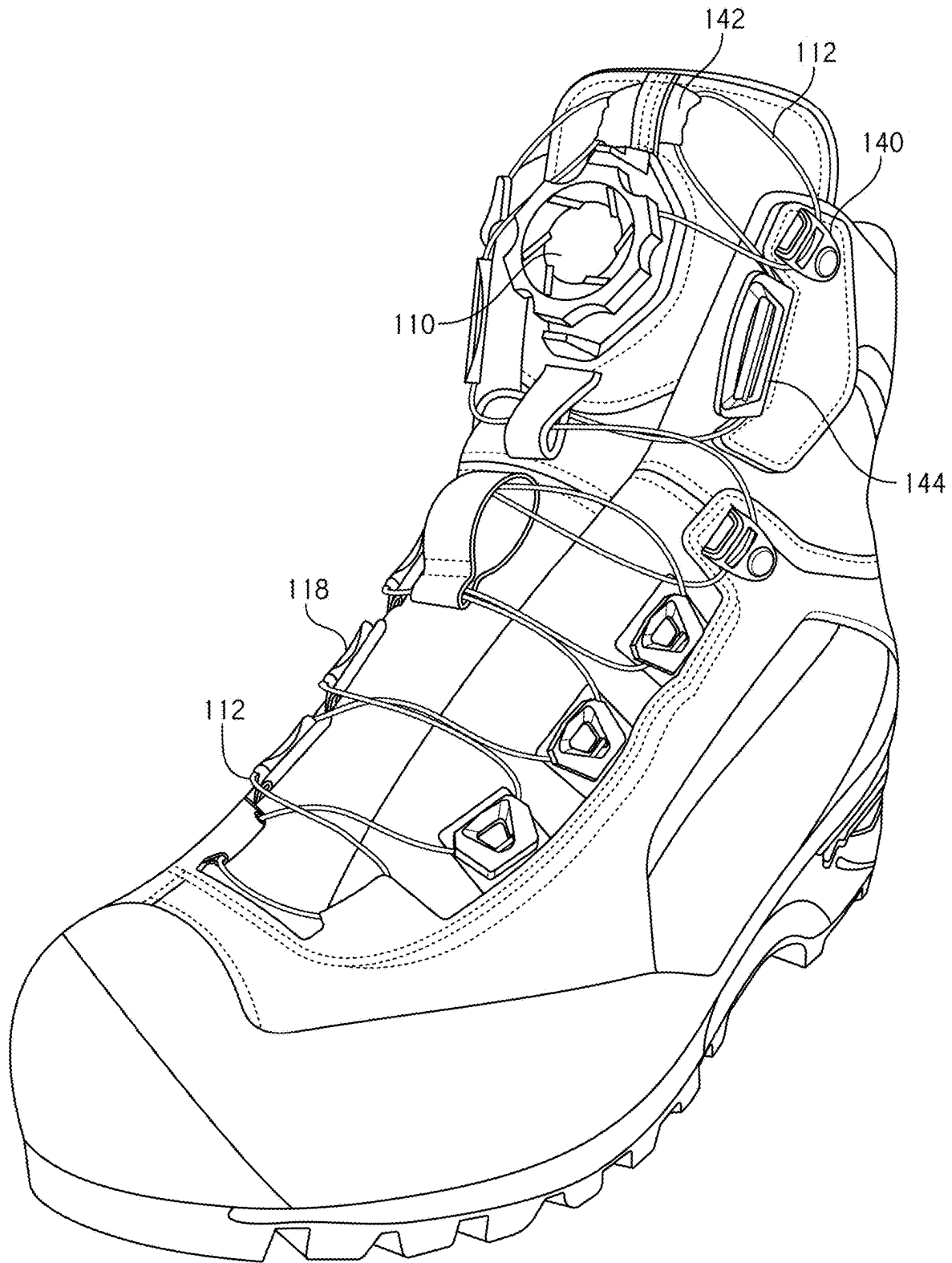


FIG. 4B

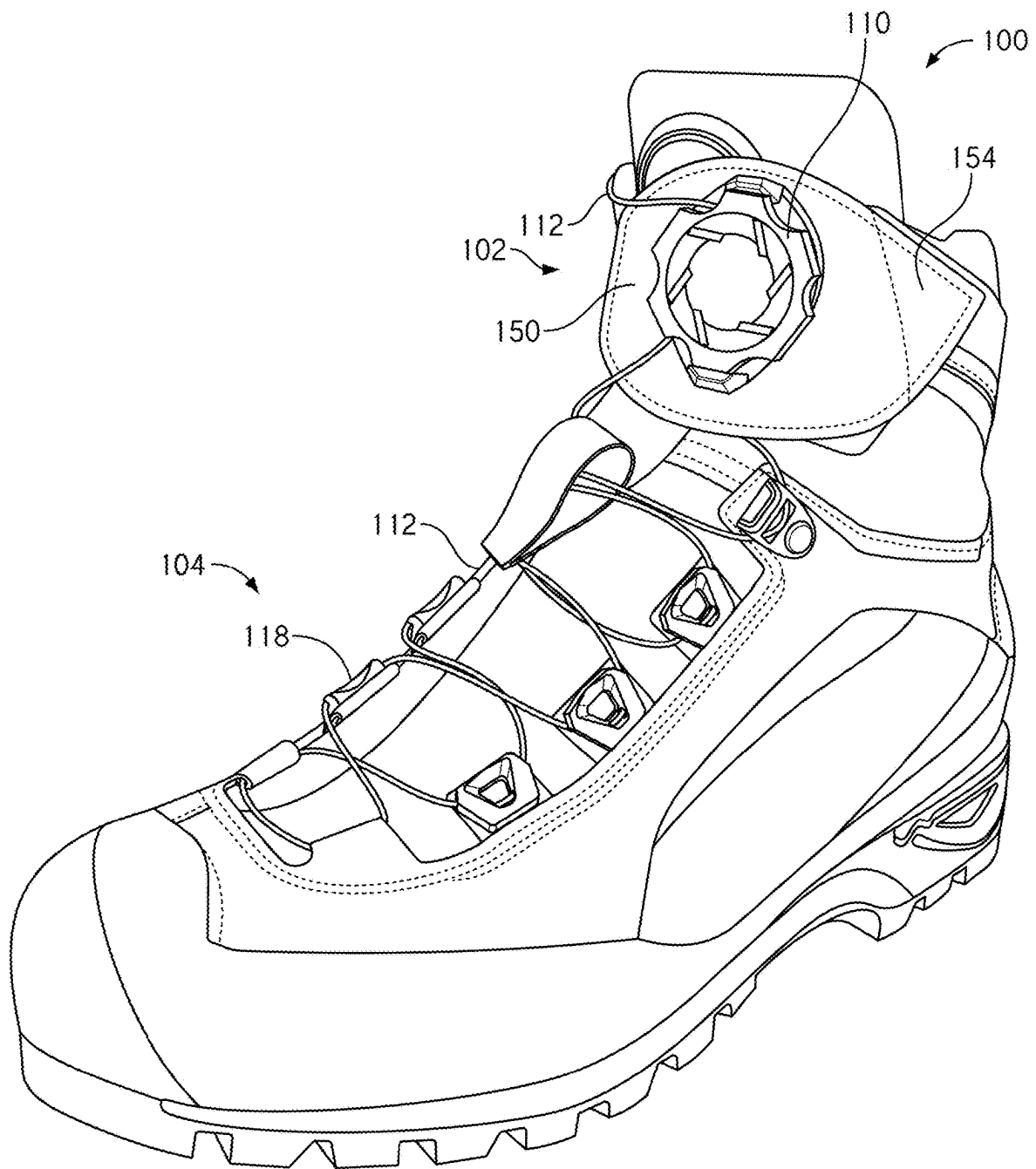


FIG. 5A

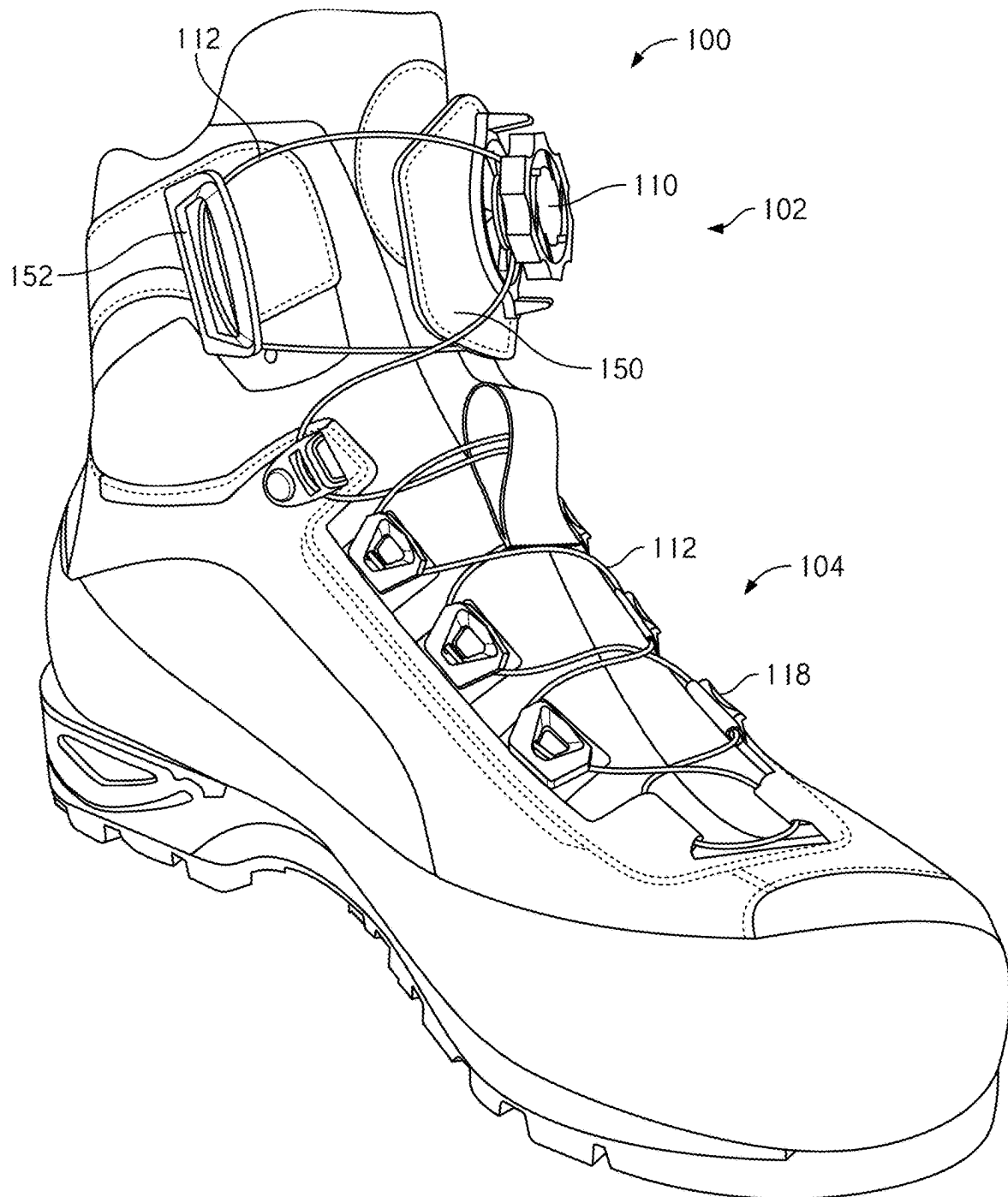


FIG. 5B

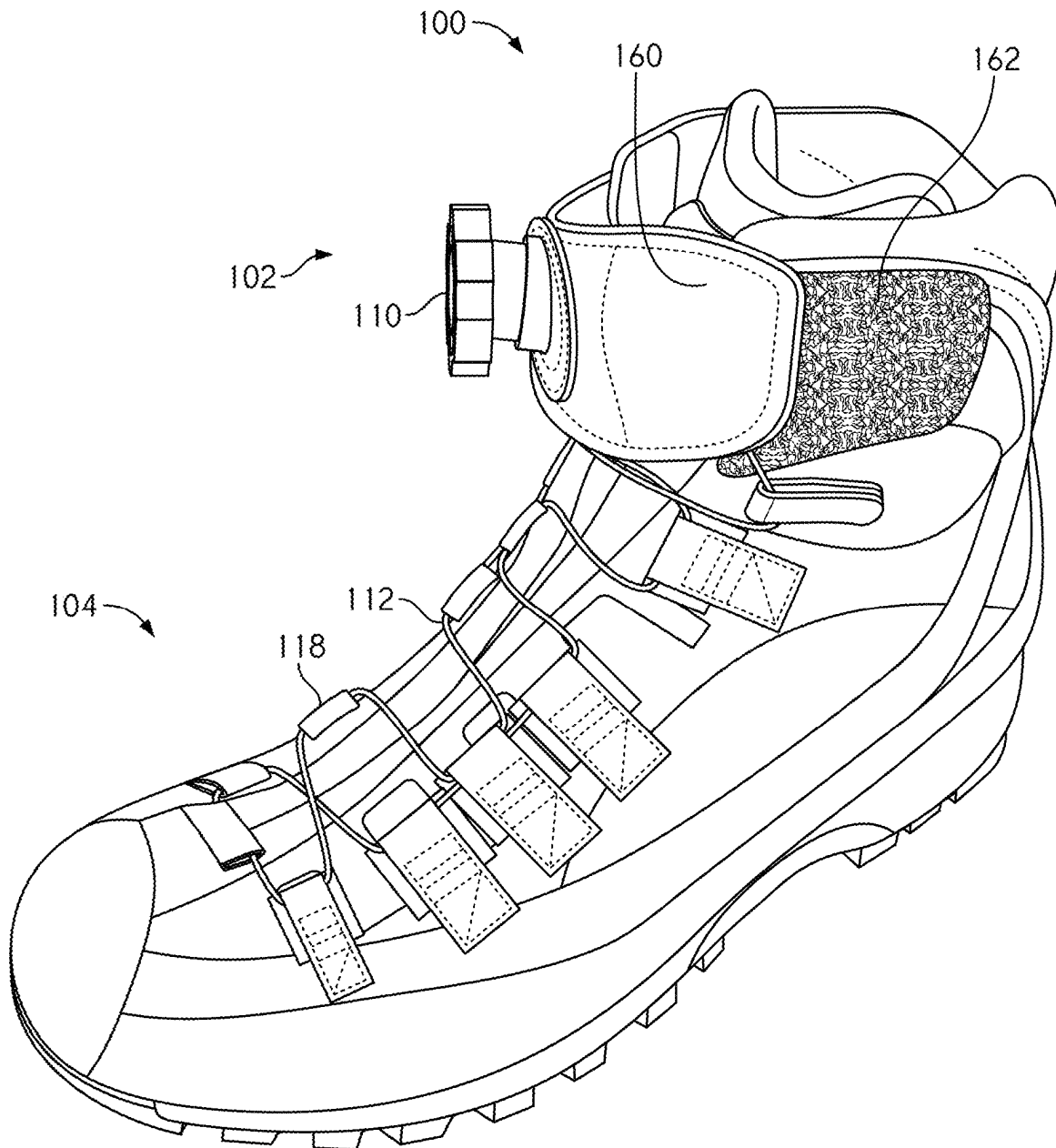


FIG. 6A

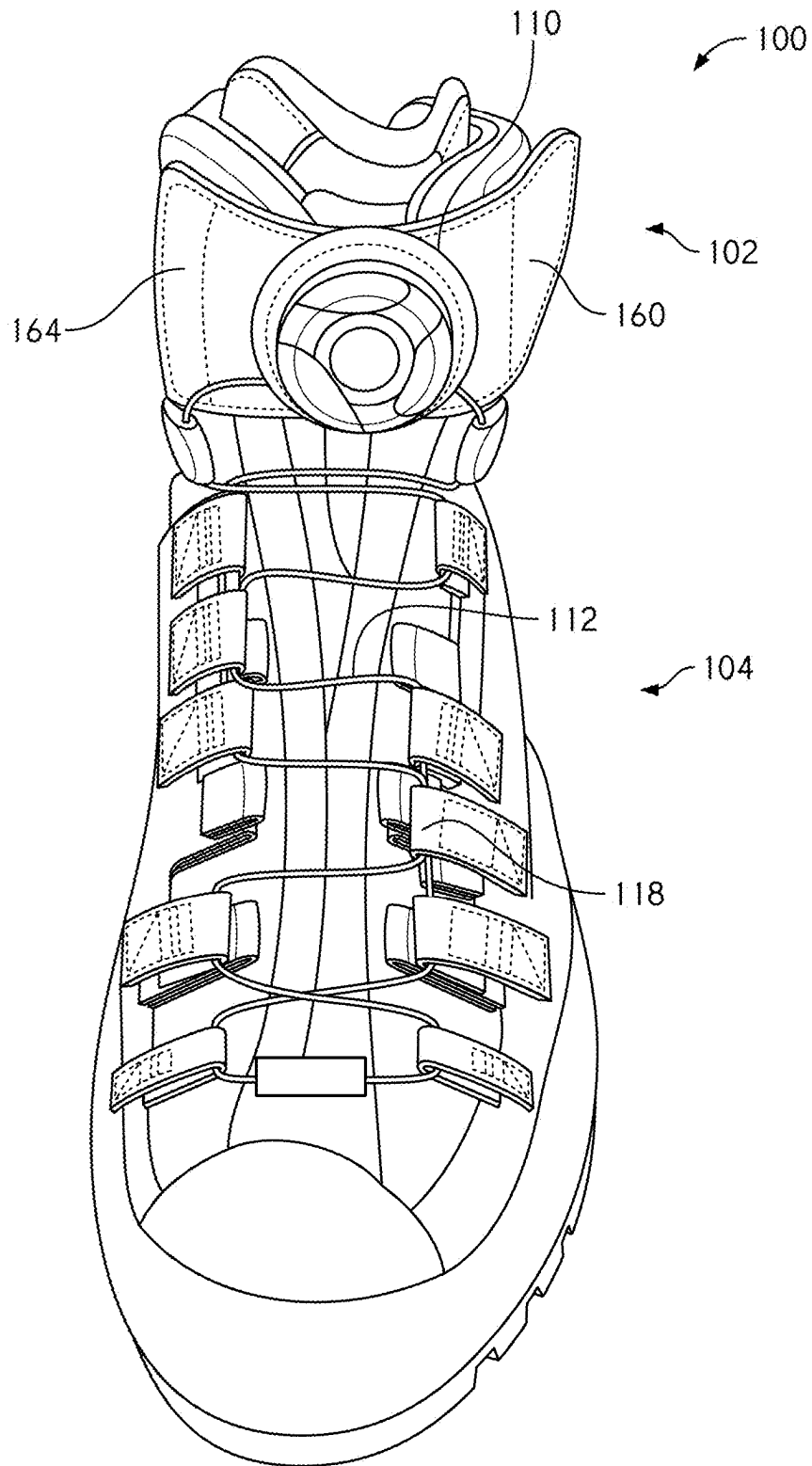


FIG. 6B

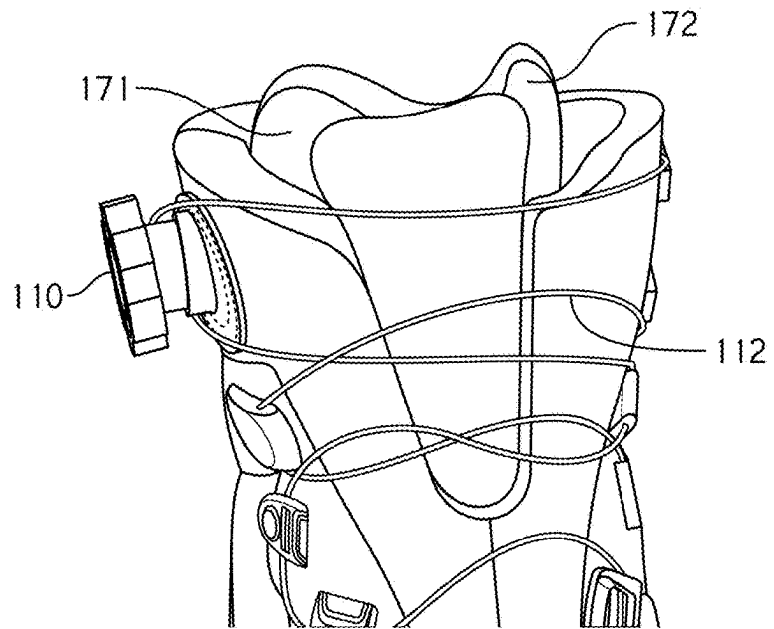


FIG. 7B

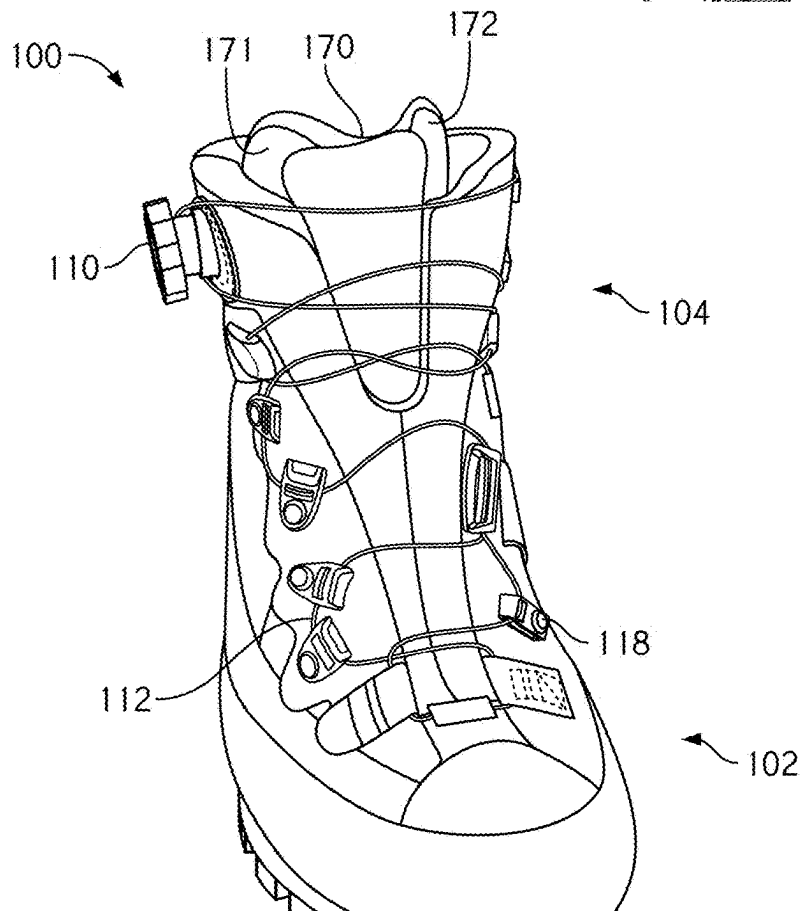


FIG. 7A

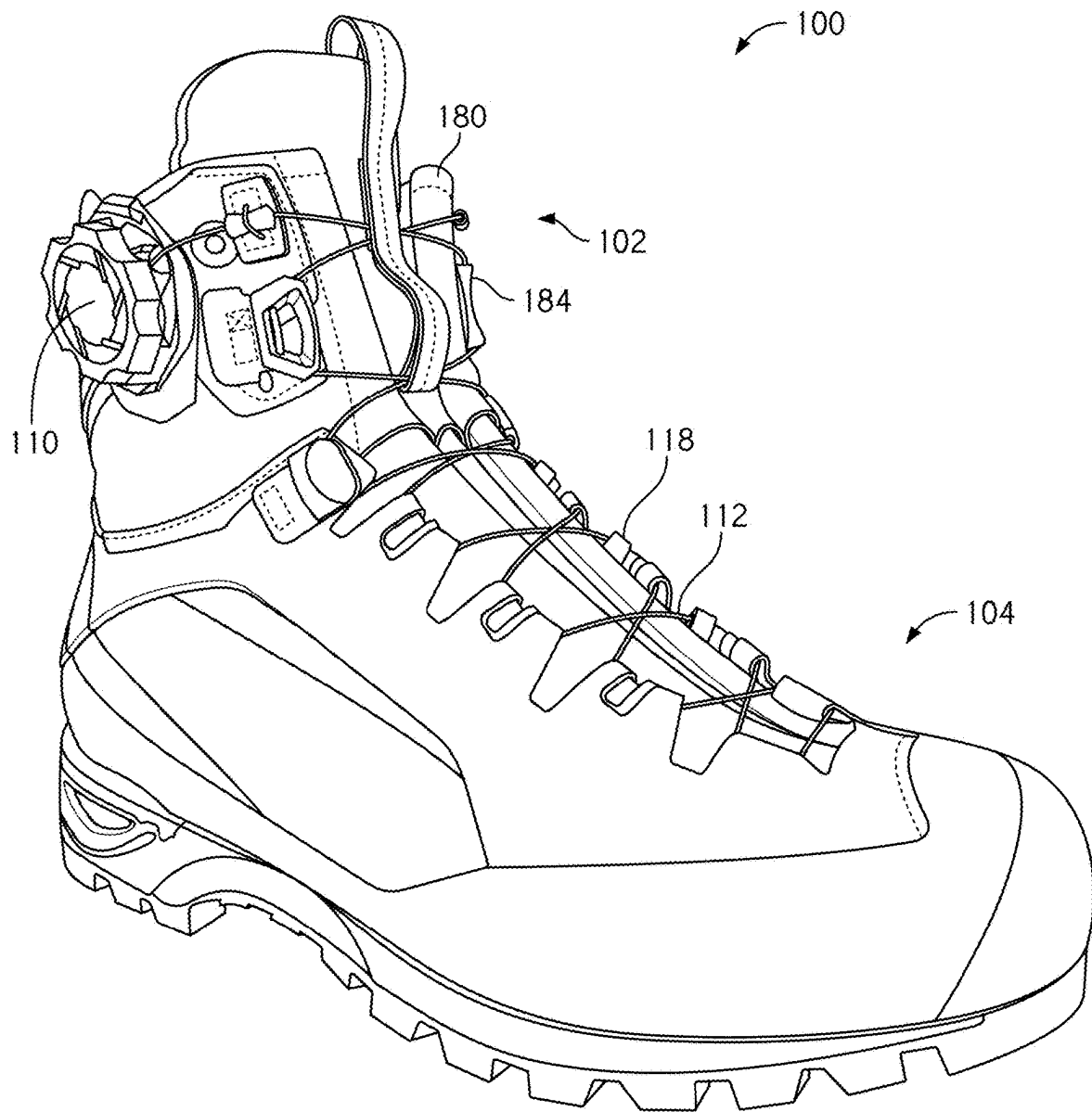


FIG. 8A

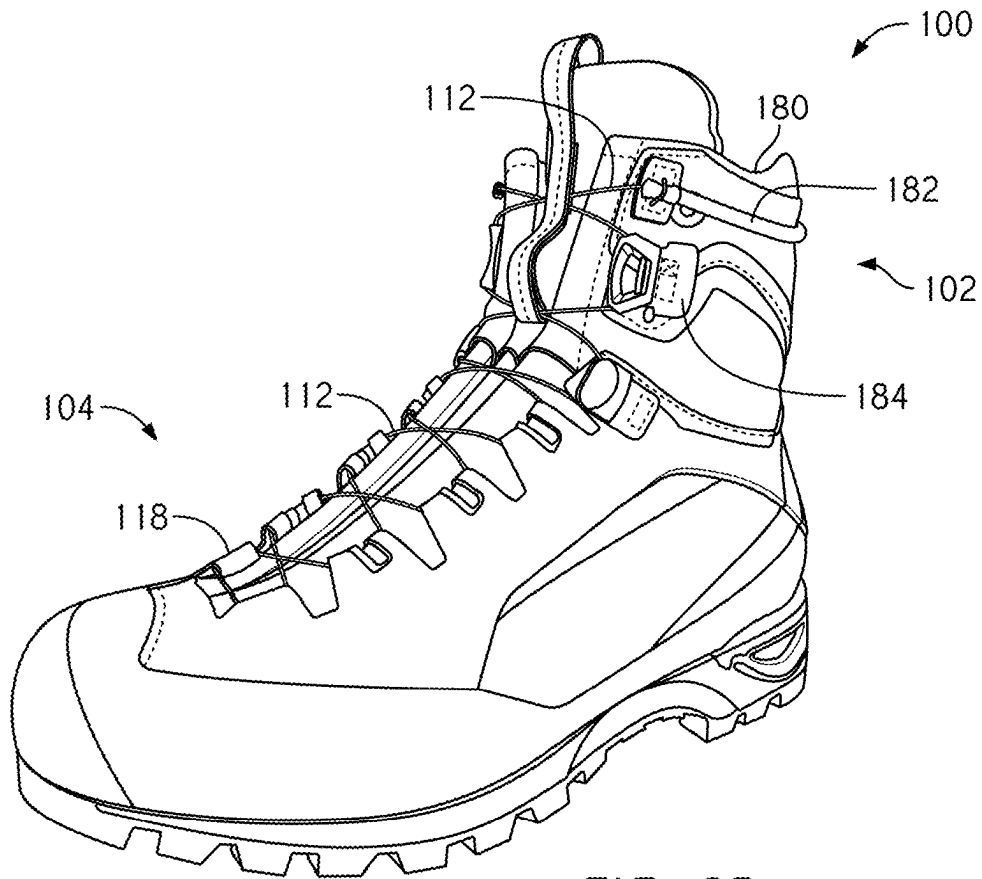


FIG. 8B

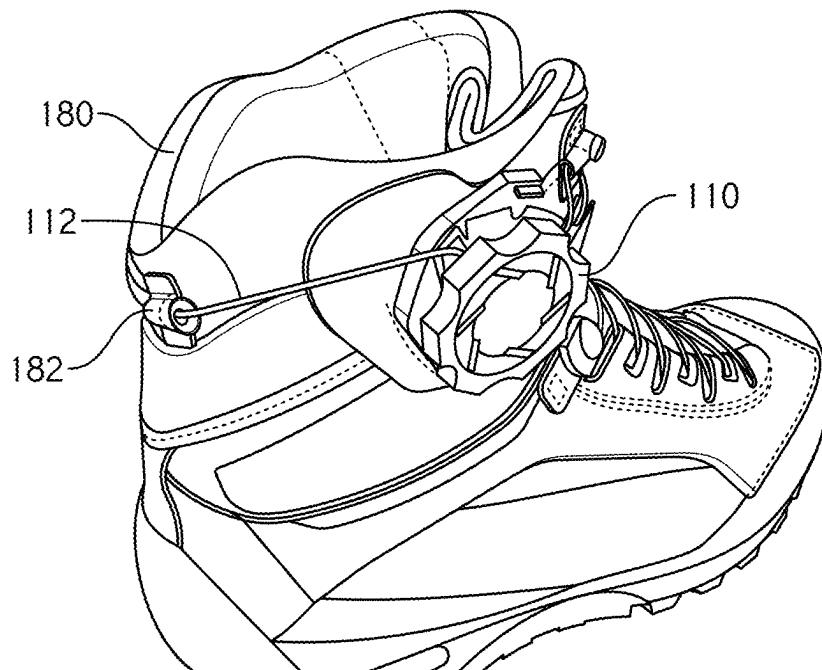


FIG. 8C

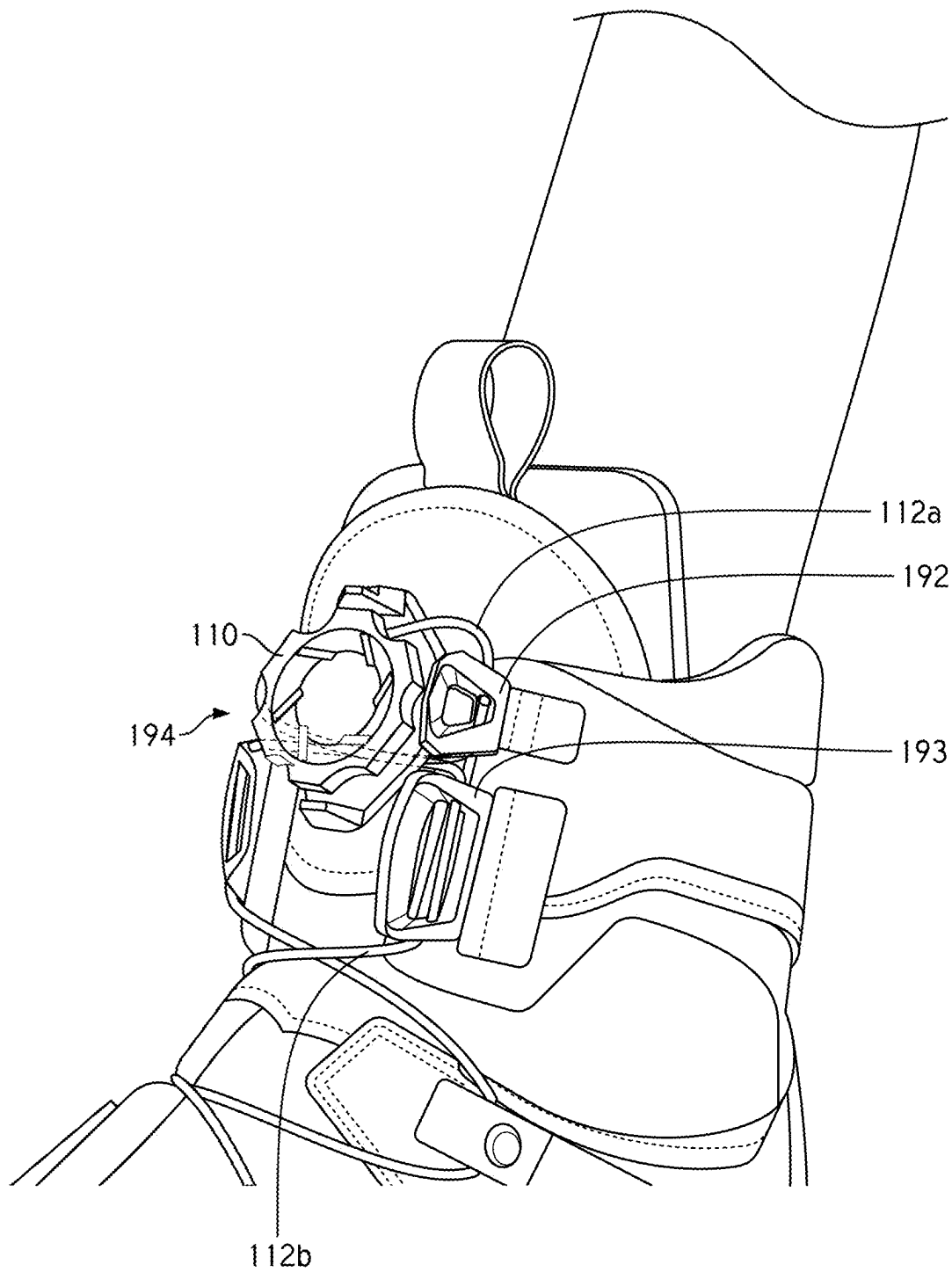
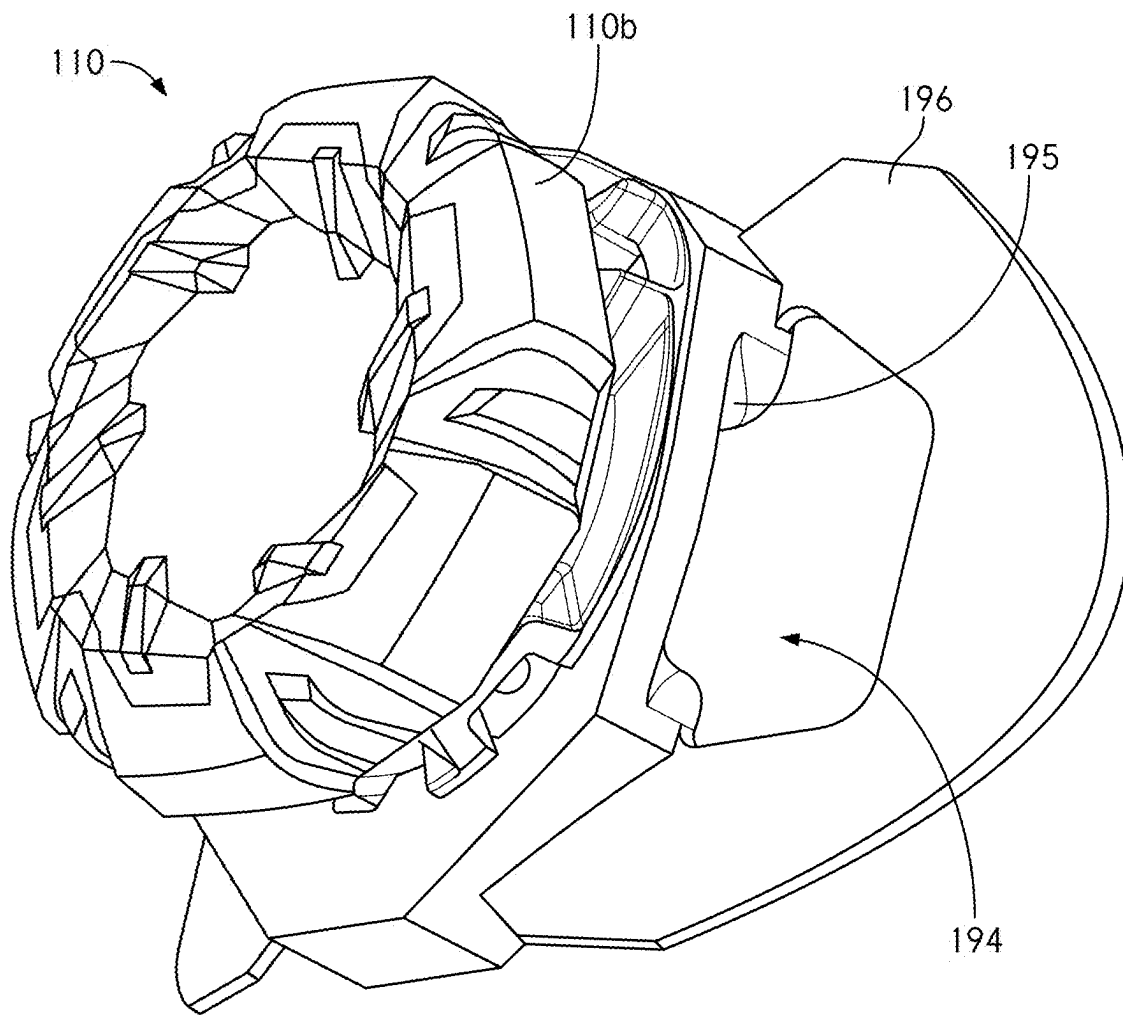


FIG. 9A

*FIG. 9B*

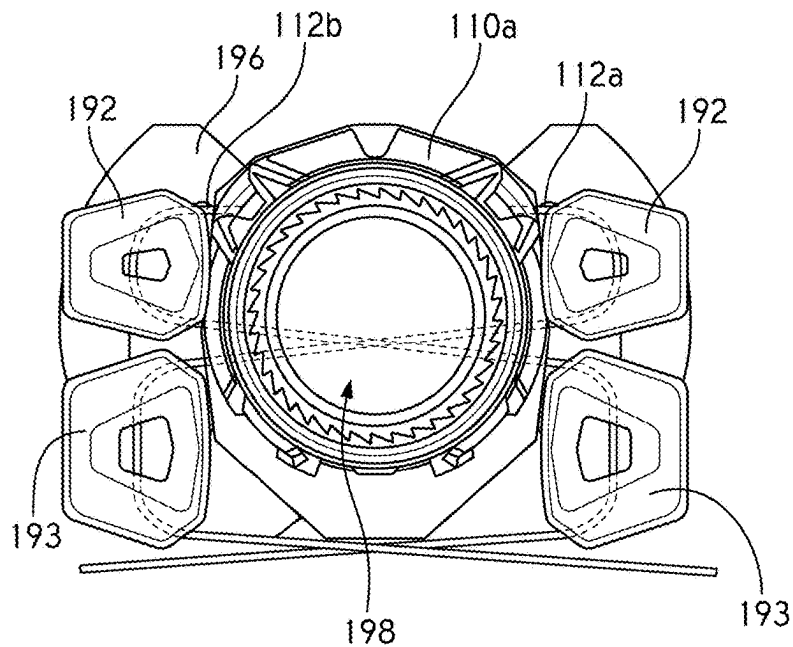


FIG. 9C

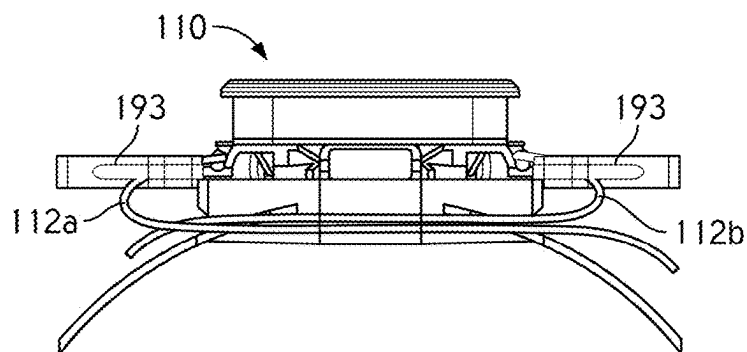


FIG. 9D

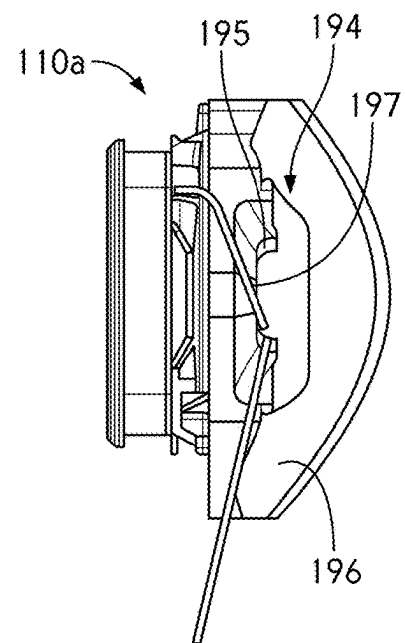
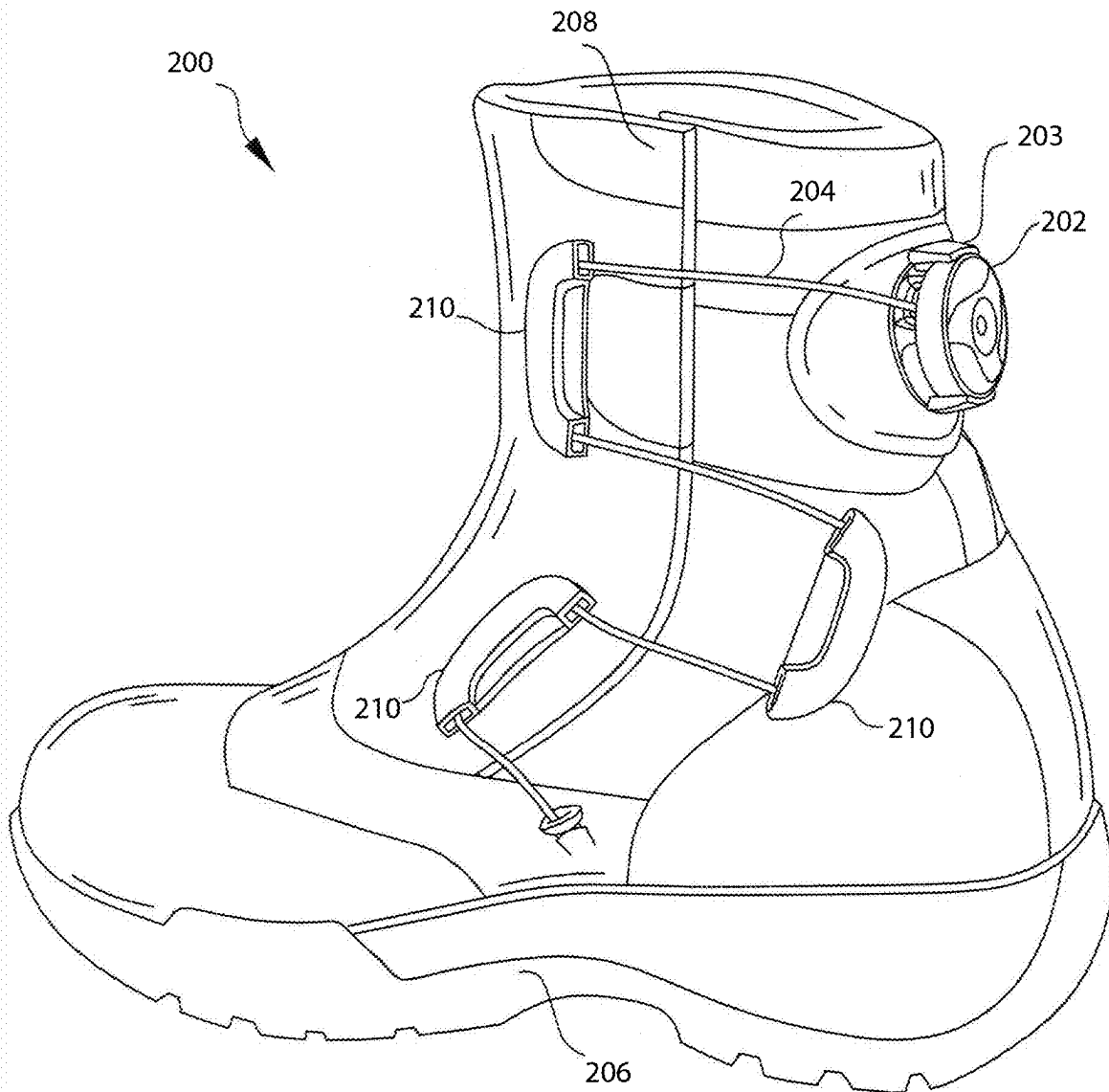
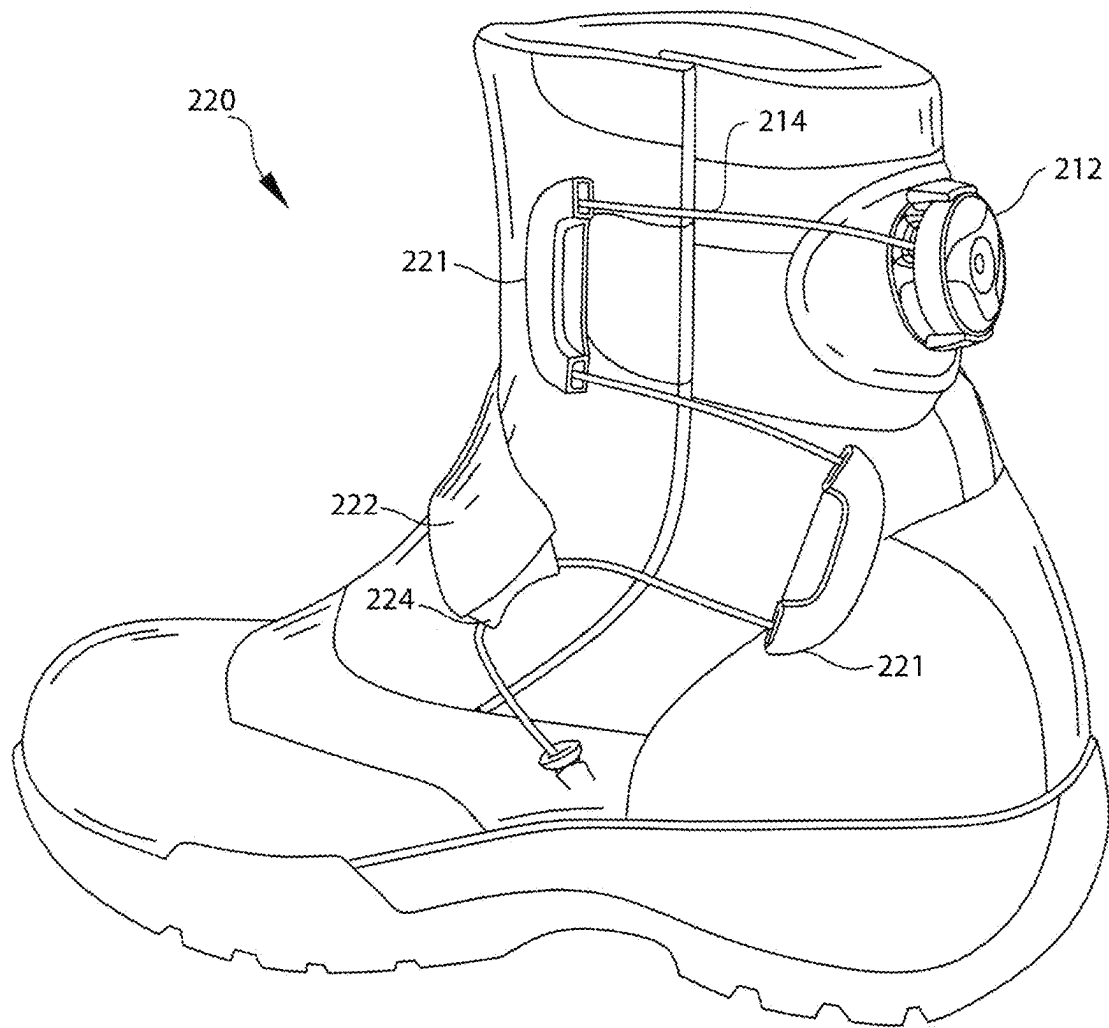
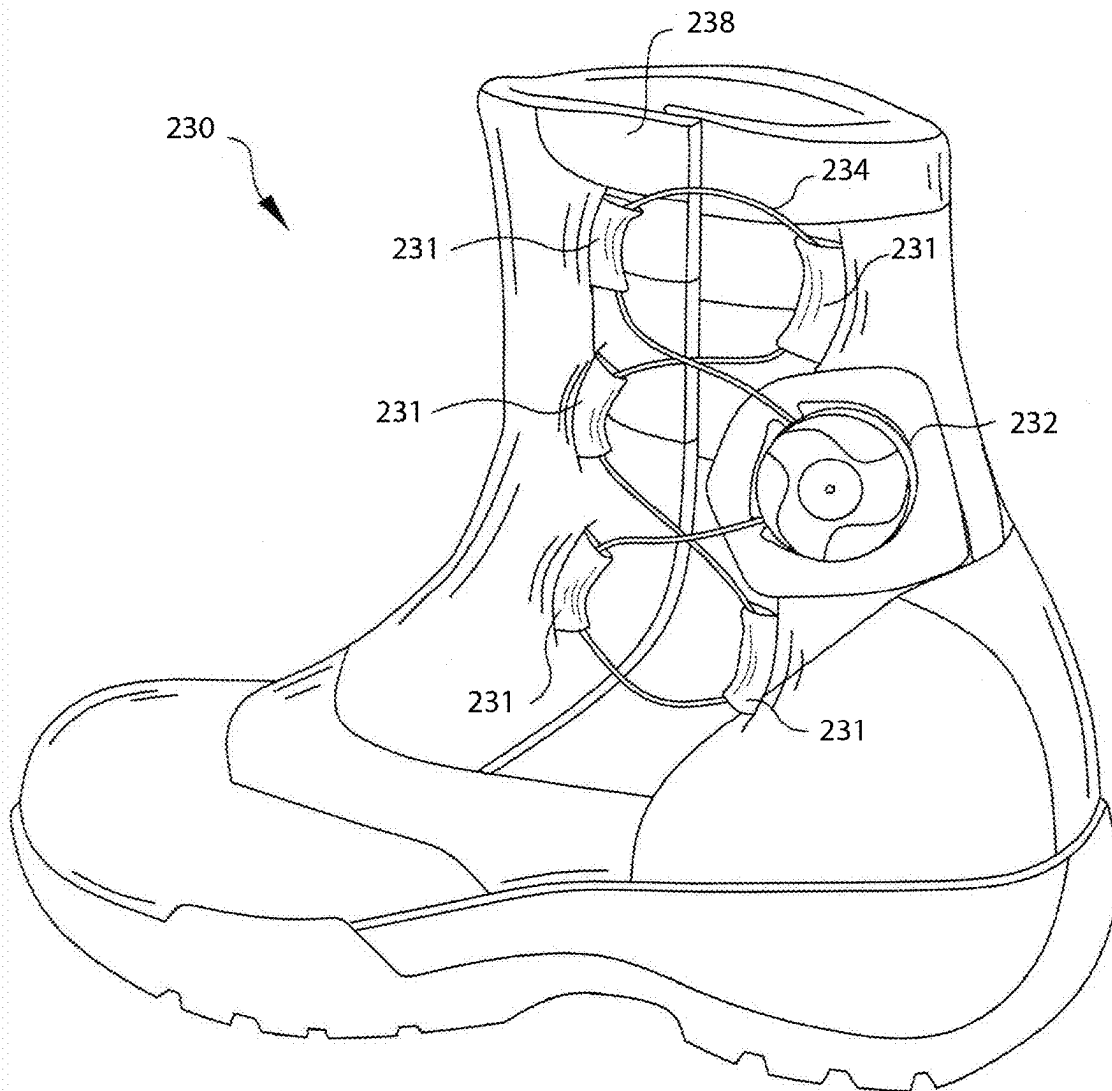
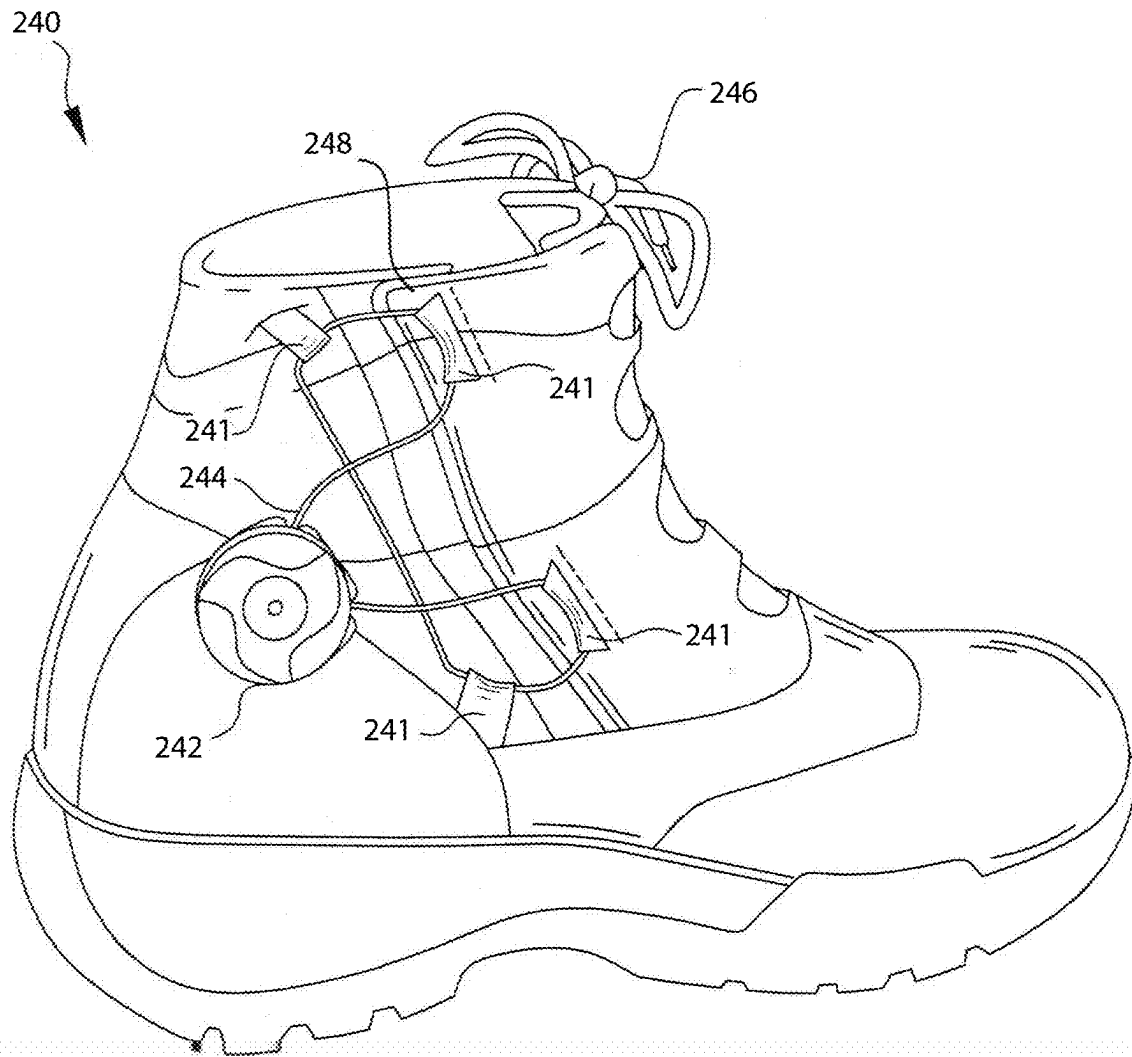


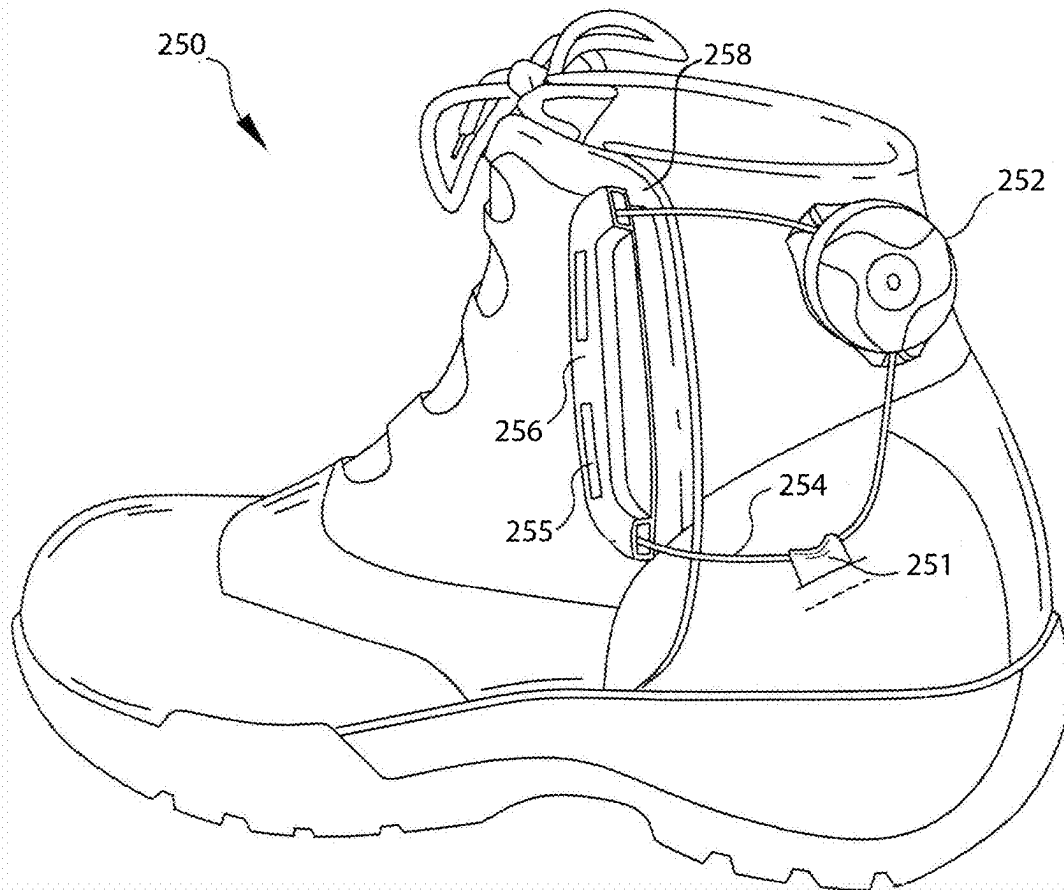
FIG. 9E

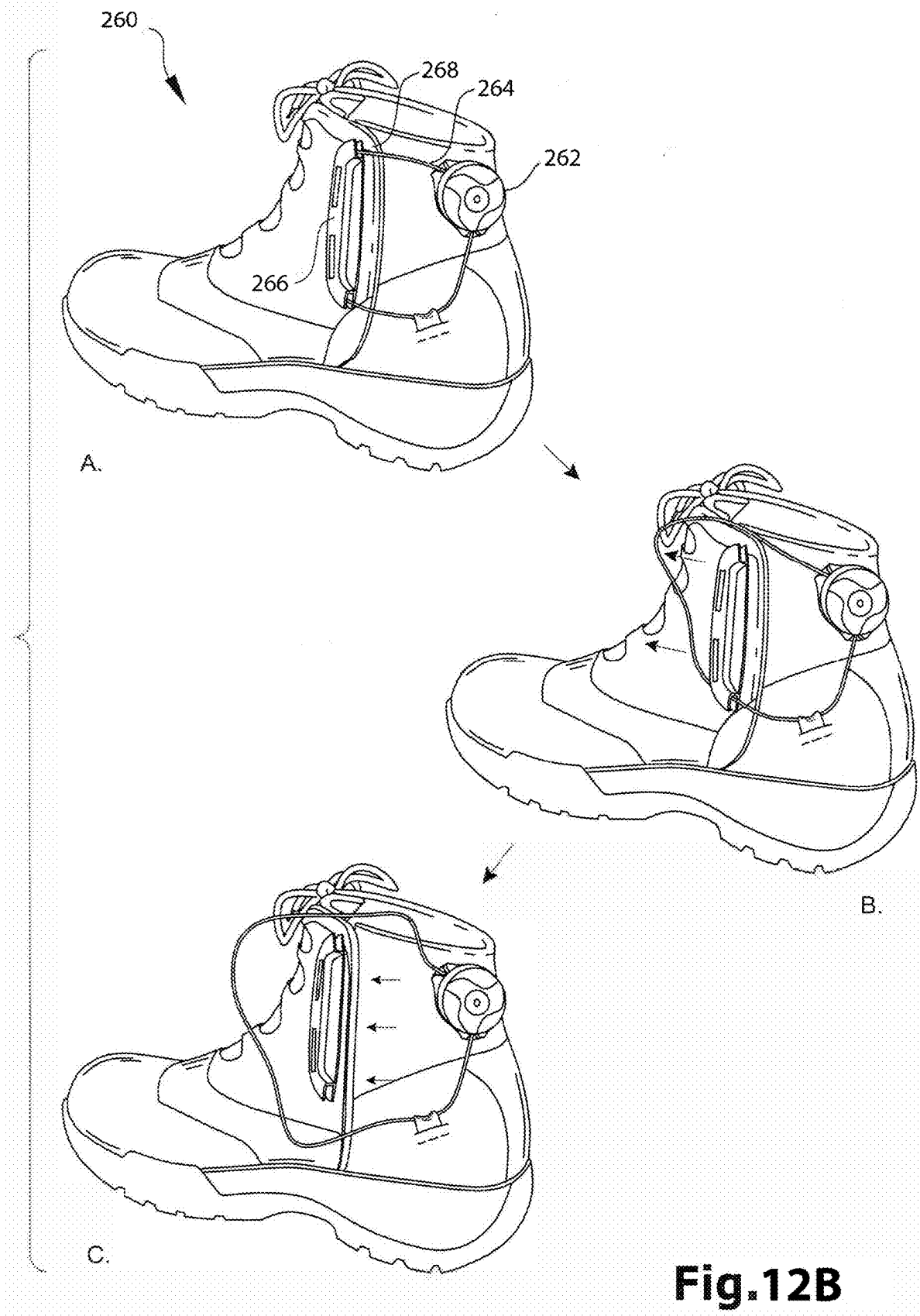
**Fig. 10A**

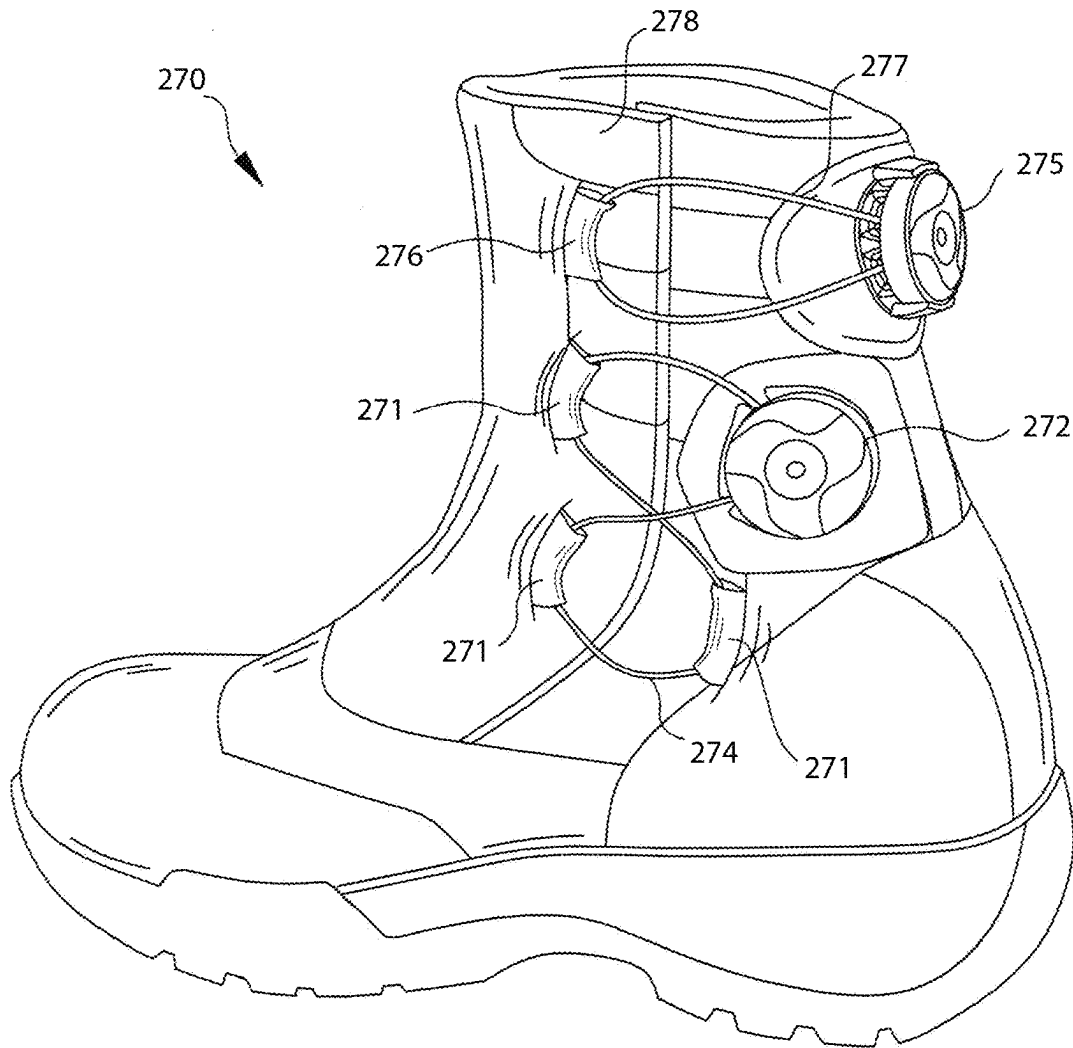
**Fig. 10B**

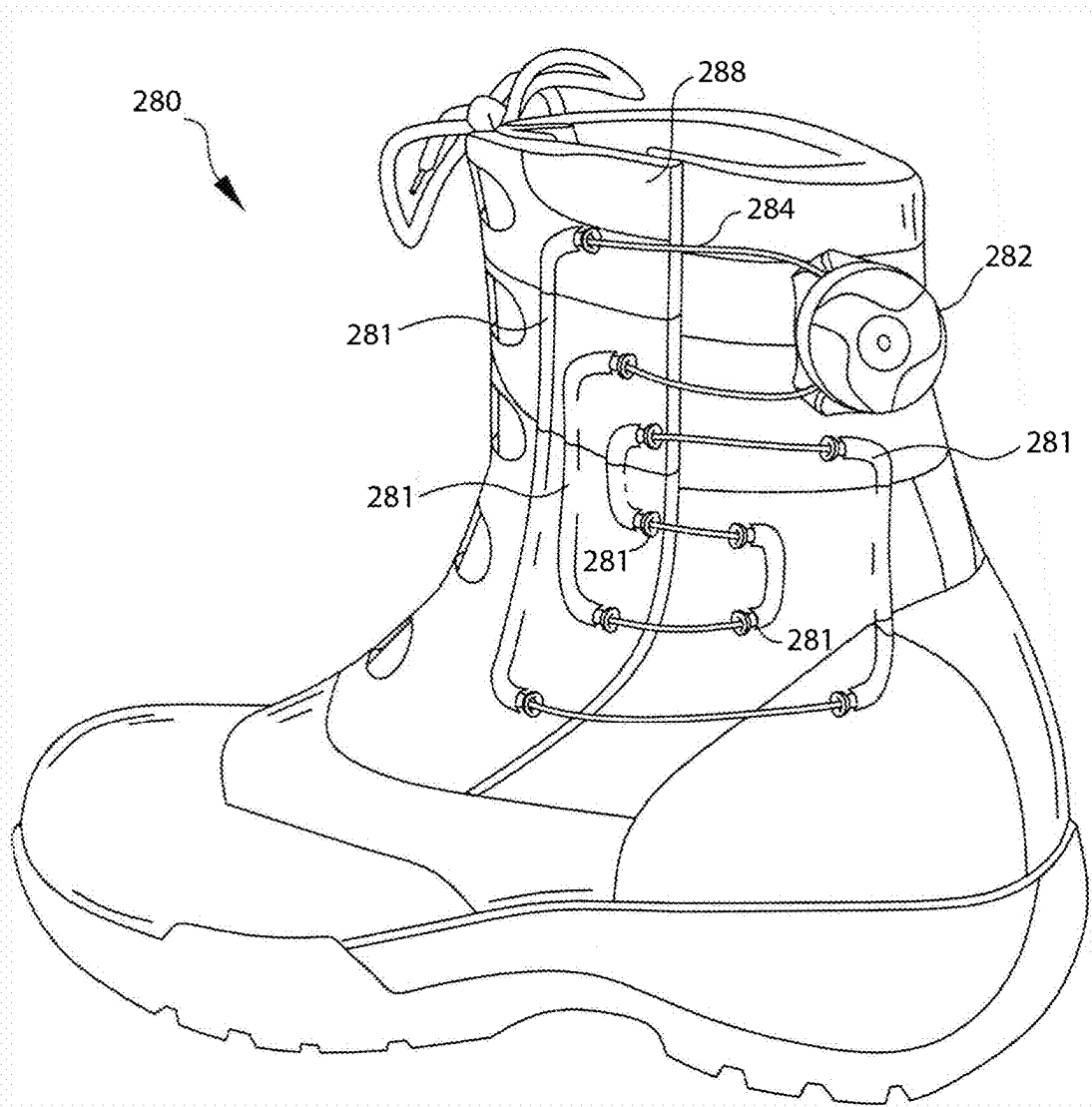
**Fig. 11A**

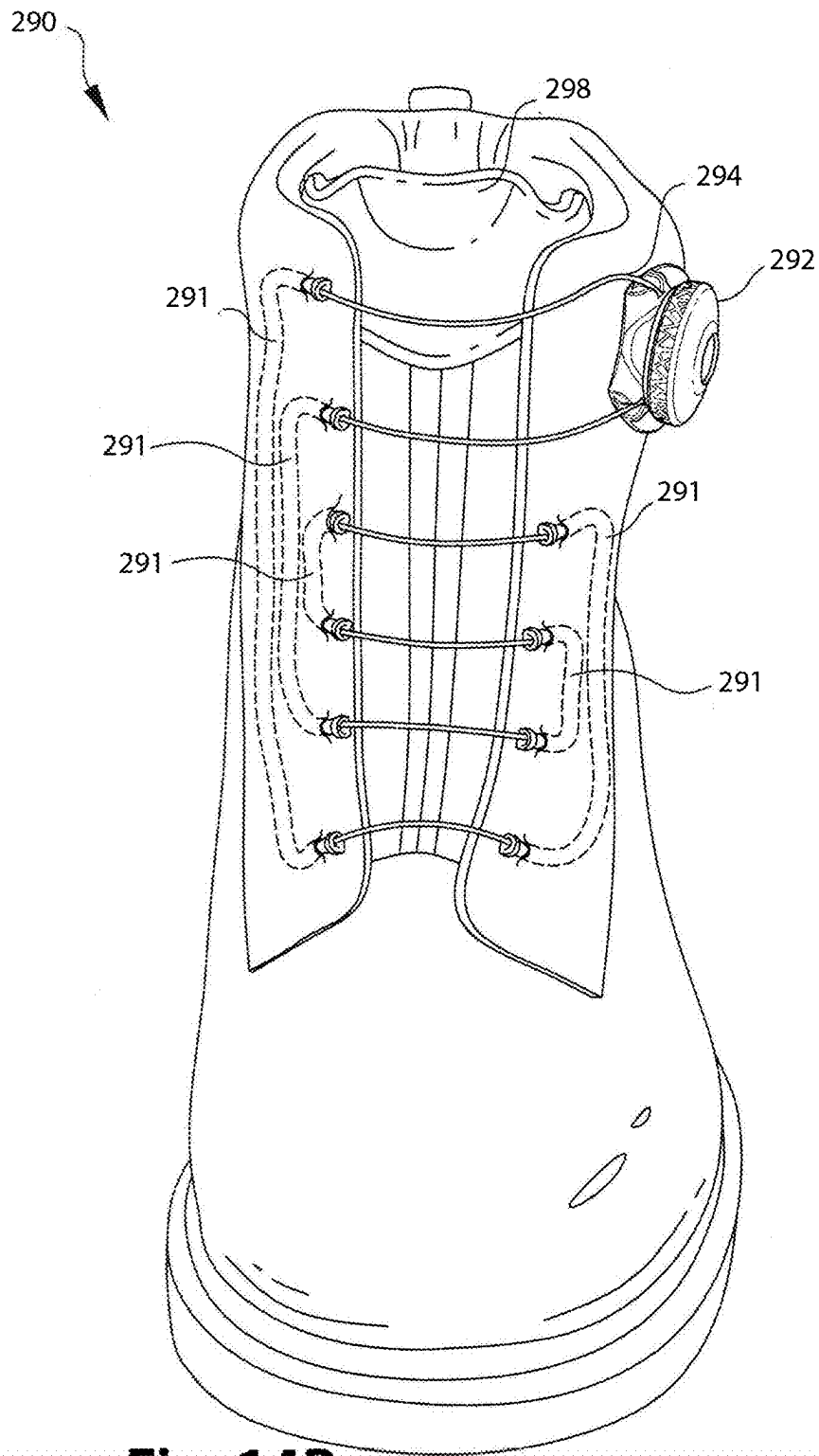
**Fig. 11B**

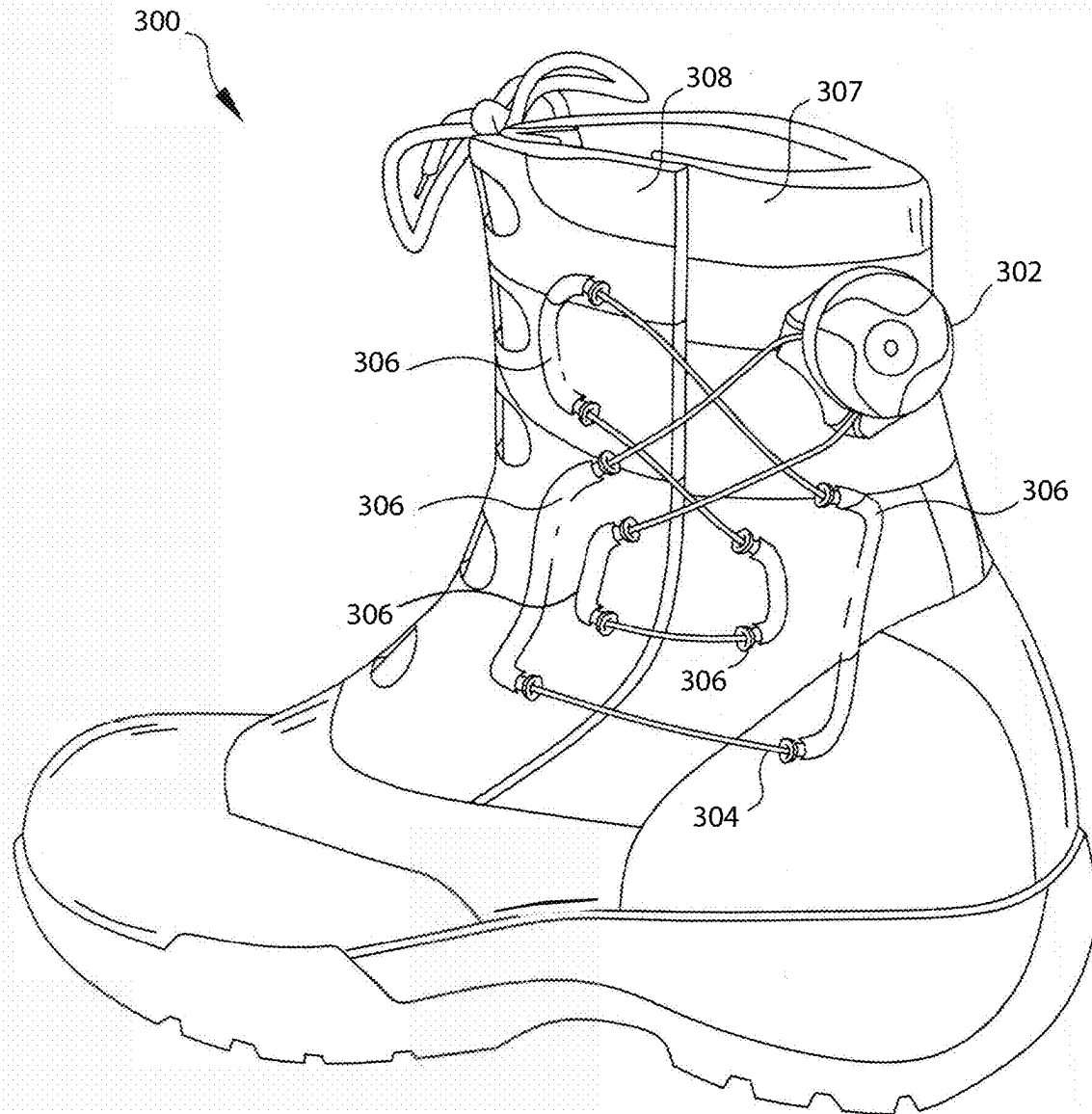
**Fig. 12A**



**Fig. 13**

**Fig. 14A**

**Fig. 14B**

**Fig. 15A**

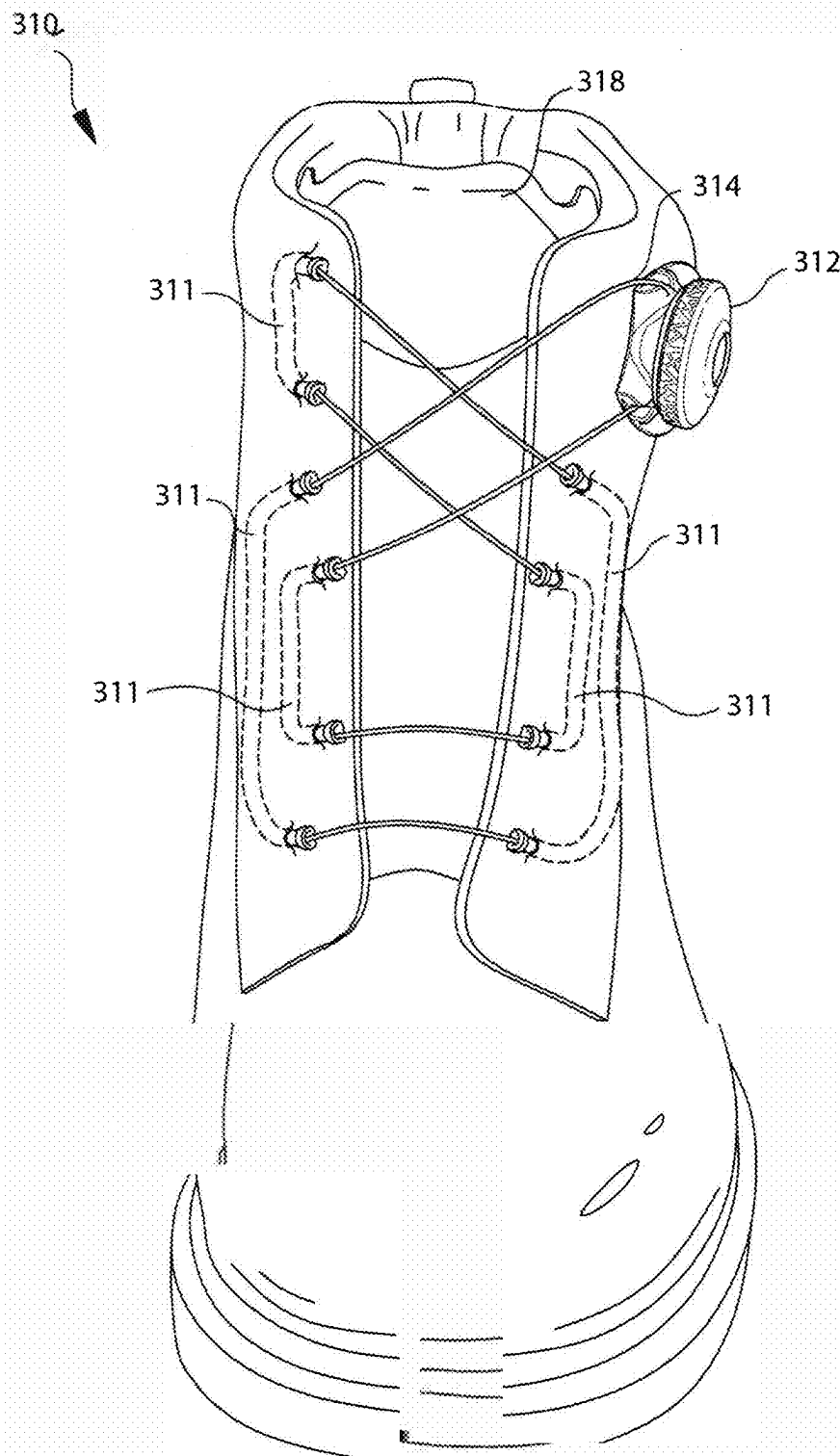


Fig. 15B

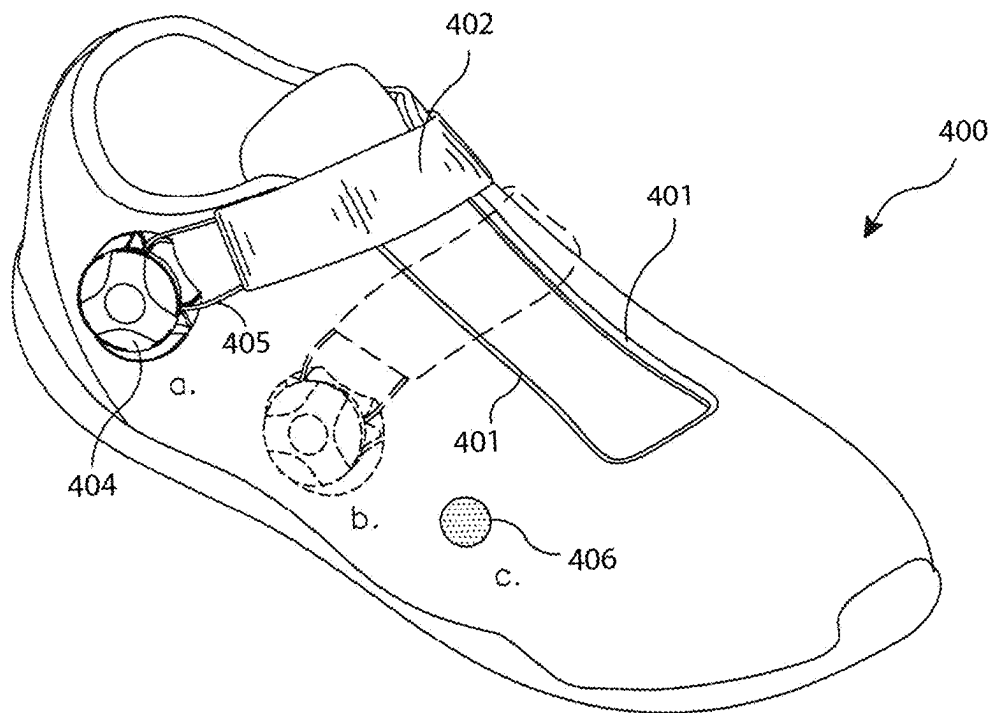


Fig. 16

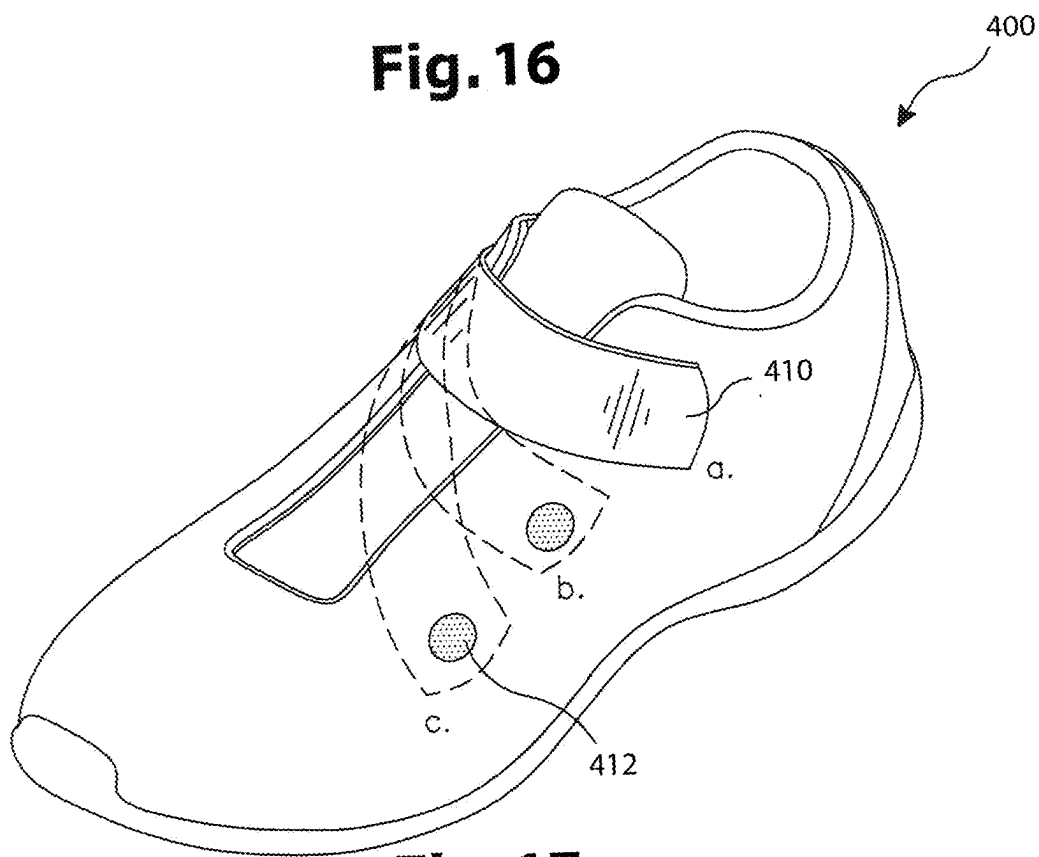
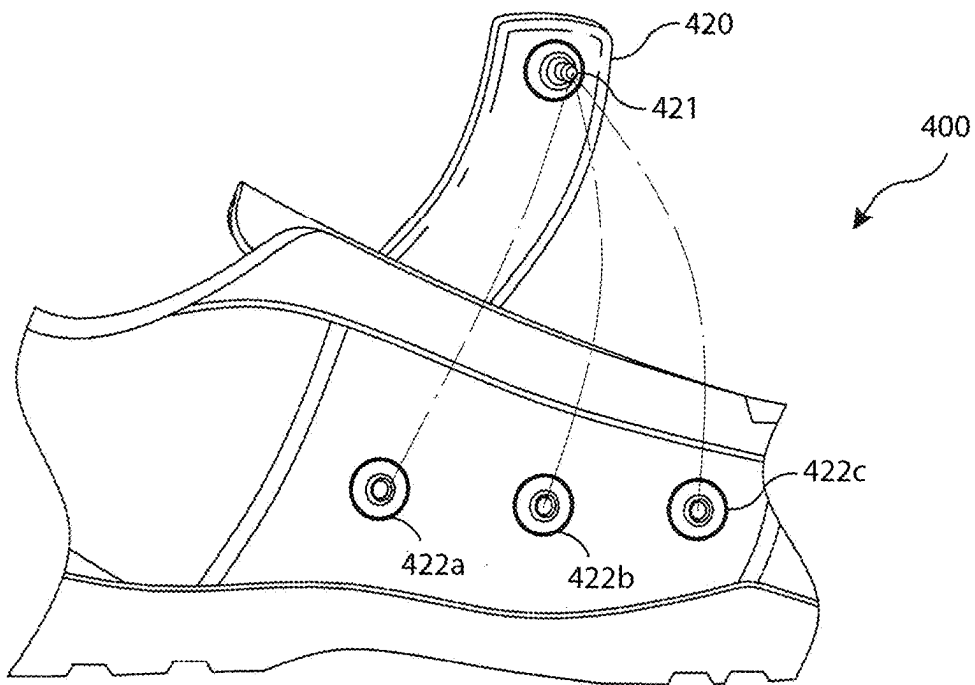
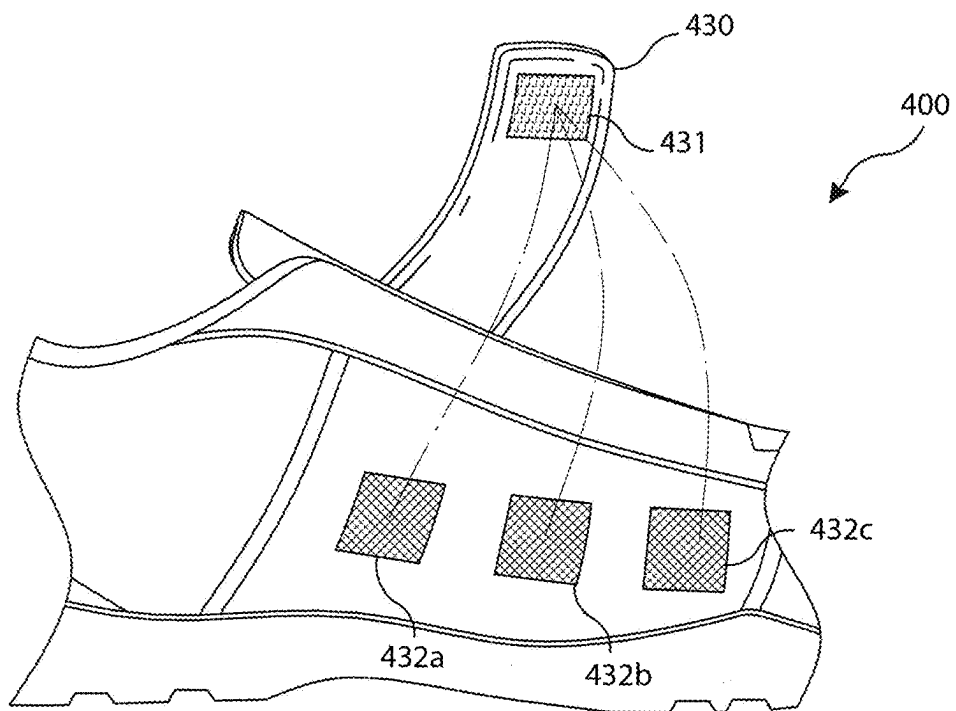


Fig. 17

**Fig. 18A****Fig. 18B**

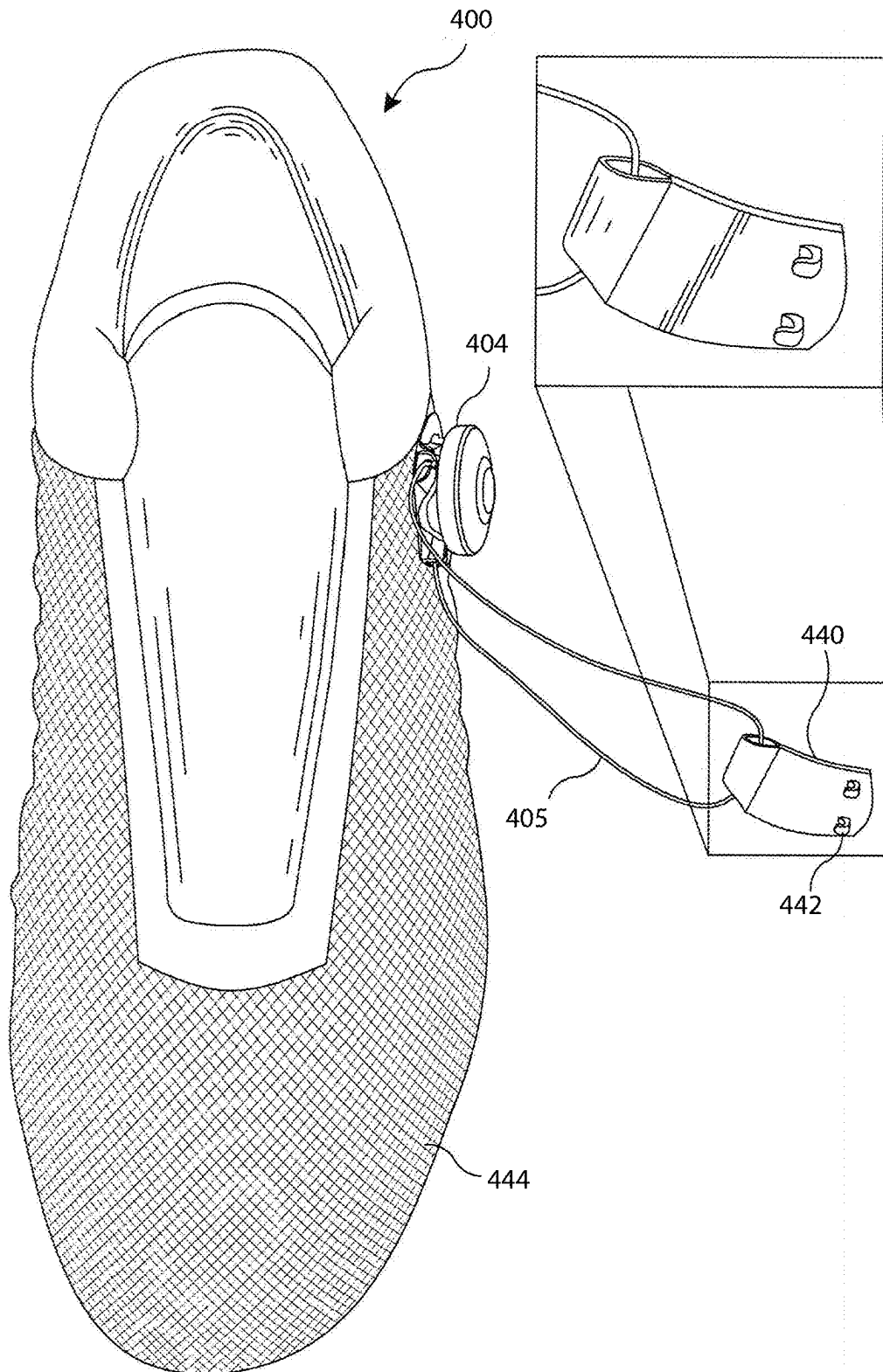


Fig. 19A

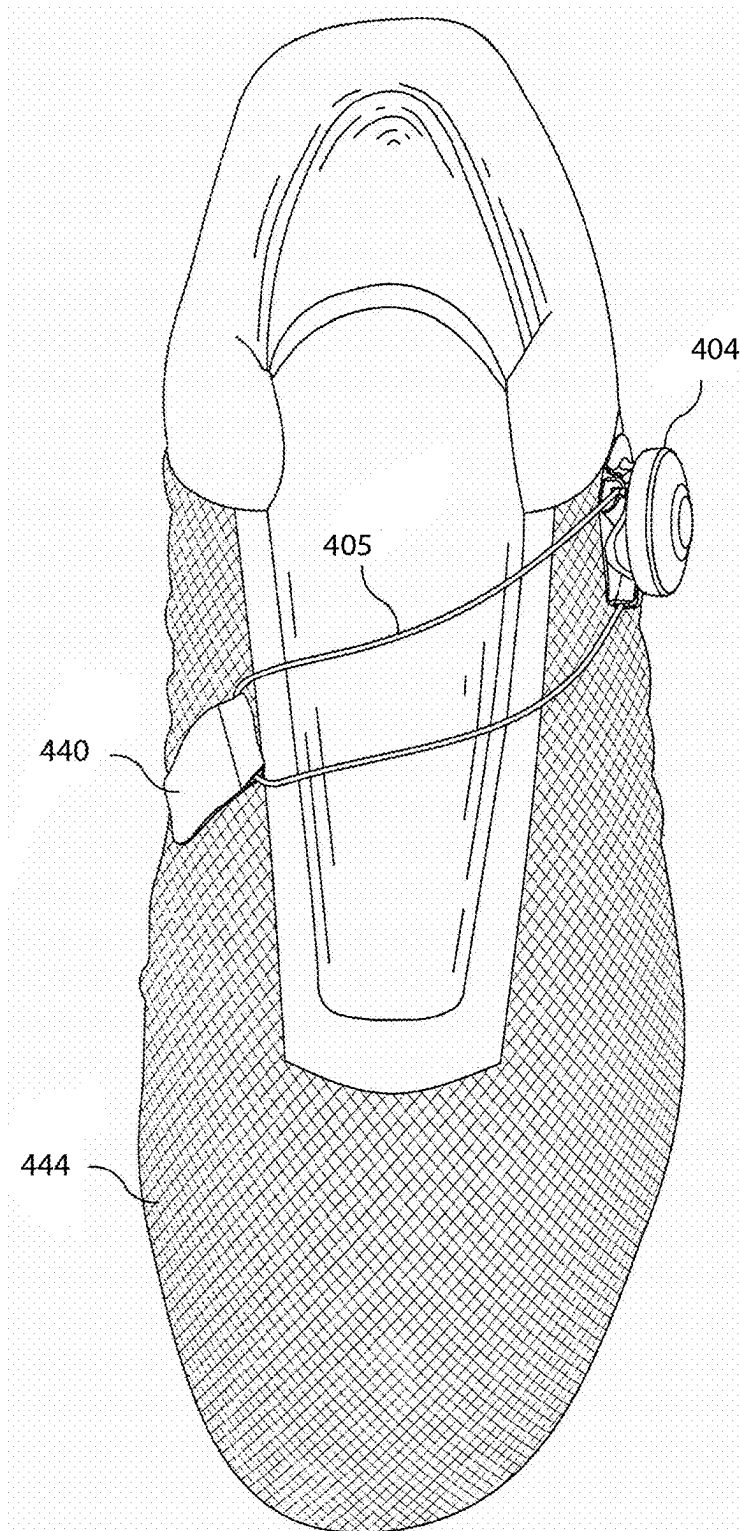


Fig. 19B

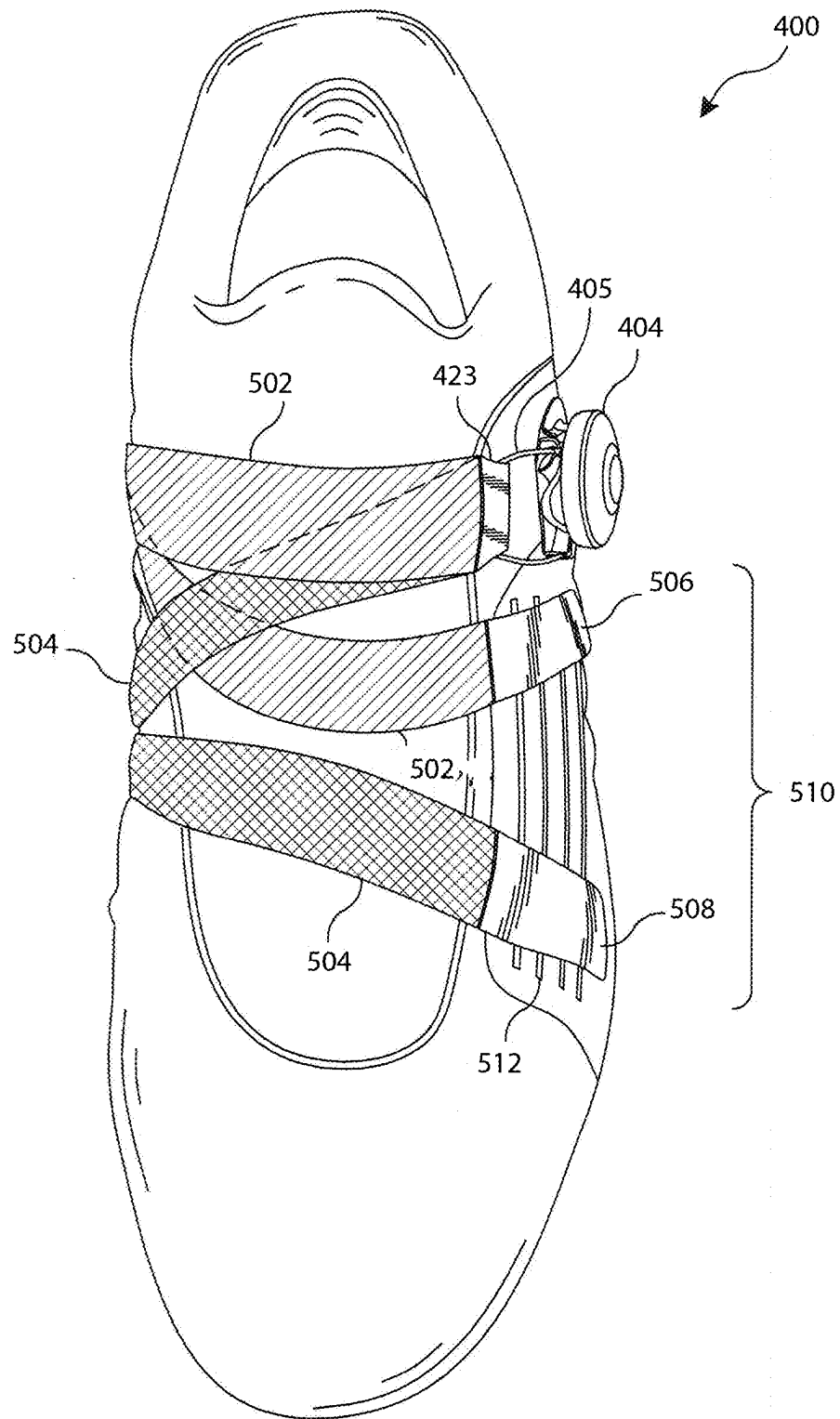


Fig. 20A

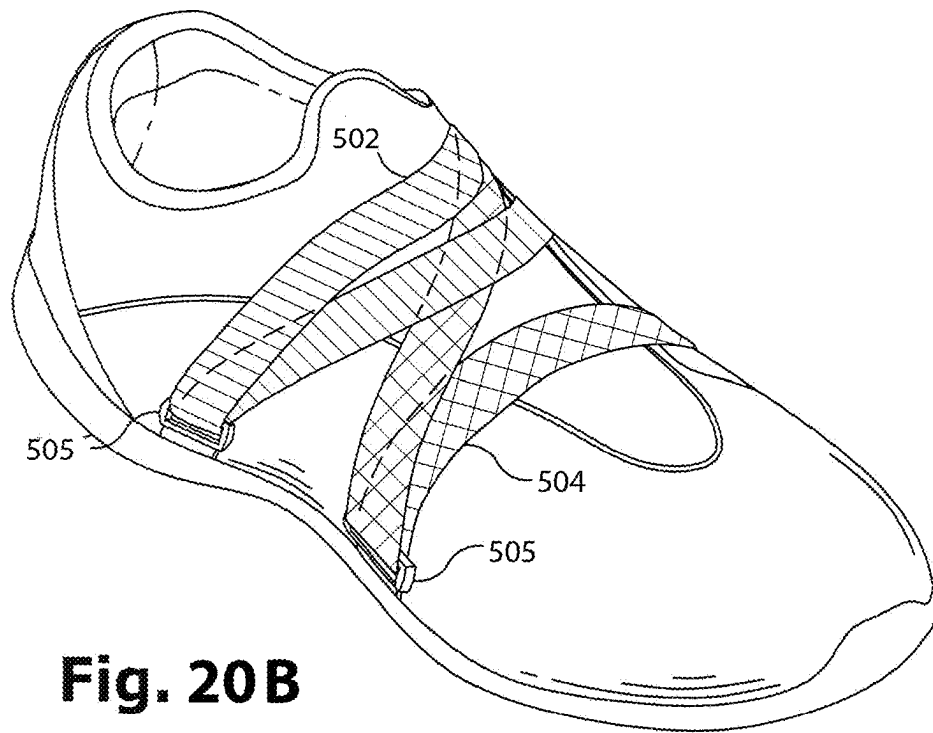


Fig. 20B

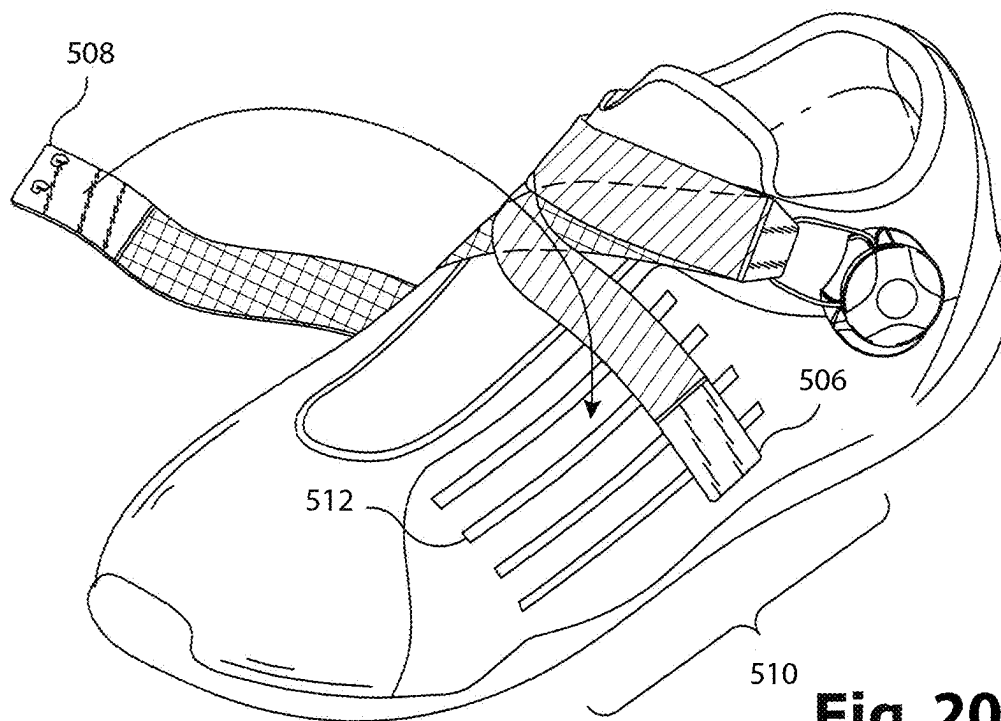
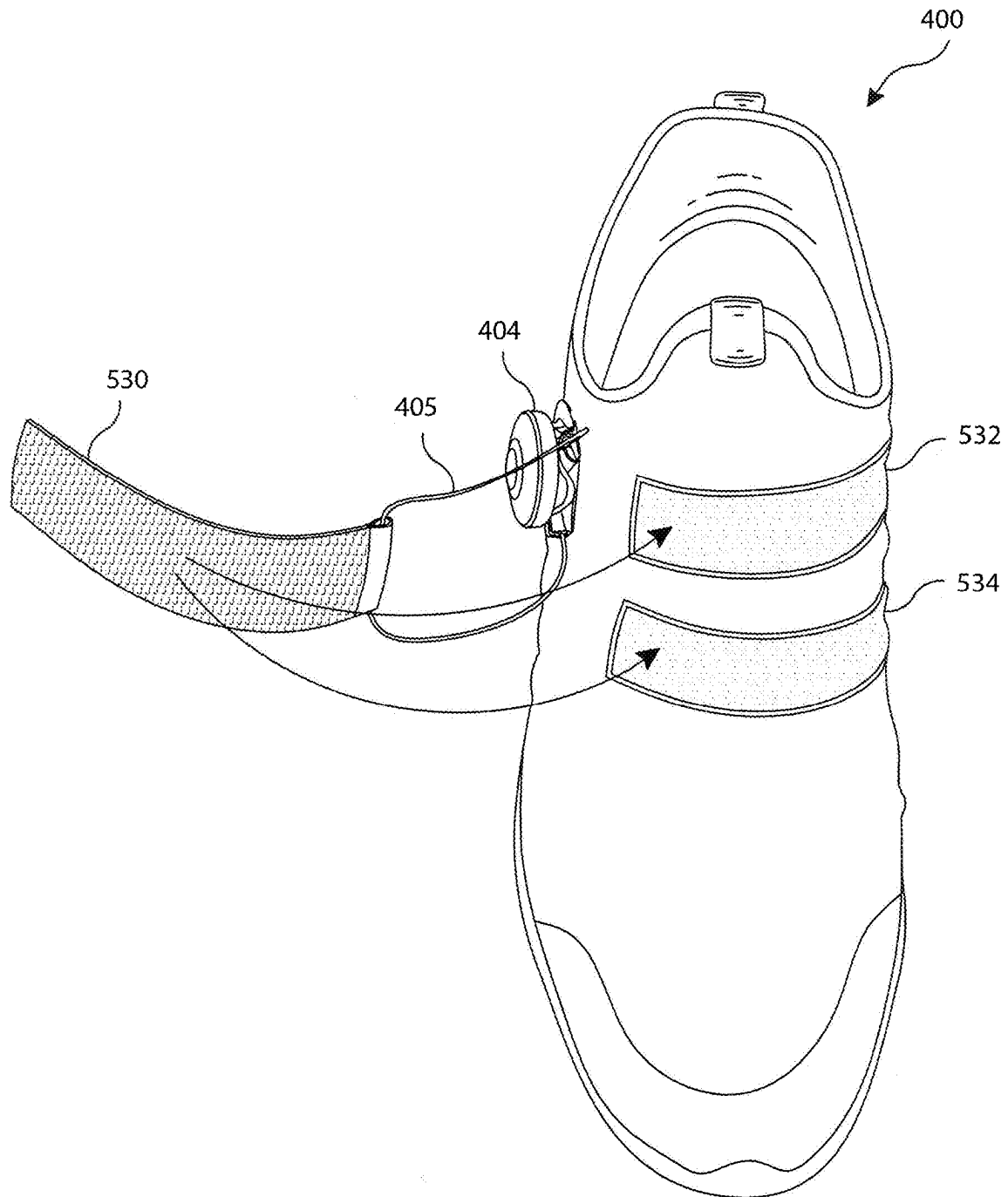


Fig. 20C

**Fig. 21A**

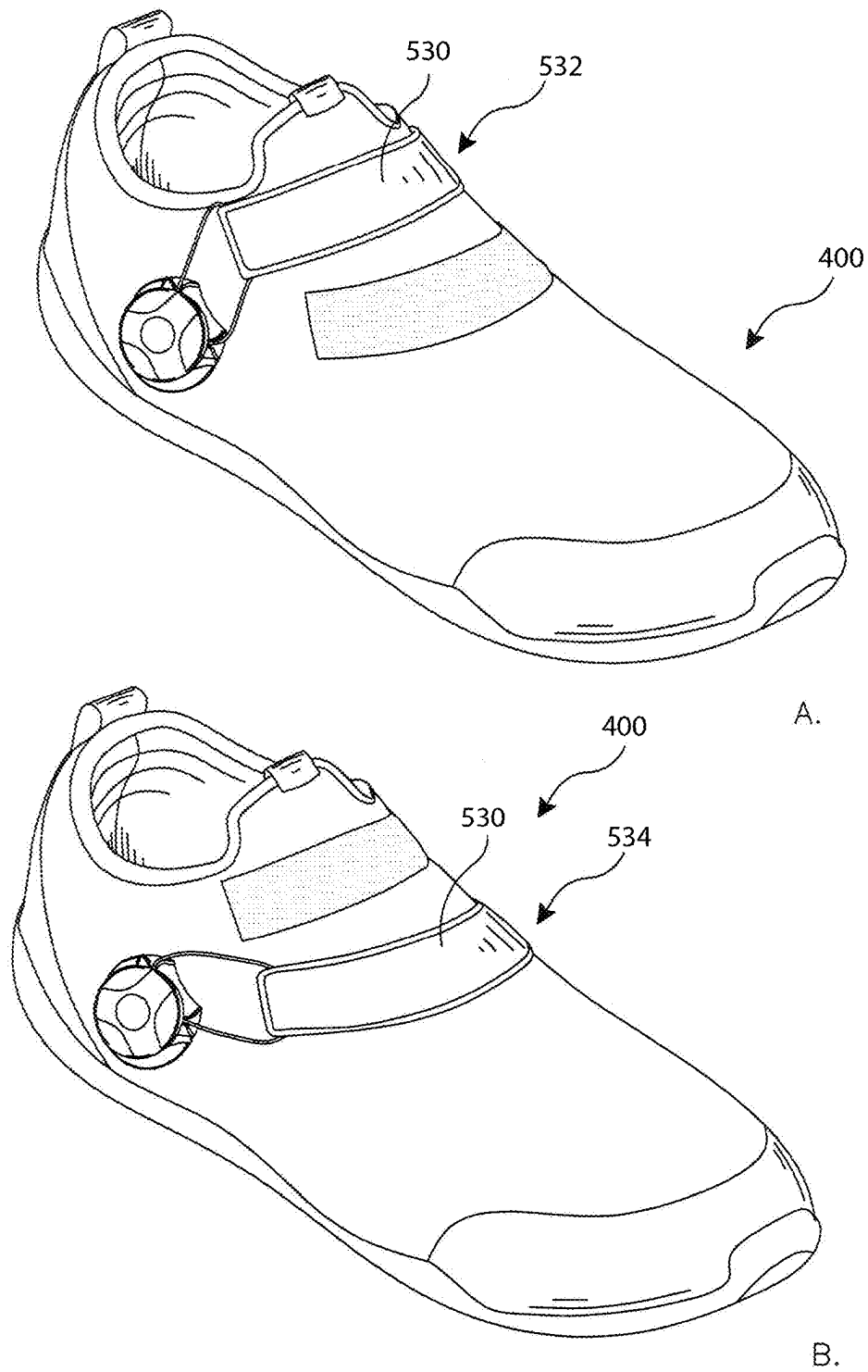
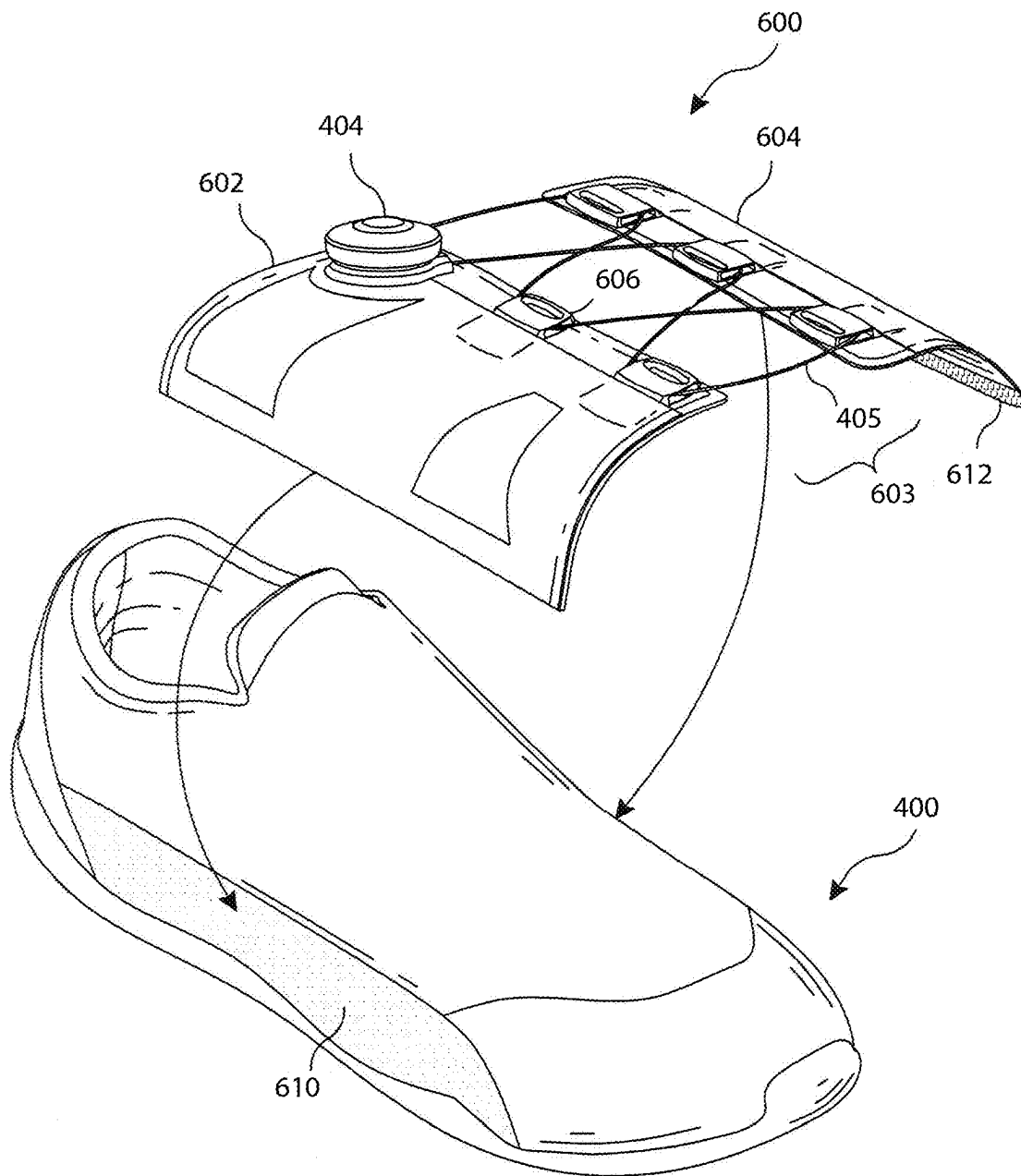
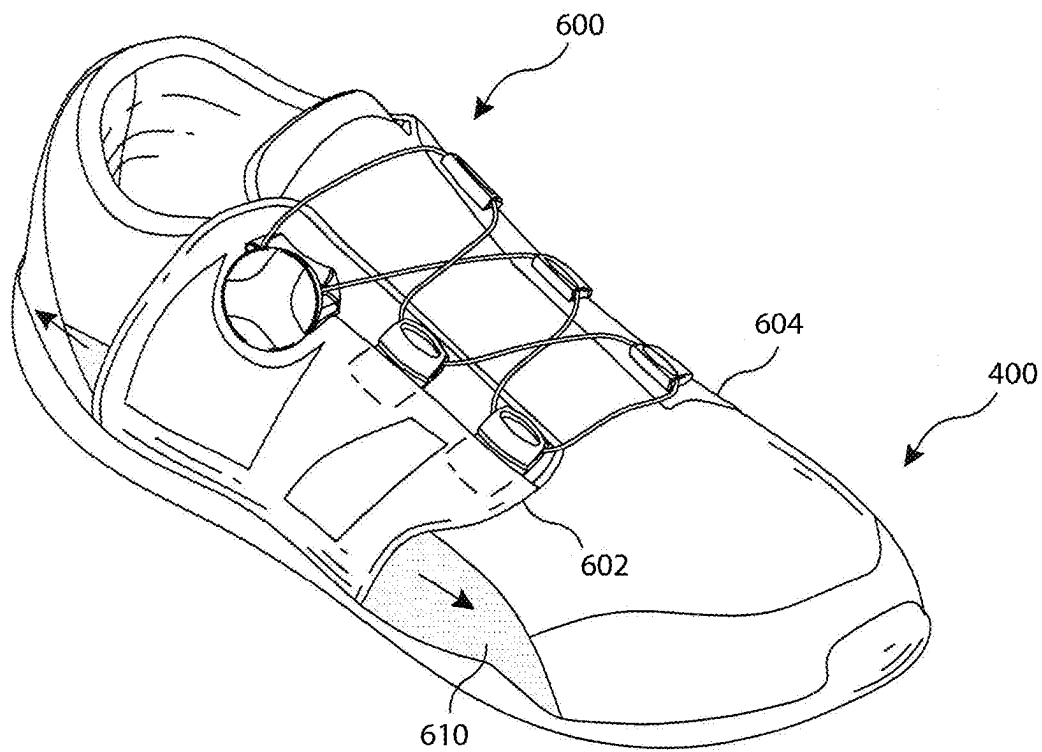
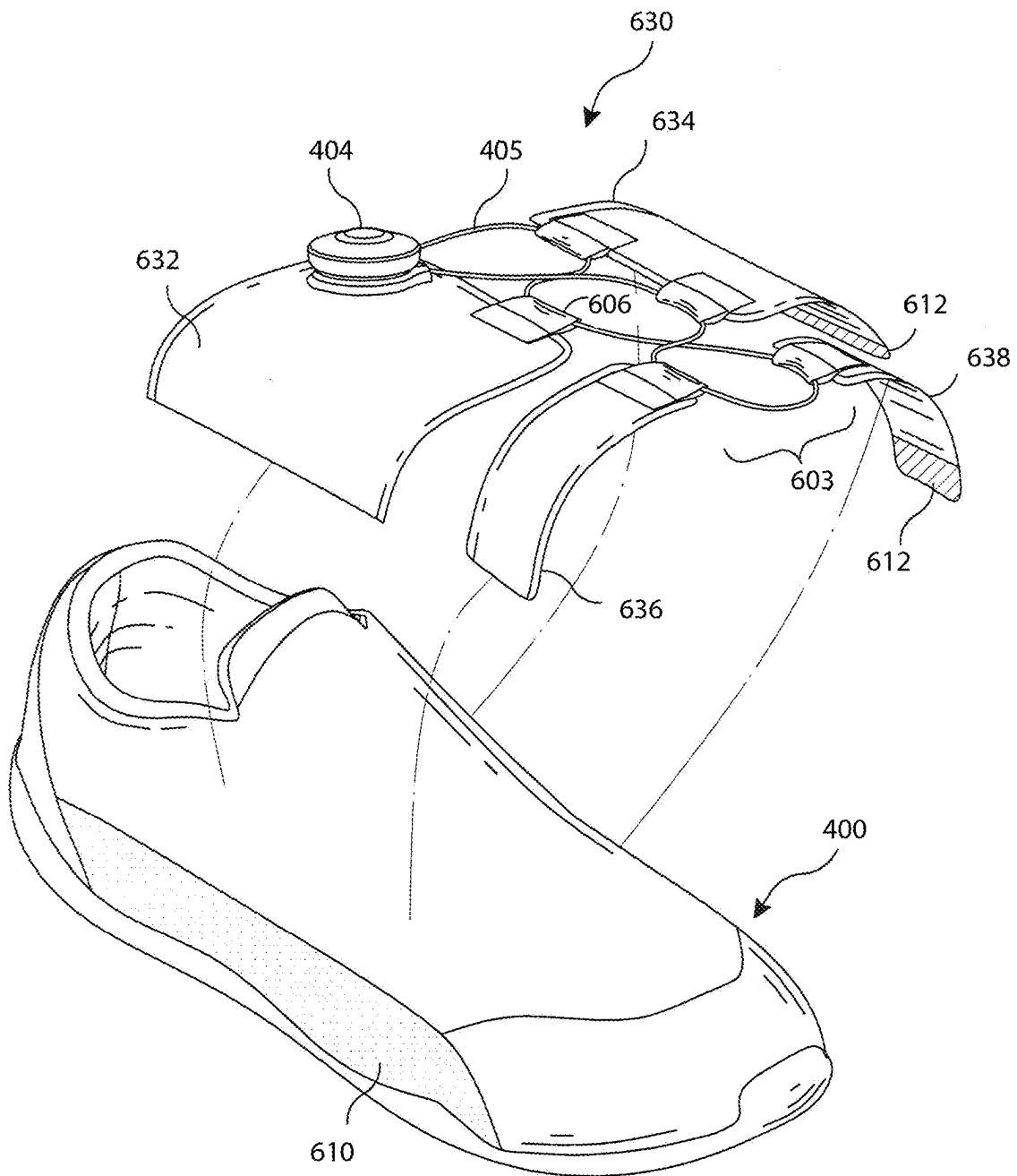


Fig. 21B

**Fig. 22A**

**Fig. 22B**

**Fig. 23A**

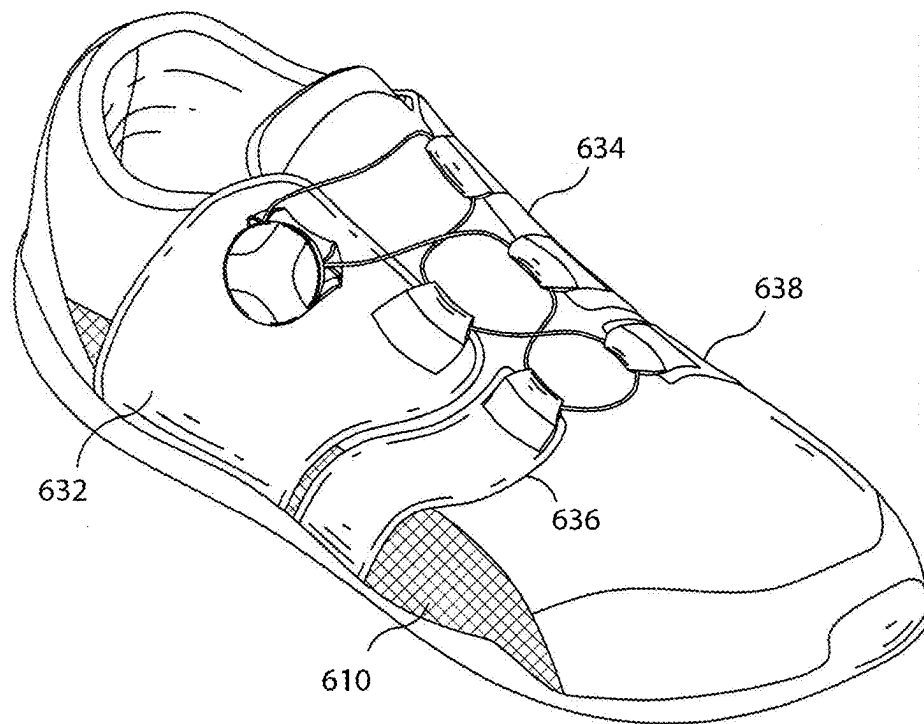


Fig. 23B

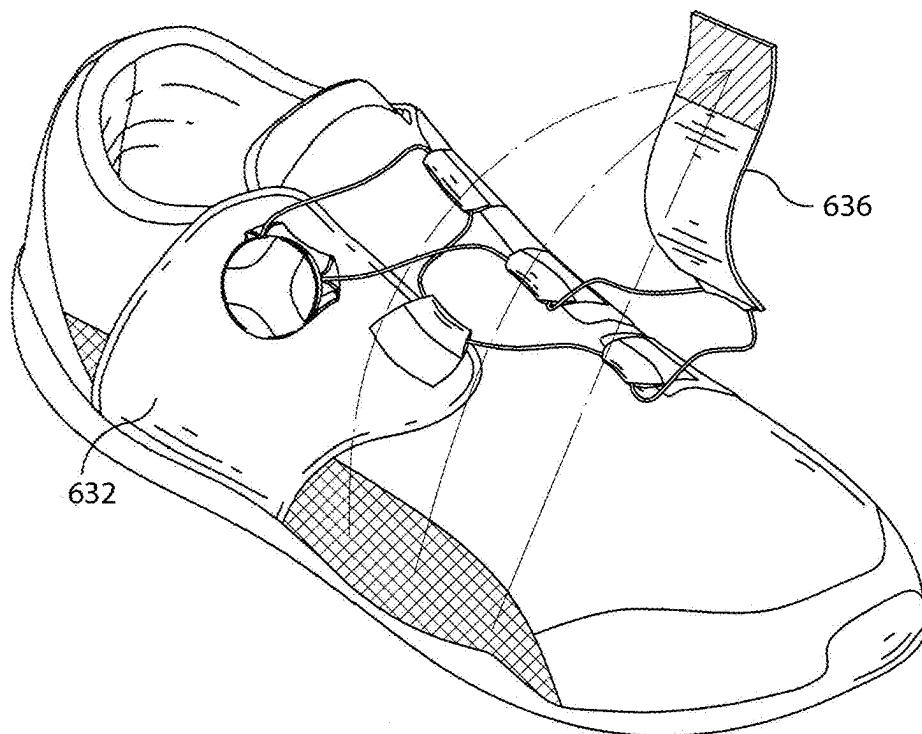
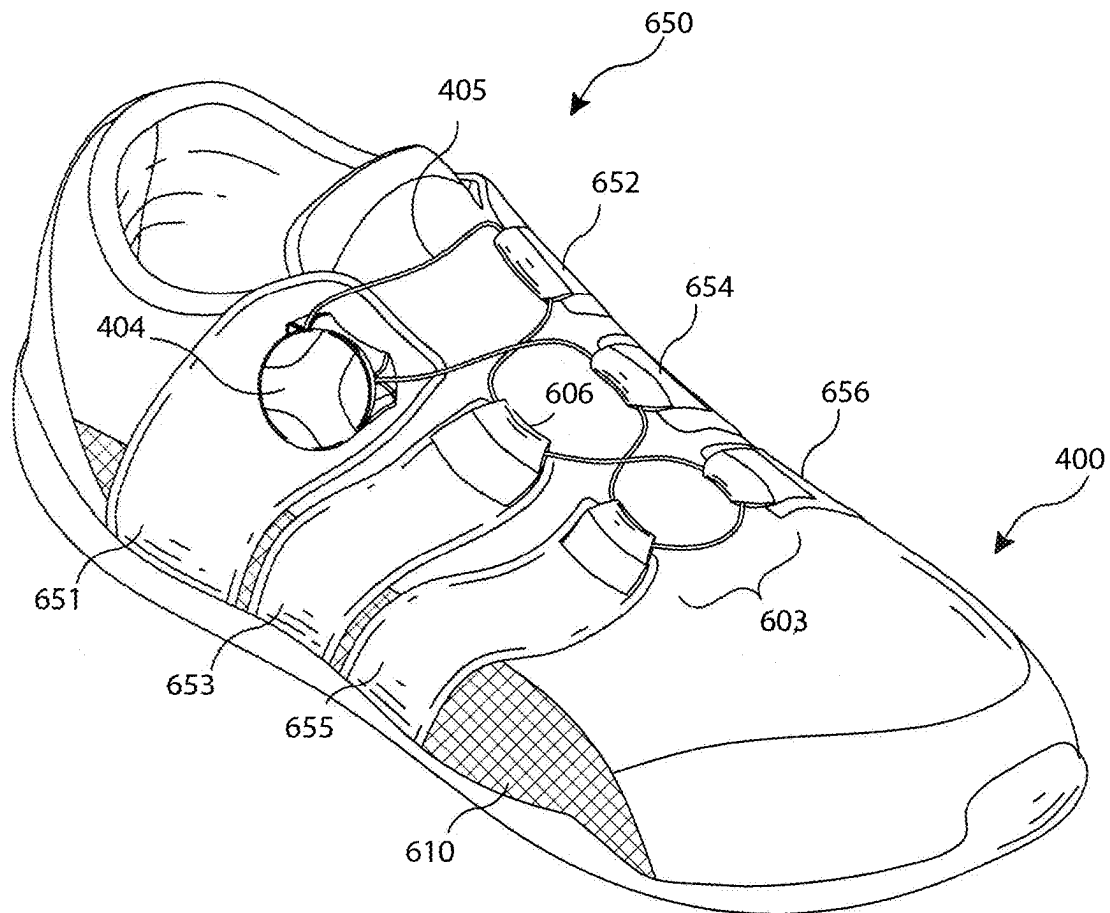
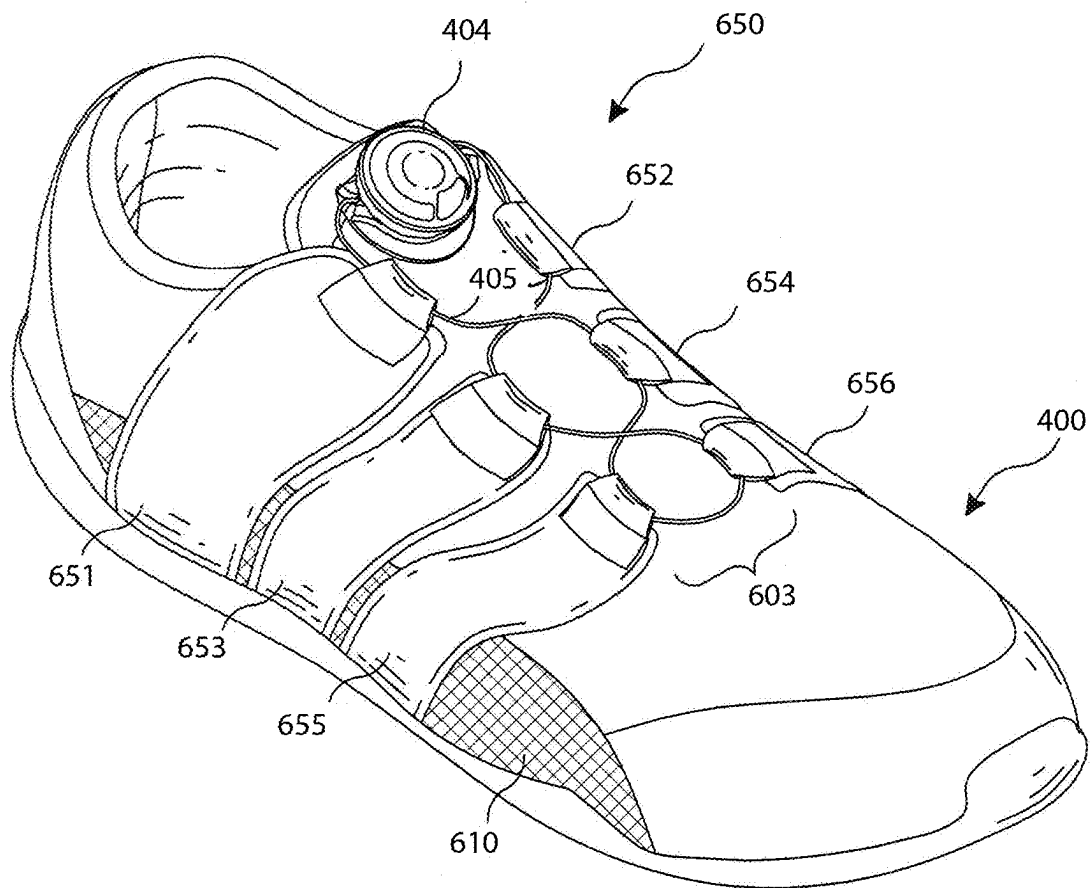


Fig. 23C

**Fig. 24A**

**Fig. 24B**

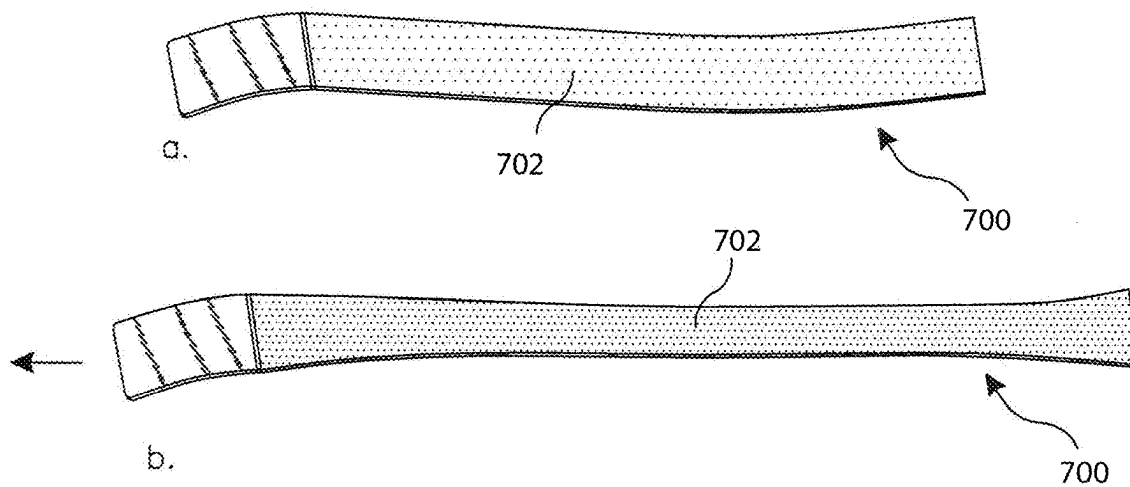


Fig. 25A

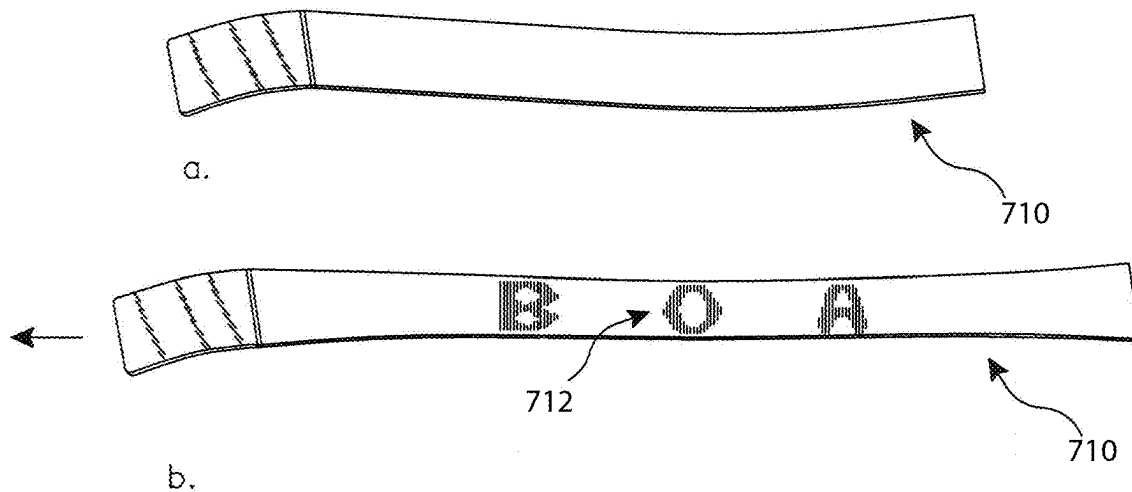


Fig. 25B

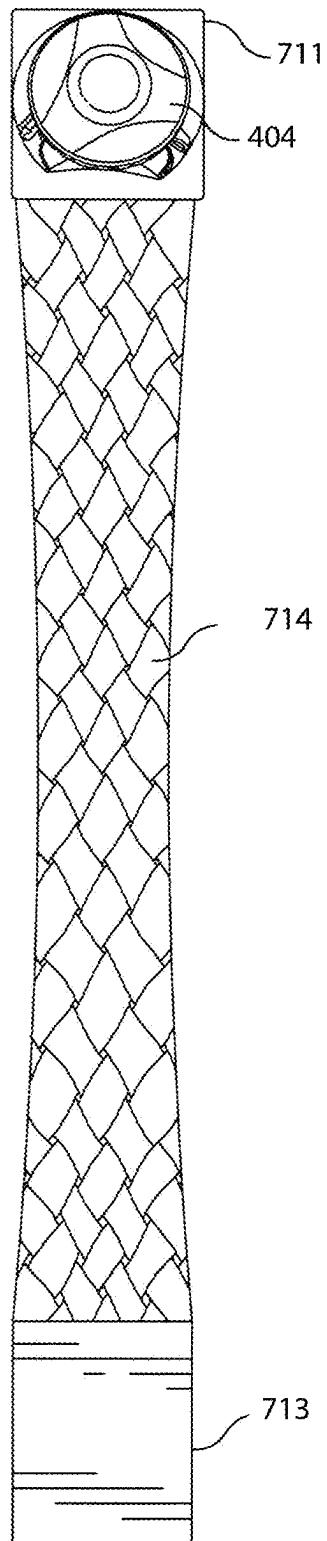


Fig. 26B

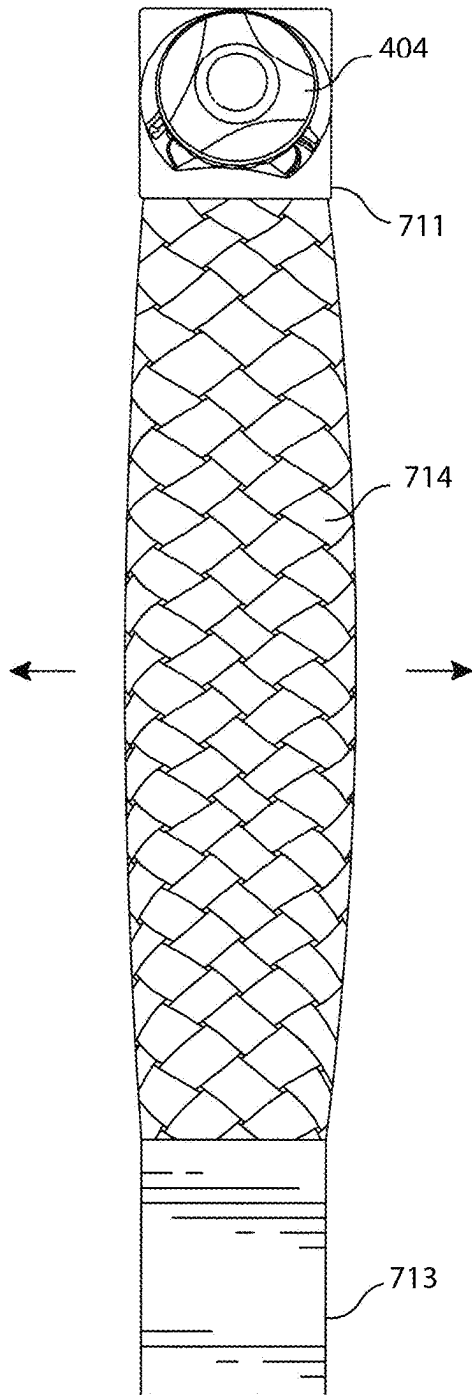
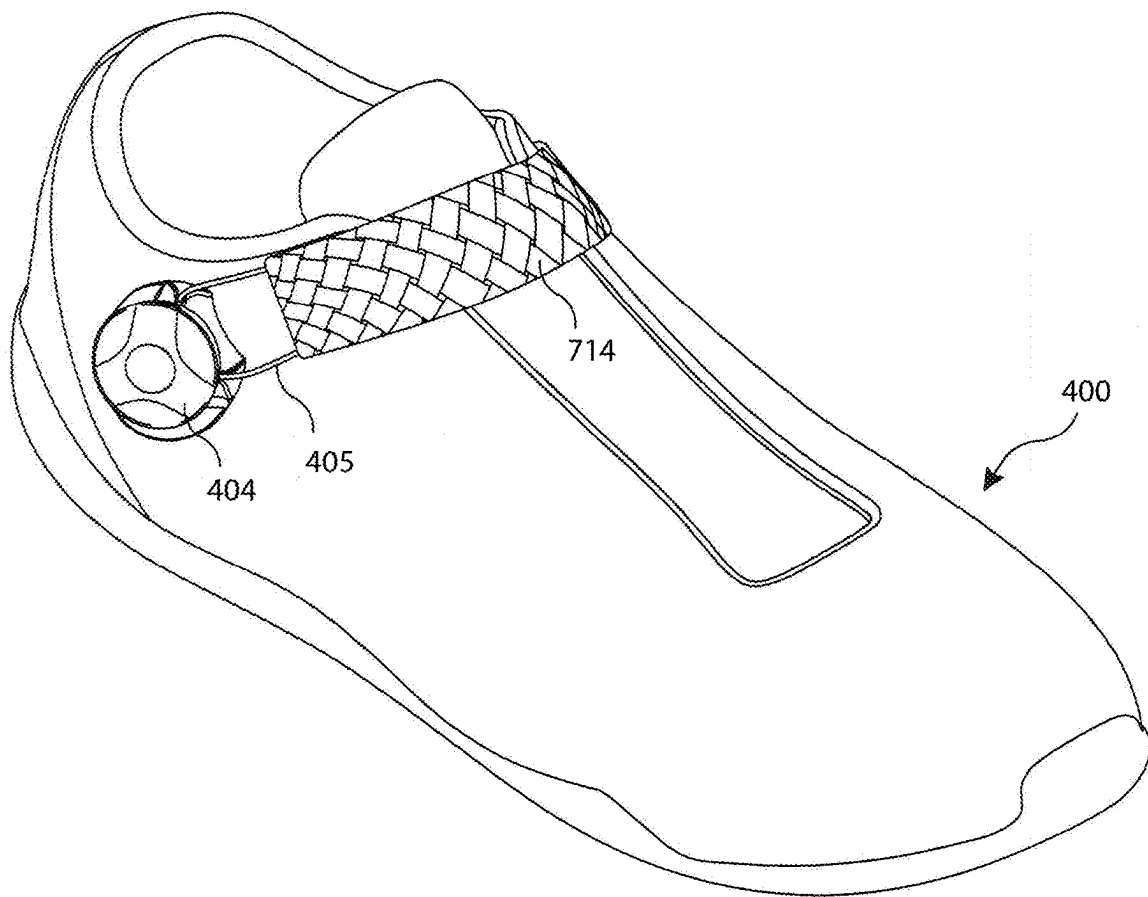


Fig. 26A

**Fig. 26C**

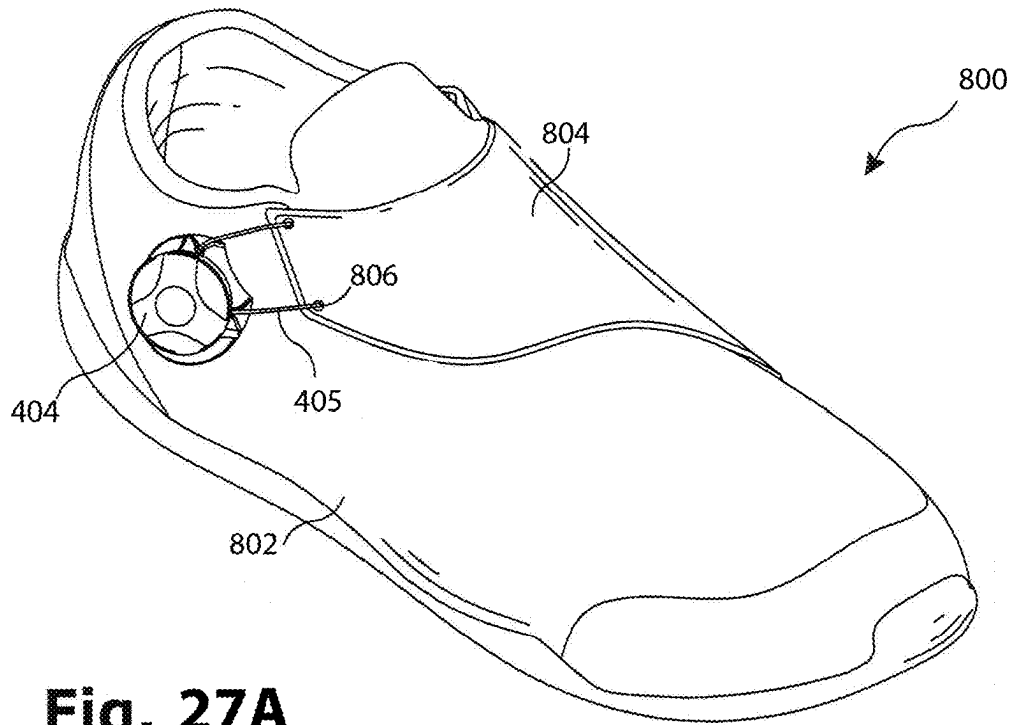


Fig. 27A

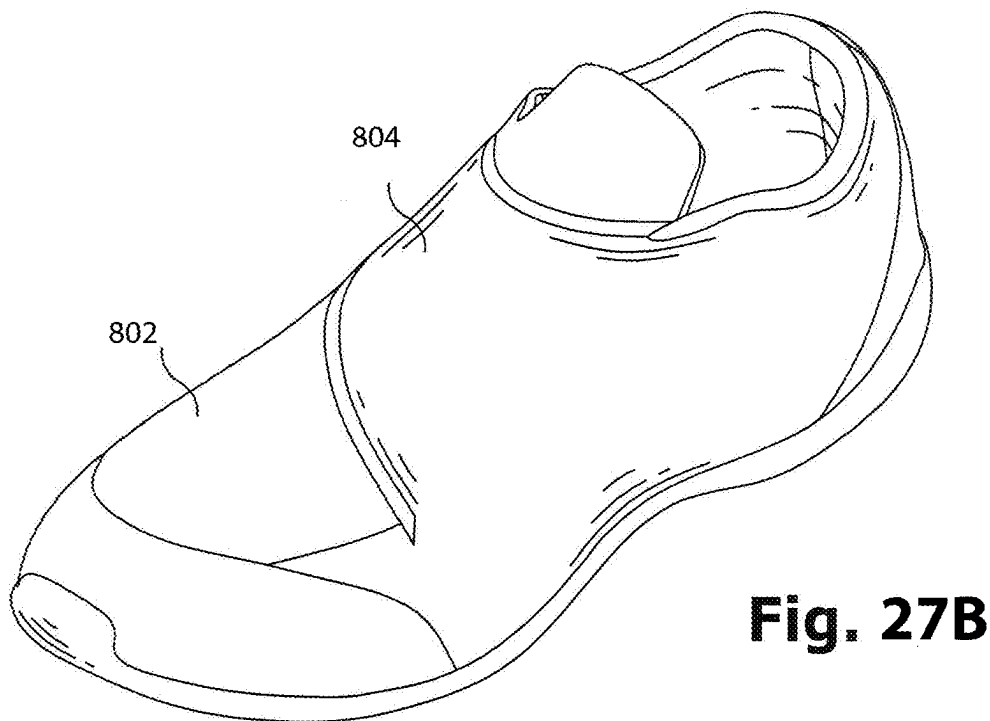
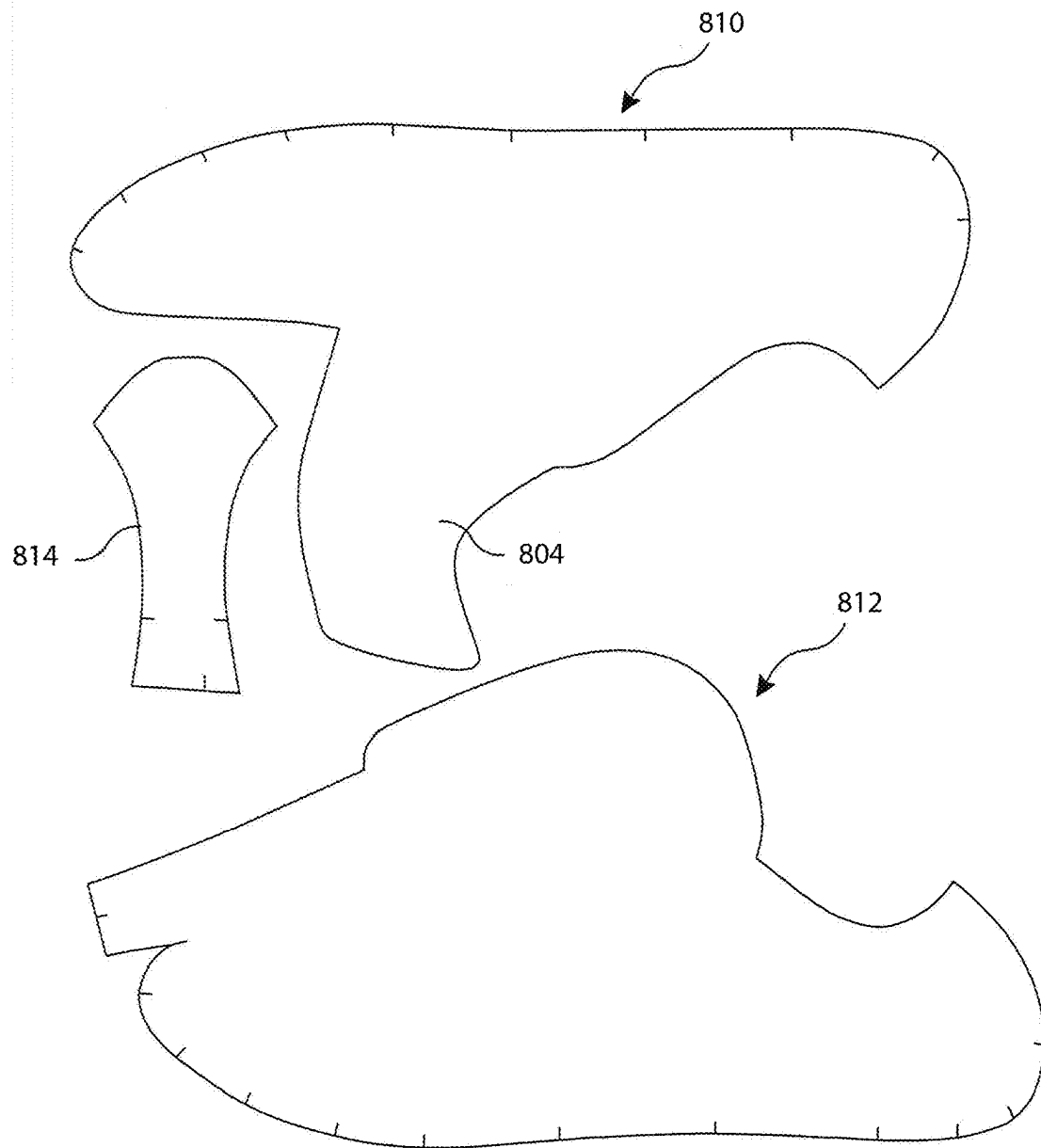


Fig. 27B

**Fig. 27C**

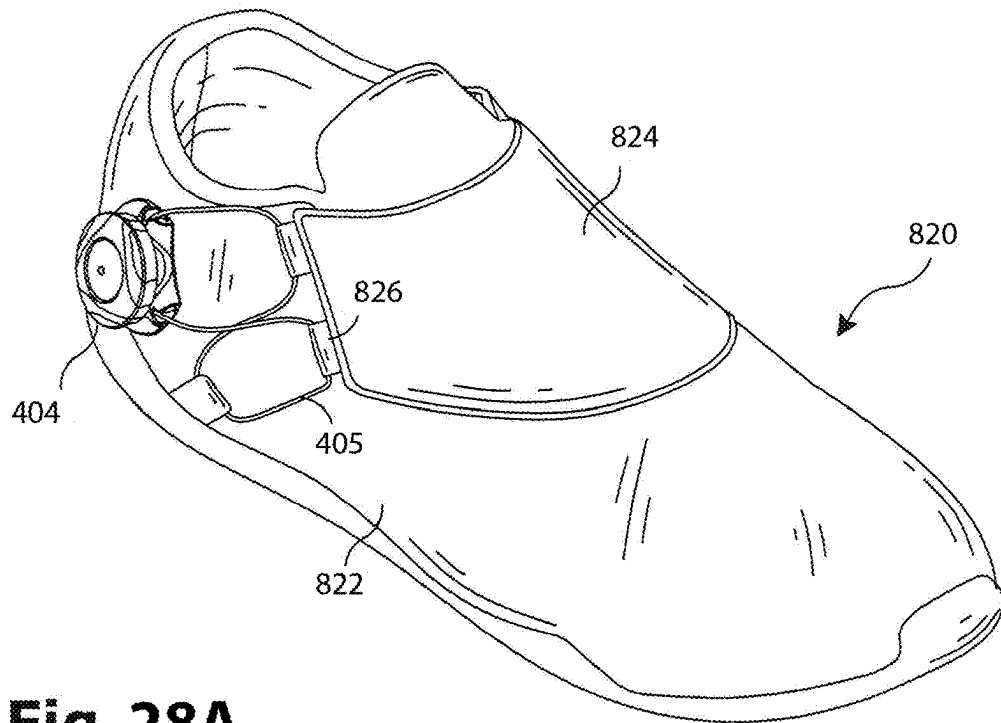


Fig. 28A

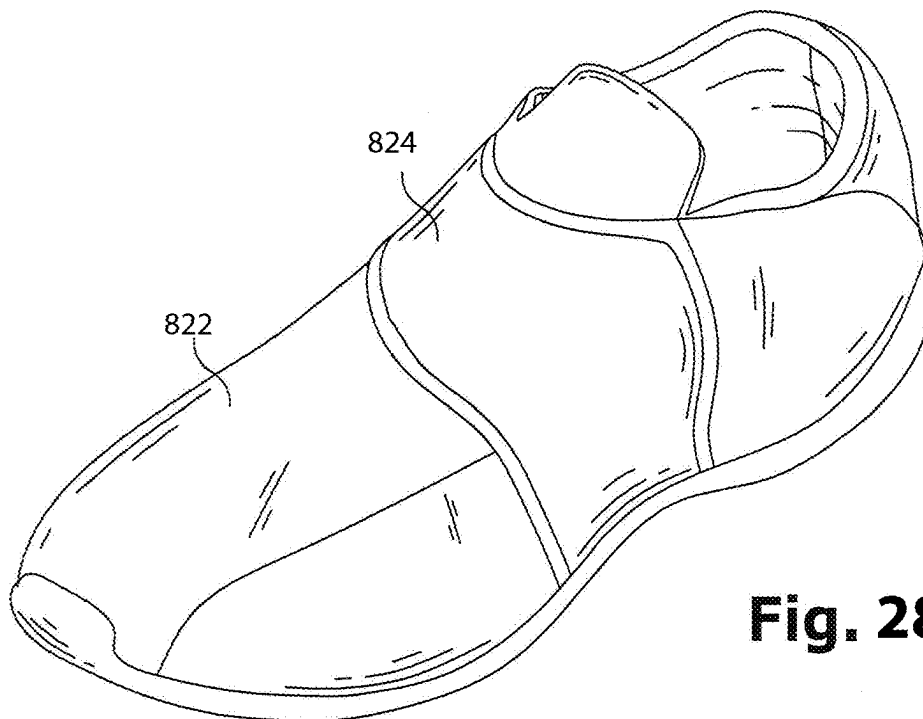
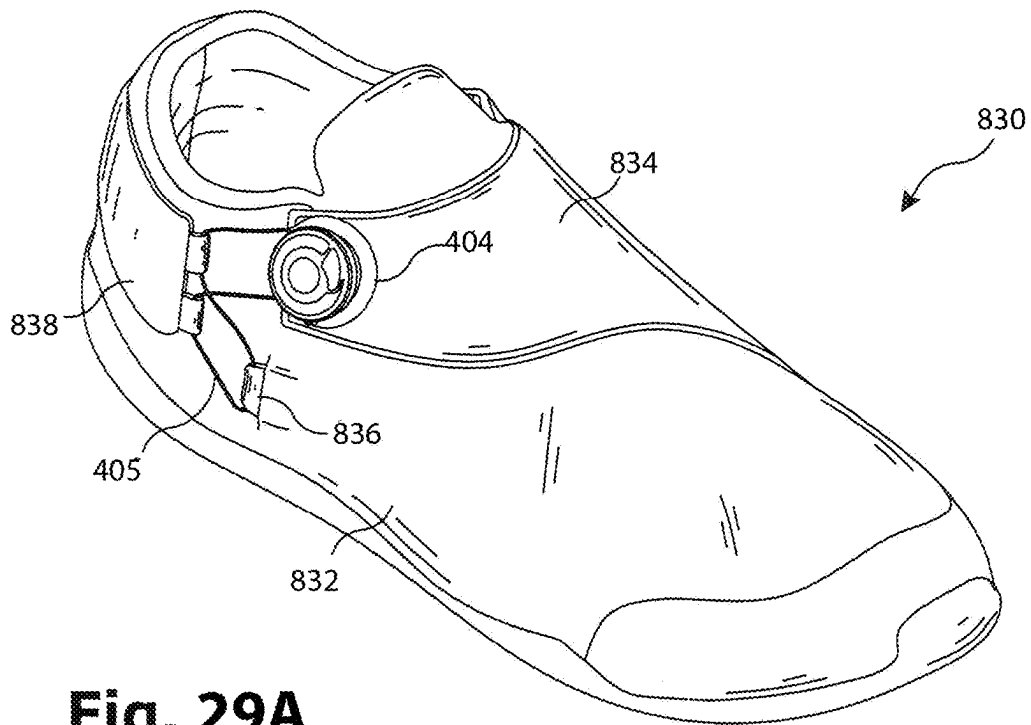
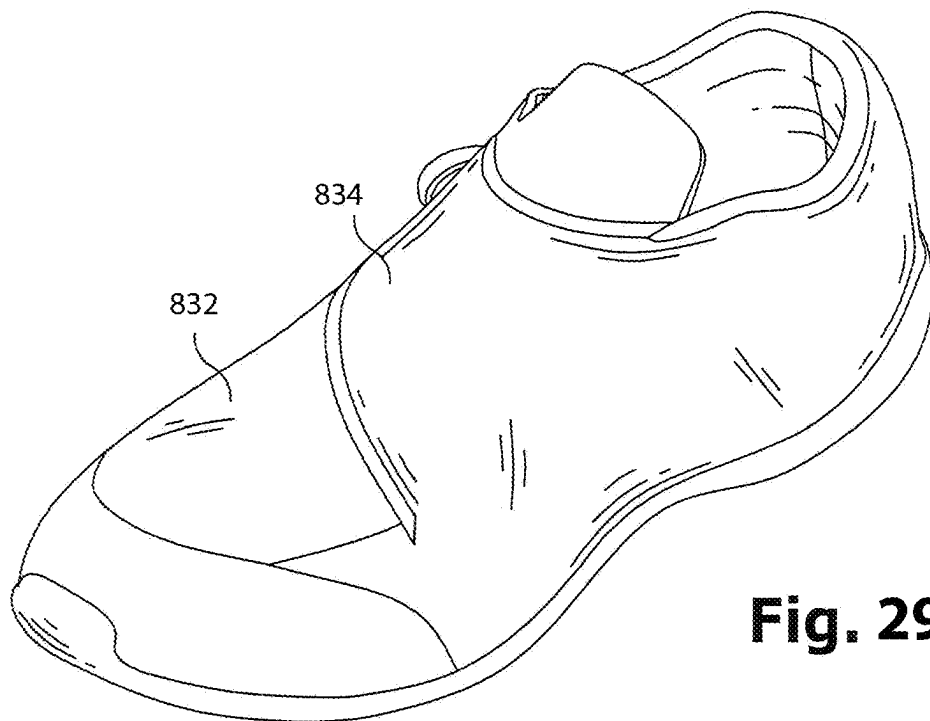
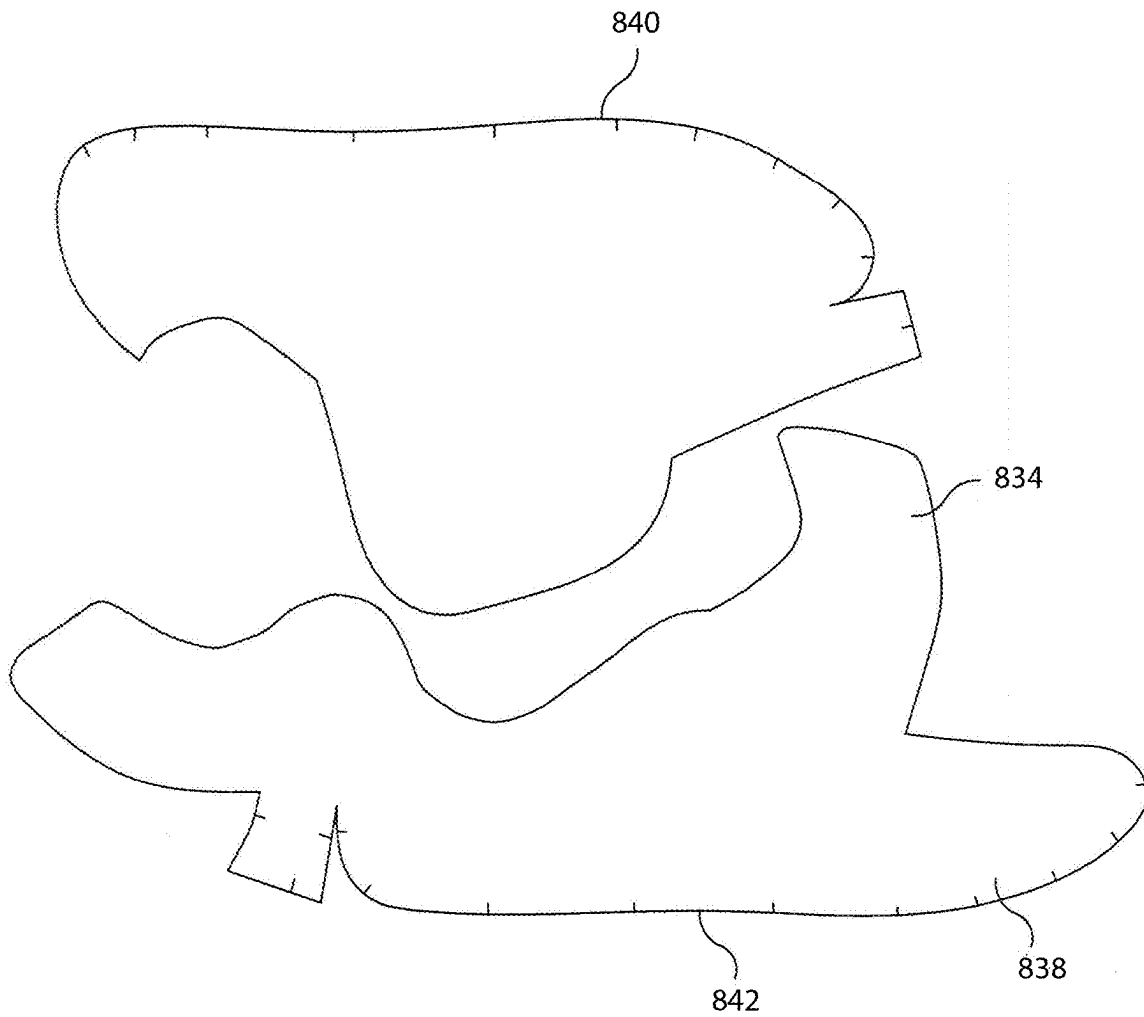


Fig. 28B

**Fig. 29A****Fig. 29B**

**Fig. 29C**

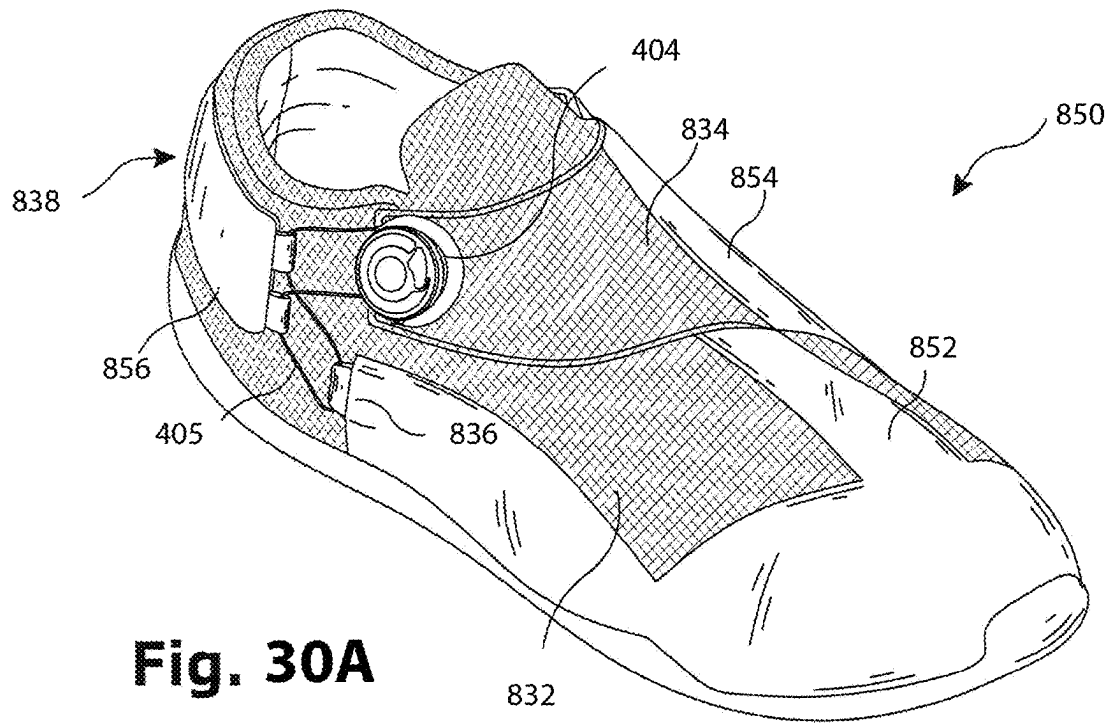


Fig. 30A

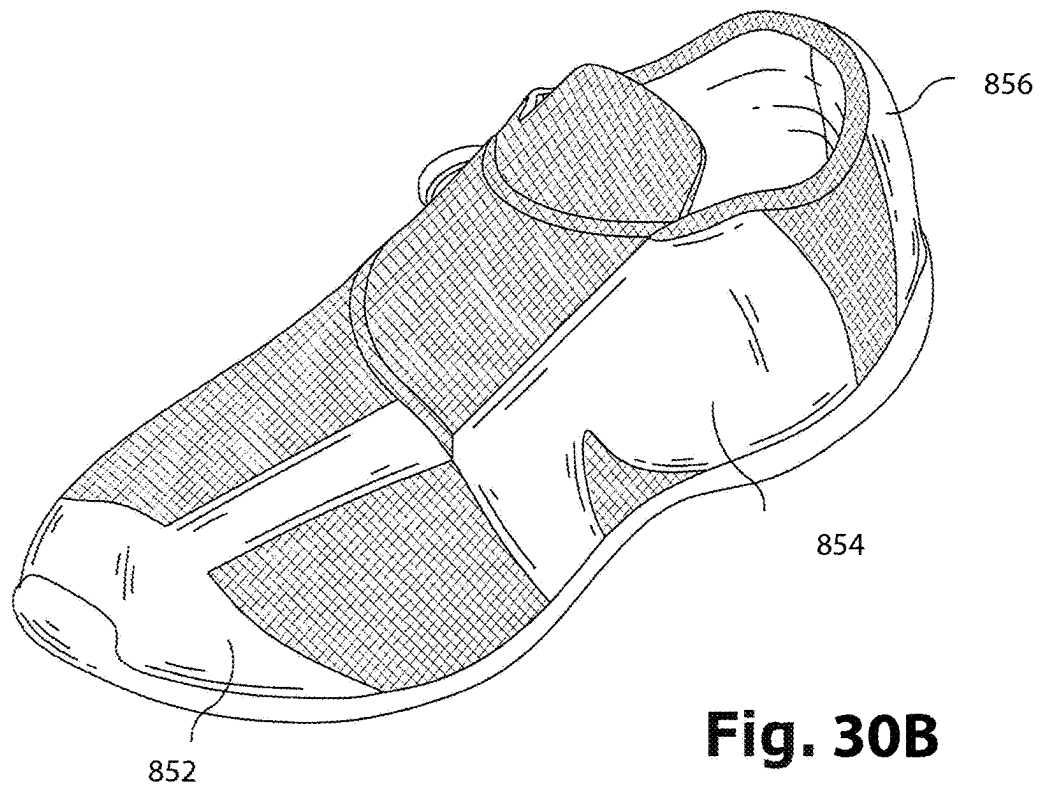
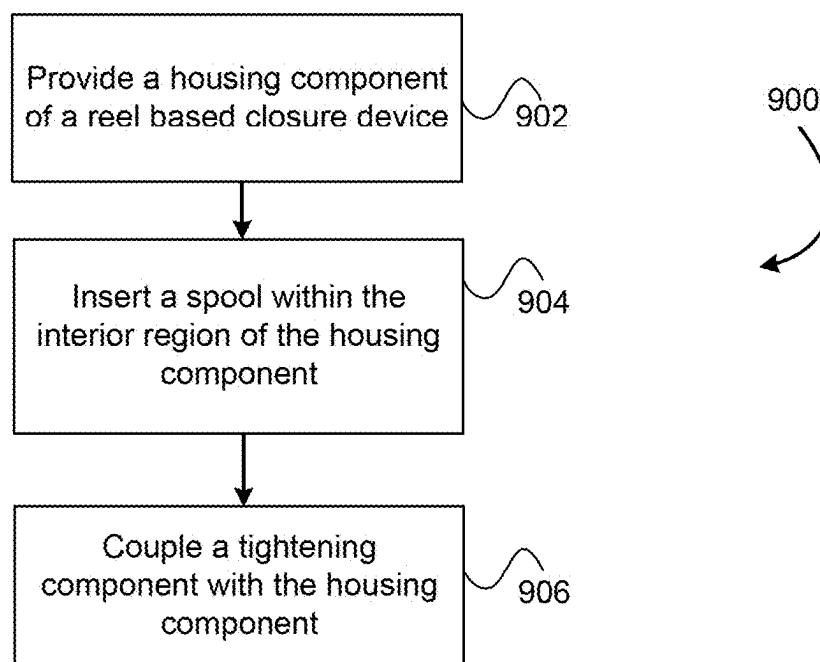


Fig. 30B

**Fig. 31**

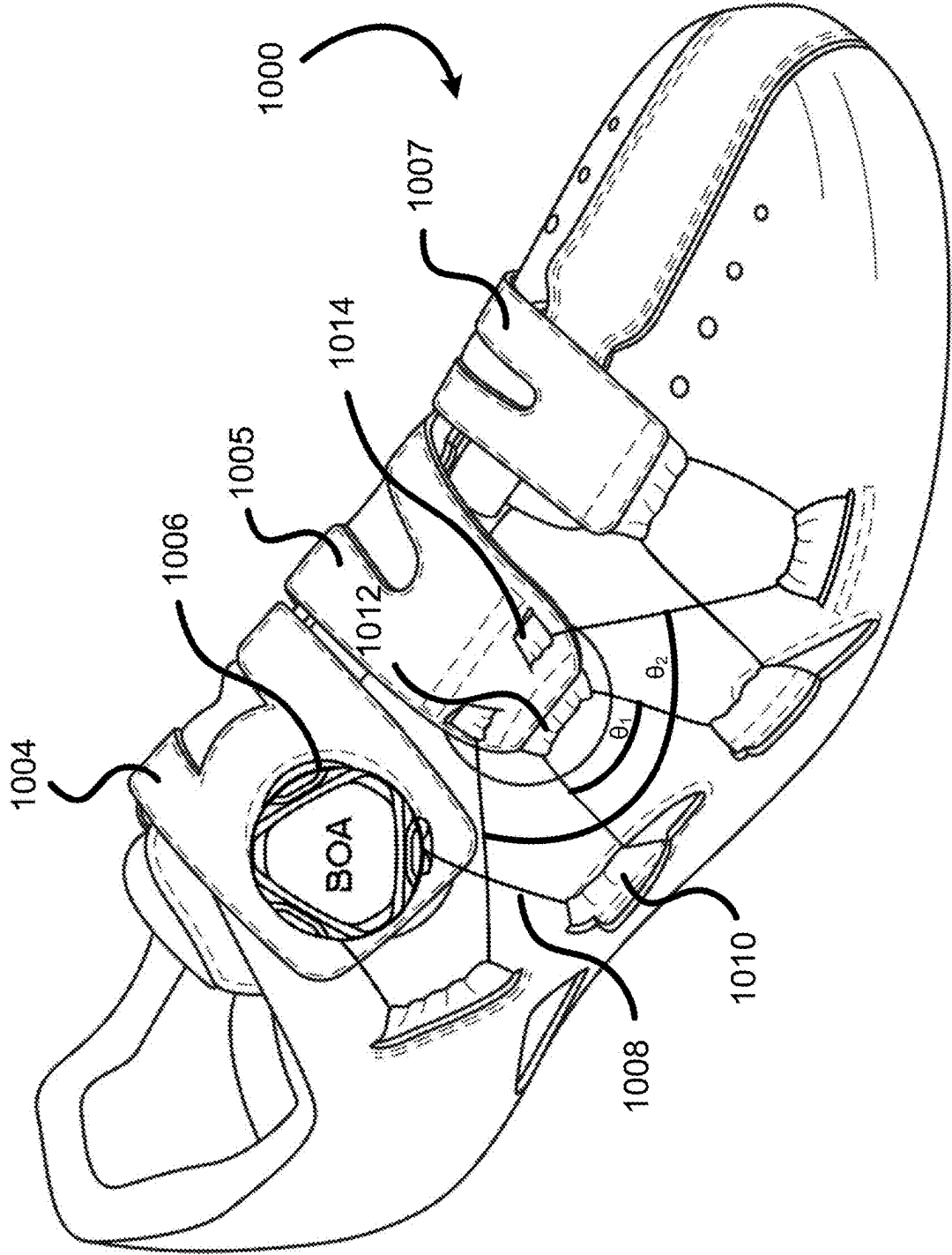
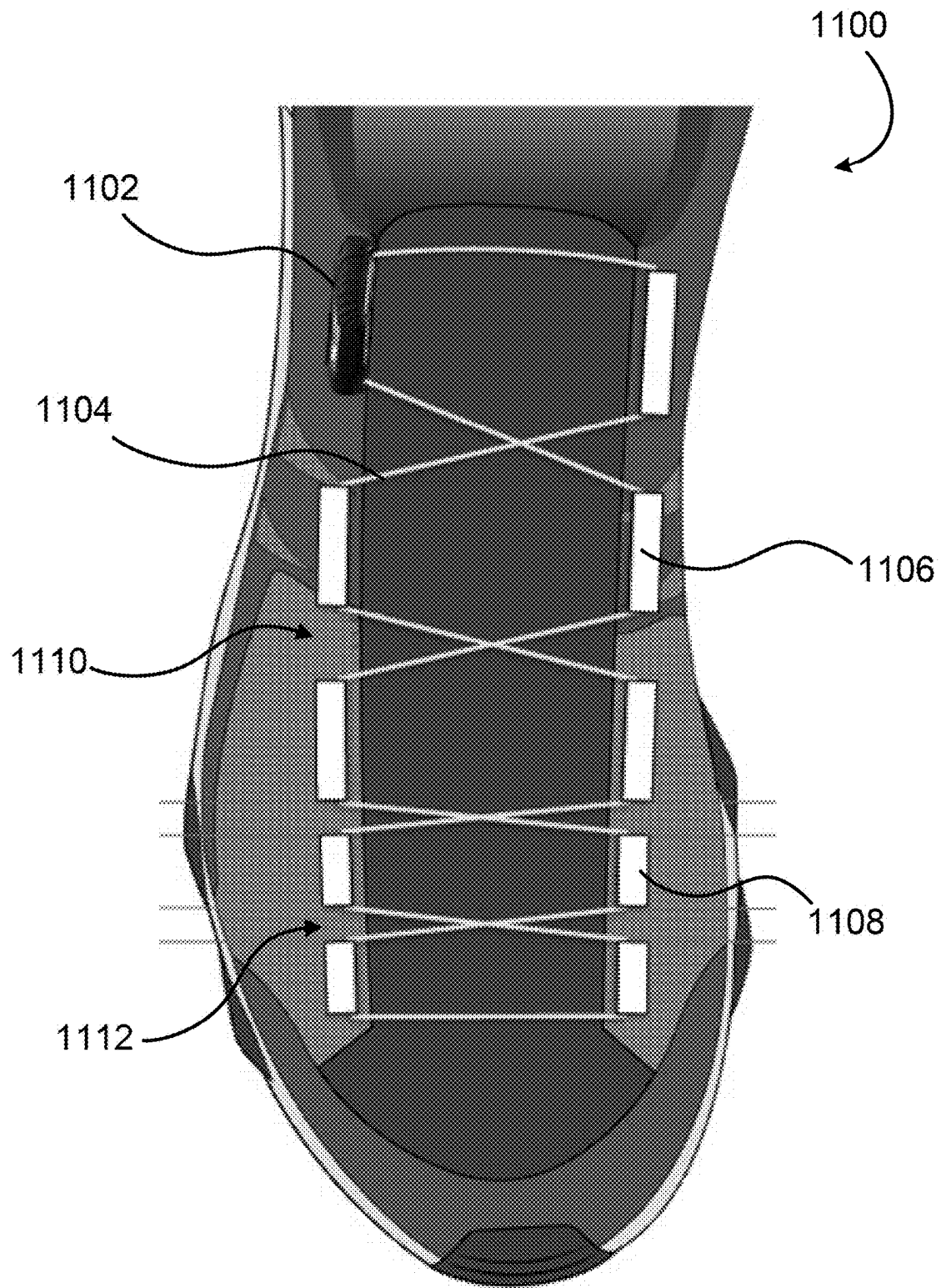
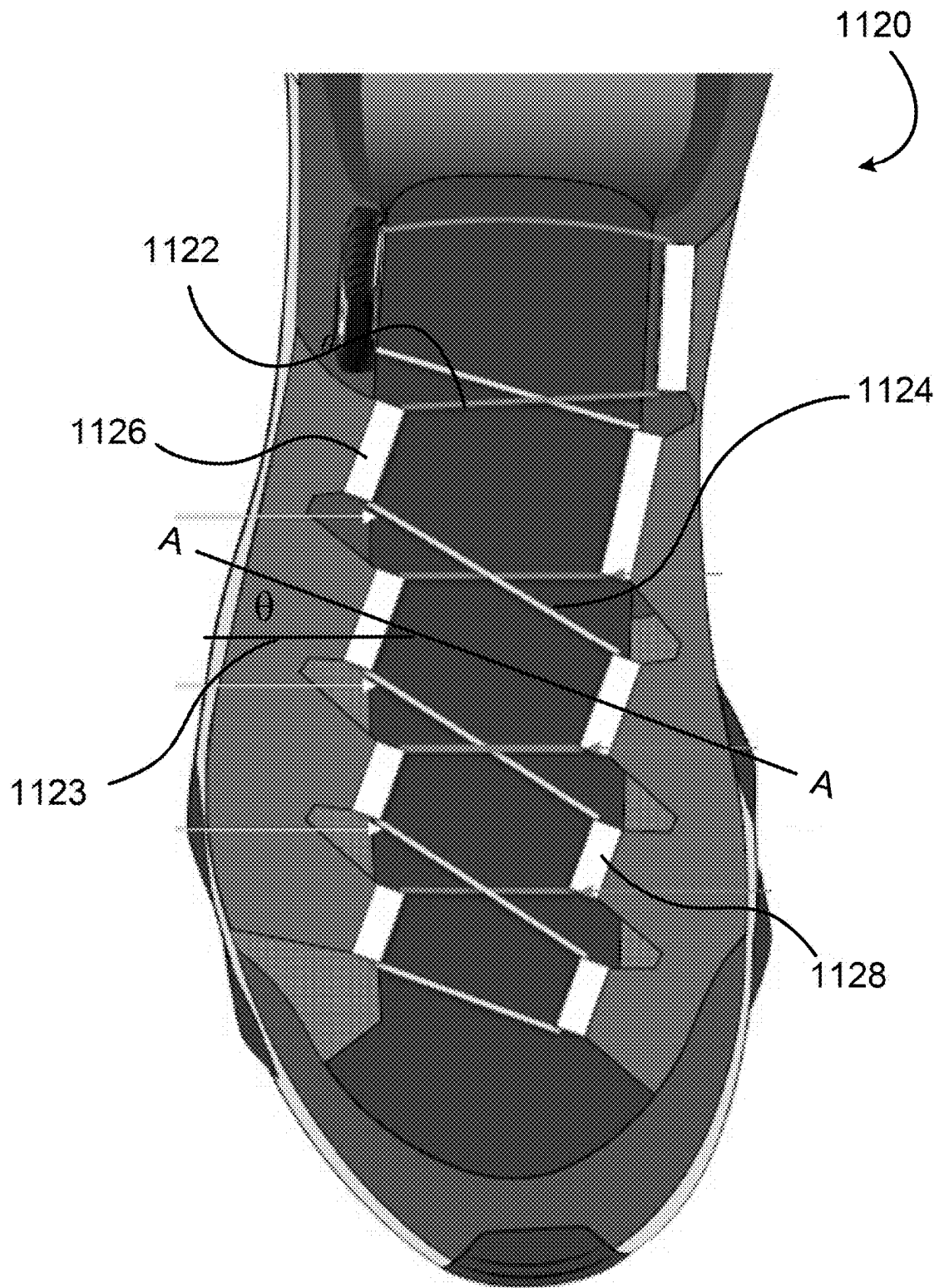


Fig. 32

**Fig. 33**

**Fig. 34**

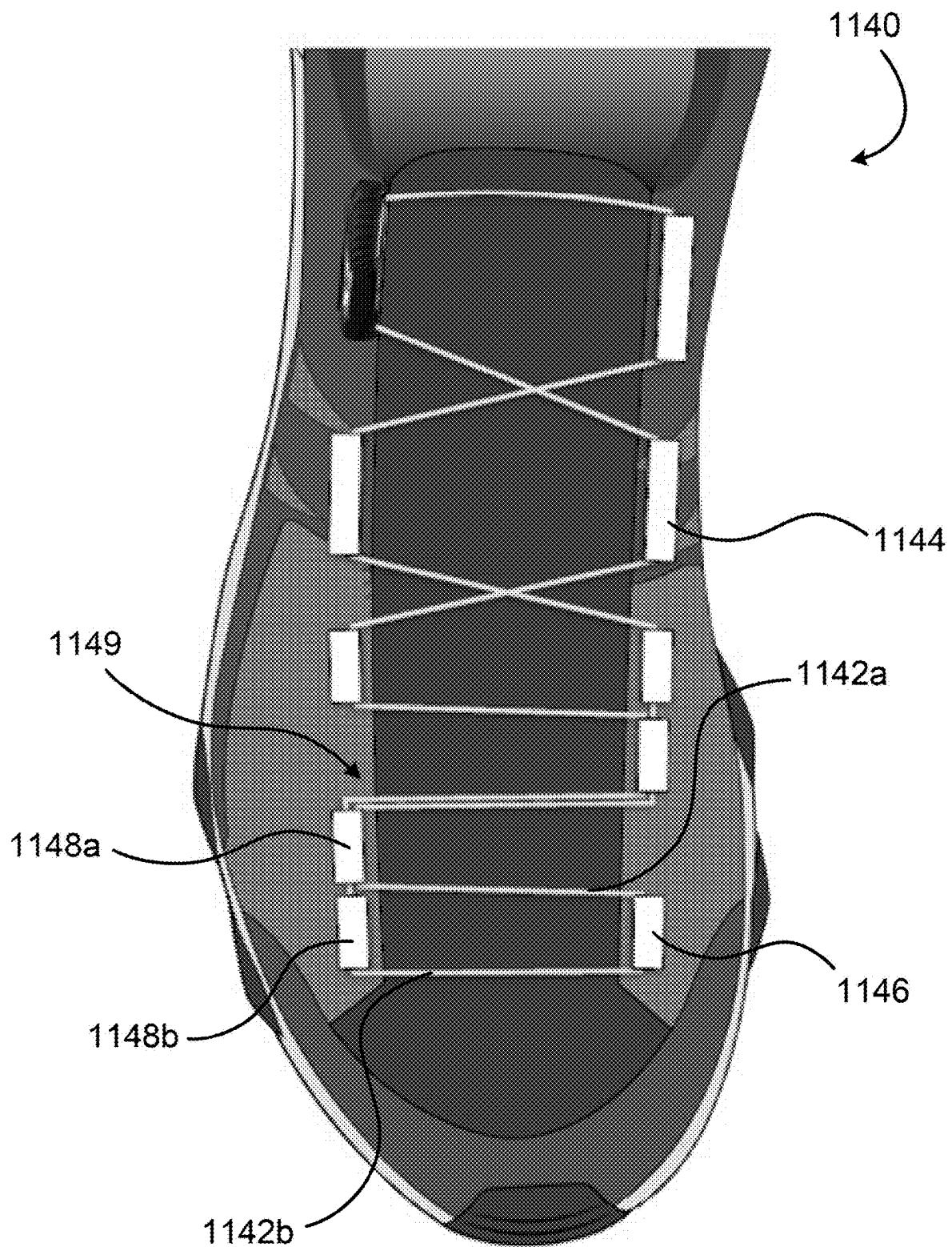


Fig. 35

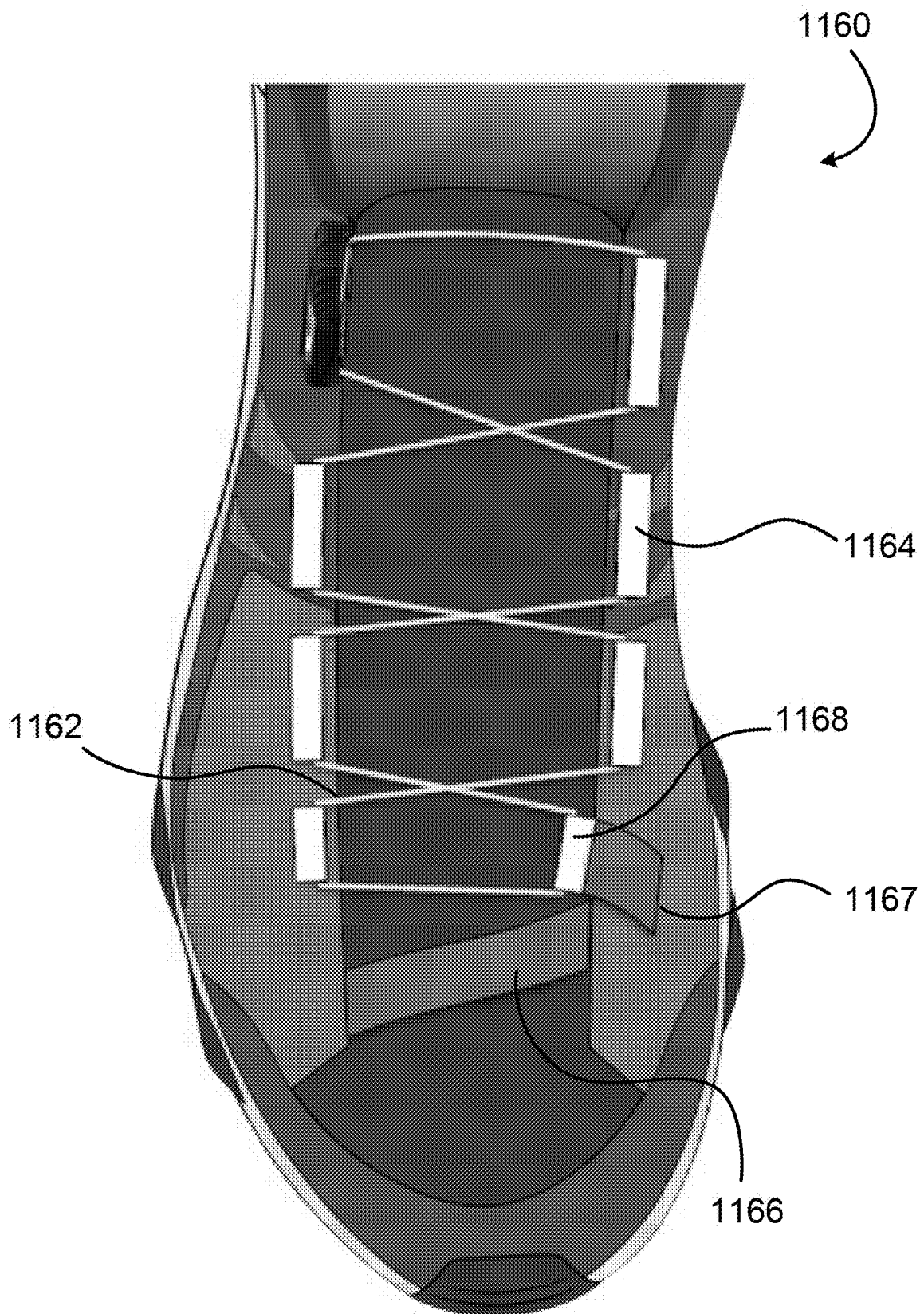


Fig. 36

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CONFIGURATIONS FOR FOOTWEAR EMPLOYING REEL BASED CLOSURE SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation-in-Part of U.S. application Ser. No. 16/038,940 filed Jul. 18, 2018, entitled "System and Methods for Minimizing Dynamic Lace Movement", which claims priority to Provisional U.S. Patent Application No. 62/622,719 filed Jan. 26, 2018, entitled "System and Methods for Minimizing Dynamic Lace Movement;" Provisional U.S. Patent Application No. 62/534,105 filed Jul. 18, 2017, entitled "Alternative Boot Closure System;" and Provisional U.S. Patent Application No. 62/562,161 filed Sep. 22, 2017, entitled "Variable Footwear Configurations." This application also claims priority to Provisional U.S. Patent Application No. 62/873,681 filed Jul. 12, 2019, entitled "Panel Configuration for Tightening an Article of Footwear". The entire disclosure of all of the aforementioned U.S. patent applications are hereby incorporated by reference, for all purposes, as if fully set forth herein.

BACKGROUND

The present disclosure is related to reel based closure devices and articles that employ a reel based closure device, such as braces, medical devices, shoes, clothing, apparel, and the like. Such articles typically include some closure system, which allows the article to be placed about a body part and closed or tightened about the body part. The closure systems are typically used to maintain or secure the article about the body part. For example, shoes are typically placed over an individual's foot and a shoelace is tensioned and tied to close and secure the shoe about the foot. Conventional closure systems have been modified in an effort to increase the fit and/or comfort of the article about the body part. For example, shoe lacing configurations and/or patterns have been modified in an attempt to increase the fit and/or comfort of wearing shoes. Conventional closure systems have also been modified in an effort to decrease the time in which an article may be closed and secured about the body part. These modifications have resulted in the use of various pull cords, straps, and tensioning devices that enable the article to be quickly closed and secured to the foot.

BRIEF DESCRIPTION

The embodiments herein describe reel based closure devices and various shoe configurations that employ reel based closure devices. According to a first aspect, a shoe includes a sole and an upper that is attached to the sole and that is configured to fit around the foot of a user. The upper has a medial side and a lateral side that fit around the medial and lateral side of the foot respectively. The shoe also includes a strap or panel having a proximal end that is attached to the shoe on the medial or lateral side and a distal end that extends toward an opposite side of the shoe. A tension member is guided or directed about a path along the shoe and a tightening mechanism is operable with the tension member to effect tensioning of the tension member upon operation of the tightening mechanism. The strap or panel includes a first guide and a second guide and the tension member is routed through both the first guide and the second guide of the strap or panel.

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In some instances, the path of the tension member is positioned on either the medial or lateral side of the shoe so that the tension member is positioned along the medial or lateral side of the shoe without being positioned atop the shoe. In such instances, the tightening mechanism may also be positioned on the medial or lateral side of the shoe adjacent to the path of the tension member. The first guide may be attached to the strap or panel so that the first guide is positioned proximally of the second guide on the strap or panel. The second guide may be attached to the strap or panel so that the second guide extends distally of a distal end of the strap or panel. The first guide may be configured so that the tension member exits from opposing sides of the first guide and forms a first angle and the second guide may be configured so that the tension member exits from opposing sides of the second guide and forms a second angle. In such instances, the first angle may be greater than the second angle. The strap or panel may be one of a plurality of straps or panels. In a specific instance, the shoe may include three straps and the strap or panel may be a middle strap of the three strap configuration.

According to another aspect, a shoe includes a sole, an upper that is attached to the sole, a first strap or panel that is attached to the shoe, a second strap or panel that is attached to the shoe, a tension member, and a tightening mechanism. The tension member is guided or directed about a path along the shoe and is coupled with the first strap or panel and the second strap or panel. The tightening mechanism is operable with the tension member to effect tensioning of the tension member upon operation of the tightening mechanism. The first strap or panel includes a first guide and a second guide and the tension member is routed through both the first guide and the second guide of the first strap or panel.

The first guide may be attached to the first strap or panel so that the first guide is positioned proximally of the second guide on the first strap or panel. The second guide may be attached to the first strap or panel so that the second guide extends distally of a distal end of the first strap or panel. The first guide may be configured so that the tension member exits from opposing sides of the first guide and forms a first angle and the second guide may be configured so that the tension member exits from opposing sides of the second guide and forms a second angle. In such instances, the first angle may be greater than the second angle. The shoe may also include a third strap or panel.

According to another aspect, a strap or panel for use in tightening a shoe includes a proximal end that is attachable to a first side of the shoe, a distal end that extends toward an opposite side of the shoe when the strap or panel is attached to the shoe, a first guide through which a tension member is insertable, and a second guide through which the tension member is insertable. The first guide is configured to route or direct the tension member along a first path about the shoe and the second guide is configured to route or direct the tension member along a second path about the shoe. The first path is different than the second path.

The first guide may be formed from a strip of material that is folded to form a loop through which the tension member is insertable. The second guide may also be formed from a strip of material that is folded to form a loop through which the tension member is insertable. The first guide may be attached to the strap or panel so that the first guide is positioned proximally of the second guide on the strap or panel. The second guide may be attached to the strap or panel so that the second guide extends distally of a distal end of the strap or panel. The first guide may be configured so

that when attached to the shoe, the tension member exits from opposing sides of the first guide and forms a first angle and the second guide may be configured so that when attached to the shoe, the tension member exits from opposing sides of the second guide and forms a second angle. In such instances, the first angle may be greater than the second angle. A middle portion of the first guide may be covered by material of the strap or panel.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention is described in conjunction with the appended figures:

FIGS. 1A-B illustrate an embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 2A-B illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 3A-C illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 4A-B illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 5A-B illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 6A-B illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 7A-B illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 8A-C illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 9A-E illustrate another embodiment of a lace guide configuration that may be employed to reduce dynamic lace movement in footwear.

FIGS. 10A-B illustrate an embodiment of footwear that is configured to allow easy entry/exit for a user's foot.

FIGS. 11A-B illustrate another embodiment of footwear that is configured to allow easy entry/exit for a user's foot.

FIGS. 12A-B illustrate an embodiment of footwear that includes a lace path having a square or D-shaped configuration.

FIG. 13 illustrates an embodiment of footwear that is configured to allow easy entry/exit for a user's foot and that includes multiple tightening components.

FIGS. 14A-B illustrate an embodiment of footwear in which a lace path is designed to ensure a more even or uniform tensioning and tightening of the footwear along an edge of a flap of the footwear.

FIGS. 15A-B illustrate another embodiment of footwear in which a lace path is designed to ensure a more even or uniform tensioning and tightening of the footwear along an edge of a flap of the footwear.

FIGS. 16-17 illustrate footwear that is configured so that a tension or fit of the footwear about a user's foot may be quickly and easily adjusted.

FIGS. 18A-B illustrate alternative means of coupling a strap with a side of a footwear.

FIGS. 19A-B illustrate other alternative means of coupling a strap with a side of a footwear.

FIGS. 20A-C illustrate an embodiment in which a reel based closure device is coupled with a plurality of straps so

that operation of the reel based closure device simultaneously tensions each of the straps.

FIGS. 21A-B illustrate an embodiment in which an adjustability of a footwear's fit is constrained to pre-designated or defined areas.

FIGS. 22A-B illustrate an embodiment in which a separate tensioning panel or member is attached to footwear.

FIGS. 23A-C illustrate an embodiment similar to FIGS. 22A-B except that a tensioning panel includes additional panel members.

FIGS. 24A-B illustrate a tensioning panel similar to those of FIGS. 22A-B and FIGS. 23A-C except that the tensioning panel includes six separate and individual panel members.

FIGS. 25A-B illustrate embodiments of straps that may be employed to visually indicate a level of tension that is imposed on the strap due to operation of a tensioning mechanism.

FIGS. 26A-C illustrate an embodiment of a strap that is designed to distribute a force that is imposed on an article due to tensioning of a lace or tension member.

FIGS. 27A-C illustrate an embodiment of footwear that is constructed so that the means of closing and tightening the footwear, via a reel based closure device, are built or manufactured into the footwear.

FIGS. 28A-B illustrate another embodiment of footwear that is constructed so that the means of closing and tightening the footwear, via a reel based closure device, are built or manufactured into the footwear.

FIGS. 29A-C illustrate an embodiment of footwear that is constructed so that the means of closing and tightening the footwear, via a reel based closure device, are built or manufactured into the footwear.

FIGS. 30A-B illustrate a means of forming panels on footwear.

FIG. 31 illustrates a method of assembling a reel based closure device.

FIG. 32 illustrates a panel or strap configuration for an article of footwear.

FIGS. 33-36 illustrate shoe configurations that minimize, or compensate for, shifting of a tension member within a toe region of the shoe.

In the appended figures, similar components and/or features may have the same numerical reference label. Further, various components of the same type may be distinguished by following the reference label by a letter that distinguishes among the similar components and/or features. If only the first numerical reference label is used in the specification, the description is applicable to any one of the similar components and/or features having the same first numerical reference label irrespective of the letter suffix.

DETAILED DESCRIPTION OF THE DRAWINGS

The ensuing description provides exemplary embodiments only, and is not intended to limit the scope, applicability or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing one or more exemplary embodiments. It being understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention as set forth in the appended claims.

Some of the embodiments described herein provide footwear configurations that may be employed to minimize the amount of "dynamic movement" of a lower friction lace or tension member about an article. The lower friction lace or

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tension member that is described herein is typically used in reel based closure devices or systems, but may also be used in other tensioning systems, such as pull-cord systems, automated or motorized closure systems, and the like. The lower friction lace or tension members of such systems may beneficially allow the article to be easily closed and/or tightened, but may easily slip or move after such tightening, which in some instances is not desired.

The term dynamic movement as used herein refers to the ability of the lace or tension member to shift and move between zones of the article. The lace or tension members is an elongate structure that is typically a cord, rope, wire, or other similar material that is capable of being tensioned, often to close and/or tighten the article. The disclosure herein focuses primarily on the configuration of "lace guides" or lace routing configurations that are able to minimize the dynamic movement of the lace or tension member (hereinafter tension member) about the article. A lace guide is a component that is positioned about the article and that is employed to direct or route the tension member about a path along the article. Common lace guides include eyelets, hooks, bosses, looped material segments, or other components about which the tension member may be positioned. In a particular embodiment, the lace guide may be a component that includes a channel within which the tension member is positioned. The lace guide may be formed of various plastic and/or metallic materials. The minimization of the dynamic movement in the system may be due to the introduction of additional lace guides that frictionally engage with the tension member to prevent movement of the tension member about the article. The position of the lace guides in the system may further increase the frictional engagement of the tension member with the lace guides, thereby reducing the ability of the tension member to move about the article.

The embodiments described herein that are directed to reducing dynamic lace movement are particularly useful in footwear (e.g., boots and shoes), and particularly in footwear that includes a high ankle or cuff. In footwear, the lower friction tension member may move between the ankle portion of the footwear and the toe portion of the footwear due to movement of the user's foot within the footwear. For example, walking or running in the footwear may cause the lower leg and ankle to press against the ankle portion of the footwear, which may increase the tension in tension member near the ankle. The increased tension in the tension member may cause the tension member to move or slide within the lace guides, which may cause the ankle portion of the footwear to open or widen and may simultaneously cause the toe portion to constrict about the user's toes. This dynamic movement of the tension member may be more pronounced when the footwear includes a high ankle or cuff because the high ankle or cuff is fit around the lower leg. The dynamic movement of the tension member may result in the user feeling like the footwear is not fully tightened or closed about the foot.

Other embodiments described herein provide alternative closure systems, mechanisms, and/or configurations for improving the fit and comfort of footwear, such as a boot. Specifically, the alternative closure systems/configurations improve the ease of entry into and/or exit from the footwear. For example, an individual may find it difficult to enter or exit a boot having a relatively tall ankle portion or a tight fitting collar. An alternative closure mechanism situated, for example, on the medial side of the boot can allow the collar of the boot to expand and provide for quick and easy entry or exit from the boot. The alternative closure systems/

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configurations also provide additional opportunities for adjusting the tightness or fit of at least one portion of footwear. For example, the alternative closure system can be affixed with specific lacing configurations that enable a user to adjust the tightness or comfort and fit of one or more specific regions of the footwear.

In some embodiments the alternative closure systems/configurations may enable the collar and/or ankle portion of the footwear to expand or open up. Expansion may be accomplished by use of an opening and a flap located on a side and/or a back of the footwear. For example, the medial side of a boot may have an opening and flap that overlaps the ankle portion. The opening and flap allow expansion of the ankle section, which may otherwise be constrictive and difficult to fit about a foot. The opening and flap may minimize the difficulty in donning the footwear by enabling the ankle portion to open up and expand. The alternative closure system may be positioned about the opening and flap to enable a user to quickly and conveniently close and secure the footwear about a user's foot.

In one embodiment, the alternative closure system may be a reel based system for tightening a lace or tension member that is routed or guided about a lace path that spans at least a portion of the opening of the footwear. The lace path may include a bottom end that is positioned near a sole of the footwear, a top end that is positioned opposite the bottom end, and a mid-portion that is roughly equidistant from the top end and the bottom end. The reel based system includes a tensioning mechanism and a plurality of guide members that are coupled with the footwear. The reel based system also includes a lace or tension member that is operationally coupled with the tensioning mechanism and that is routed about the footwear along the lace path via a plurality of guide members. The lace includes a first portion and a second portion that may each be operationally coupled with the tensioning mechanism so that operation of the tensioning mechanism simultaneously tensions both the first portion and the second portion. In other embodiments, the first portion of the lace may be operationally coupled with the tensioning mechanism while the second portion of the lace is terminated or coupled with the footwear or another component. In such embodiment, operation of the tensioning mechanism only directly tensions the first portion of the lace and not the second portion of the lace.

Other embodiments described herein provide footwear configurations that may be quickly and easily varied to provide a desired fit about a user's foot. In particular, the position and/or orientation of various components of the footwear may be quickly and easily adjusted to vary the fit of the footwear about the user's foot. In some instances, the fit of the footwear may be adjusted to provide a desired level of support or comfort, such as when the user is engaging in a sporting event, outdoor activity, leisure activity, and the like. In other instances, the fit and/or configuration of the footwear may be adjusted to determine an optimal placement of one or more components and/or how the placement of one or more components may affect the fit and/or functionality of the shoe.

In many of the embodiments, the footwear is closed and/or tightened via a tensioning mechanism and a lace or tension member. The footwear typically includes one or more guides or components (hereinafter guides) that route or direct the tension member or lace (hereinafter lace) about a path of the footwear, such as along the footwear's tongue. The lace is tensioned via operation of the tightening mechanism. Tensioning of the lace causes the article to close and/or tighten about the user's foot. Specifically, the lace may be

routed along and across an opening of the footwear so that tensioning of the lace urges one side of the opening toward an opposite side of the opening in order to close and tighten the footwear about the foot.

The guide is generally positioned near the opening of the footwear, such as on opposing sides of an eyestay, and directs, routes, or guides the lace along and/or across the opening. The guide may be made of a low friction material that minimizes frictional engagement of the lace and guide. In some instances, the guides may be formed of a fabric or webbing type materials that is folded over to form a loop and/or made of one or more plastic or rigid materials.

As briefly described above, the lace is tensioned via operation of the tensioning mechanism. In a specific embodiment, the tensioning mechanism is a reel based closure system. The reel based closure system includes a knob that may be grasped and rotated by a user to tension the lace. Exemplary embodiments of reel based closure devices are further described in U.S. patent application Ser. No. 13/098,276, filed Apr. 29, 2011, titled "Reel Based Lacing System", U.S. patent application Ser. No. 14/328,521, filed Jul. 10, 2014, titled "Closure Devices Including Incremental Release Mechanisms and Methods Therefor," and U.S. patent application Ser. No. 12/623,362, filed Nov. 20, 2009, titled "Reel Based Lacing System", the entire disclosures of which are incorporated by reference herein.

In another embodiment, the tensioning mechanism may be a motorized device or mechanism that tensions the tension member or lace. An exemplary embodiment of a motorized mechanism that may be used to tension the lace is further described in U.S. patent application Ser. No. 14/015,807, filed Aug. 30, 2013, titled "Motorized Tensioning System for Medical Braces and Devices", the entire disclosure of which is incorporated by reference herein.

In yet other embodiments, the tensioning mechanism may be a pull cord type device that is configured to be grasped and pulled by a user to tension the lace. Exemplary pull cord devices are further described in U.S. patent application Ser. No. 14/166,799, filed Jan. 28, 2014, and titled "Lace Fixation Assembly and System", the entire disclosure of which is incorporated by reference herein. For ease in describing the various embodiments herein, the tensioning mechanism will be referred to generally as a "reel assembly" or "reel based closure device".

In some embodiments, multiple alternative systems may be employed to allow the user to customize the tightness and fit of the footwear, such as for comfort. Specifically, multiple reel based systems may be used to tighten and secure different portions or regions of the boot. For example, a user may desire that the ankle portion of the footwear be relatively loose in comparison with a calf or toe portion of the footwear. The use of multiple separate reel based enable these separate portions of the footwear to be differentially tightened, thereby provided a customizable and desired fit of the footwear. In other embodiments, the alternative closure system may be used in addition to traditional laces, or another closure system, that is positioned on the front of the footwear.

For convenience in describing the various embodiments, the remainder of the description will focus on the article typically being an article of footwear, although it should be realized that the configurations and/or other aspects described herein may be equally applicable to other articles, such as bags, boots, packs, helmets, jackets, articles of clothing or apparel, and the like.

Having describe various aspects of the embodiments generally, additional features and aspects will be readily

recognized with reference to the description of the various drawings provided herein below.

Dynamic Lace Movement Minimization

Referring now to FIGS. 1A-B, illustrated is a shoe or footwear **100** that includes a high ankle or upper cuff **102** and a toe portion or forefoot **104**. The footwear **100** also includes a reel based closure device **110** that is operably coupled with a tension member **112** so that operation of the reel based closure device **110** (e.g., rotation of a knob) causes the tension member **112** to be tension, such as by winding the tension member **112** about an internal spool (not shown) of the reel based closure device **110**. The tension member **112** is routed or directed about a path along the footwear **100** via lace guides that are positioned on opposing sides of an opening of the footwear **100**. The lace guides include lower lace guides **118** that are positioned near the forefoot **104** and upper lace guides, **114** and **116**, that are positioned near the upper cuff **102**.

As described previously, when a user walks or runs in conventional high ankle footwear, the lower leg typically presses against the upper cuff, which may increase the tension in the tension member near the upper cuff. The increased tension in the tension member may cause the tension member to move or slide within the lace guides, which may cause upper cuff to widen and the forefoot to constrict about the user's toes. When the lower leg ceases to press against the upper cuff, the upper cuff may relax and close or constrict while the forefoot opens or widens to some degree. The result may be a constant constriction and opening of different portions of the footwear (e.g., the upper cuff and forefoot) about a user's foot, which may be somewhat uncomfortable and/or irritating to the user.

The configuration of the footwear **100** of FIGS. 1A-B minimizes this dynamic movement of the tension member **112**, which results in significantly less constriction and opening of the footwear **100** about the foot. In particular, the footwear **100** employs a lace guide configuration in the upper cuff **102** that adds friction into the system and thereby minimizes movement, shifting, or sliding of the tension member **112** about the footwear **100**. The lace guide configuration includes a first lace guide pair **114** and a second lace guide pair **116** that are each positioned on opposing sides of the reel based closure device **110** and on opposing eyestay edges of an upper of the footwear **100**. The first lace guide pair **114** and second lace guide pair **116** have a stacked configuration, which means that the second lace guide pair **116** is positioned directly laterally on the side of the first lace guide pair **114**. The first lace guide pair **114** typically has a longitudinal length that is greater than the second lace guide pair **116**, which ensures that the positioning of the first lace guide pair **114** in front of the second lace guide pair **116** does not interfere with the tension member **112** being accessible to the second lace guide pair **116**. In some embodiments, the first lace guide pair **114** and second lace guide pair **116** may be separated components while in other embodiments the first lace guide pair **114** and second lace guide pair **116** may be a unitary or single component.

In routing the tension member **112** about the lace path, the tension member **112** exits the reel based closure device **110** near a bottom end that is positioned closer to the forefoot **104**. The tension member **112** is then routed through the first lace guide pair **114** and then immediately to the second lace guide pair **116**. The tension member **112** is routed from the second lace guide pair **116** to the forefoot **104** via the lace guides **118** that are positioned near the forefoot **104**. The reel based closure device **110** is oriented so that the lace exits are near the forefoot **104** to ensure that when the tension

member 112 exits the second lace guide pair 116, the tension member 112 is positioned near the forefoot 104 rather than at the top of the upper cuff 102.

The use of the first lace guide pair 114 and second lace guide pair 116 adds an additional friction element to the system, which helps reduce movement or sliding of the tension member 112 due to the increased frictional engagement of the tension member 112 with the lace guides. In addition, the stacked configuration of the first lace guide pair 114 and second lace guide pair 116 results in the tension member 112 opposing opening or widening of the upper cuff 102 since the tension force is directed essentially opposite of a force that would widen the upper cuff 102. The multiple crossings of the tension member 112 about the upper cuff 102 also increases or amplifies the tension force in the upper cuff 102, which aids in keeping the upper cuff 102 closed and tightened about the lower leg. The stacked configuration of the first lace guide pair 114 and second lace guide pair 116 also results in a more tortuous path of the tension member 112, which renders the opening or widening of the upper cuff 102 more difficult.

Referring now to FIGS. 2A-B, illustrated is another embodiment of lace guide configurations that may be employed to reduce dynamic lace movement in the footwear 100. In particular, the footwear 100 employs the use of a second lace guide pair 122, but the second lace guide pair 122 is not in a stacked configuration with the first lace guide pair 120. Rather, the first lace guide pair 120 is positioned near the top end of the upper cuff 102 while the second lace guide pair 122 is positioned closer to the forefoot 104. The second lace guide pair 122 is also positioned on the footwear's tongue 126 rather than on the eyestay edge of the upper. The footwear 100 also employs a third lace guide pair 124 that is positioned on the eyestay edge of the upper closer to the forefoot 104 than the second lace guide pair 122.

The tension member 112 is routed from the reel based closure device 110 to the first lace guide pair 120. The lace is routed from the first lace guide pair 120 to the second lace guide pair 122 positioned on the footwear's tongue 126 and is then routed from the second lace guide pair 122 to the third lace guide pair 124. The tension member 112 is routed from the third lace guide pair 124 to the lace guides 118 that are positioned near the forefoot 104. The use of the first lace guide pair 120, the second lace guide pair 122, and the third lace guide pair 124 provides a tortuous path for the tension member 112, which increases the frictional engagement of the tension member and thereby reduces movement, shifting, or sliding of the tension member 112 about the footwear 100. The positioning of the second lace guide pair 122 on the footwear's tongue 126 may also counteract some of the forces that cause the upper cuff 102 to open or widen. For example, as the tension member 112 is tensioned and the upper cuff 102 begins to widen, the lace guide pair 122 may press the tongue 126 inward and against the user's lower leg, which may result in a sensation of the fit of the footwear 100 being relatively unchanged. As illustrated in FIGS. 2A-B, the lace exits the reel based closure device 110 may be positioned near the top of the upper cuff 102 to increase the path of the tension member 112 about the footwear 100, which may increase the frictional engagement of the tension member 112 within the system.

Referring now to FIGS. 3A-C, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear 100. In particular, the footwear 100 includes a panel 130 that is positioned over the upper cuff 102 of the footwear 100. The panel 130 may be free floating atop the upper cuff 102 or

may be fixed to the upper cuff 102 as desired. The reel based closure device 110 is positioned on the front surface of the panel 130 and the panel 130 includes lace guides that couple with the tension member 112. Specifically, a strap 132 is attached to the medial and lateral sides of the panel 130 and is fixed to the medial and lateral sides of the footwear 100 as shown in FIGS. 3B-C. The straps 132 are positioned through D-rings 137 that are fixed or anchored to the footwear 100 via stitching 138 or another means of coupling.

The straps 132 are moveable or slidable within the D-rings 137 so that tensioning of the tension member 112 causes the distal end of each strap 132 to move toward the footwear's tongue, which causes the panel 130 to move rearward and against the user's lower leg. The distal end of each strap 132 includes an elongate lace guide 134 that directs the tension member 112 from the reel based closure device toward the forefoot 104 of the footwear. The term "elongate" means that the lace guide 134 has a width and lace channel that is substantially longer than a typical lace guide, such as the lace guides 118 that are positioned near the forefoot 104. The elongate lace guide 134 is attached to the distal end of the straps 132 by forming a loop in the distal end of the straps 132 and attaching the strap to itself. Various other means of coupling may be used to attach the elongate lace guide 134 to the strap 132.

The panel 130 also includes a pull tab or guide 136 that is positioned on a portion of the panel 130 that protrudes or extends toward the forefoot 104 of the footwear 100. The pull tab or guide 136 is positioned on the panel 130 so that it is roughly centered in relation to the footwear's tongue. The pull tab or guide 136 may be formed of a plastic component or formed from folding a fabric material to form a loop. The tension member 112 is positioned through a channel that is formed in the pull tab or guide 136. The channel guides or directs the tension member 112 from the panel 130 and to the guides 118 that are positioned near the forefoot 104. The pull tab or guide 136 is also arranged so that it may be easily grasped by a user's fingers to pull the panel 130 away from the upper cuff 102 of the footwear 100, which aids in opening the footwear 100.

The panel 130 and strap 132 help close the upper cuff 102 around the lower leg and keep it from opening as the footwear 100 is flexed forward due to the user walking and the lower leg contacting the upper cuff 102. The closure of the upper cuff 102 about the lower leg is due mainly to the straps 132 being positioned around the collar of the footwear 100. The straps 132 may aid in minimizing dynamic movement of the tension member 112 by adding an additional friction element in the system. The straps 132 may be made of a material with an increased frictional coefficient so that the panel 130 frictionally engages with itself and thereby helps minimize shifting of the components of the footwear.

Referring now to FIGS. 4A-B, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear 100. In particular, the footwear 100 includes a pair of looped webbing or fabric guides 142 that are positioned in the upper cuff 102 above the reel based closure device 110. The footwear configuration is similar to that illustrated in FIG. 2A except that the fabric guides 142 are disposed above the reel based closure device 110 rather than below the device 110. The tension member 112 is routed from the reel based closure device 110 to a first lace guide 140 and is routed therefrom to the fabric guide 142. From the fabric guide, the tension member 112 is routed to a second lace guide 144 and then is routed to the forefoot 104 via the guides 118. The use of the fabric guide 142 above the reel based closure device 110

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provides a tortuous path for the tension member **112**, which increases the frictional engagement of the tension member and thereby reduces movement, shifting, or sliding of the tension member **112** about the footwear **100**. The positioning of the fabric guides **142** on the footwear's tongue and within the upper cuff **102** may also counteract some of the forces that cause the upper cuff **102** to open or widen, such as by pressing the tongue **126** inward and against the user's lower leg when the tension member **112** is tensioned.

Referring now to FIGS. **5A-B**, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear **100**. In particular, the footwear **100** includes a panel **150** that is positioned atop the tongue and within the upper cuff **102**. The Reel based closure device **110** is positioned on the panel **150**. A proximal side **154** is fixed to the footwear **100** while a distal side is foldable over the footwear's tongue. An upper portion of the tension member **112** is routed from the reel based closure device **110** to a guide member **152** that is positioned within the upper cuff **102**. The tension member **112** is routed from the guide member **152** and underneath the panel **150** to the guides **118** positioned near the forefoot **104**. A lower portion of the tension member **112** is routed from the reel based closure device **110** to an uppermost lace guide **118**.

When the tension member **112** is tensioned, the panel **150** is pulled toward and typically into contact with the guide member **152**. The panel **150** presses the footwear's tongue against a user's lower leg and also pinches or compresses the tension member **112** that is routed underneath the panel **150**. The compression or pinching of the tension member **112** by the panel **150** may increase the frictional engagement of the tension member with the footwear **100** and thereby minimize dynamic movement or shifting of the tension member **112**. The panel **150** also reduces the "crossings" of the tension member **112** across the tongue that can pull open when the user flexes their lower leg forward. The crossings of the tension member **112** are reduced because the proximal side **154** of the panel is fixed or locked to footwear **100** and the length of the tension member **112** between the free end of the panel **150** and the guide member **152** is minimized, which helps keep the upper cuff **102** closed when the footwear **100** is flexed.

Referring now to FIGS. **6A-B**, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear **100**. In particular, the footwear **100** includes a panel or strap **160** that is positioned in the upper cuff **102** of the footwear **100**. The panel **160** is fixed on a proximal end **164** to the footwear **100** while a distal end of the panel **160** is removably coupleable with the footwear **100**. In the illustrated embodiment, the distal end of the panel **160** includes a hook and loop type fastener (e.g., Velcro®) that couples with a corresponding hook and loop type fastener **162** that is positioned on the upper cuff **102** of the footwear **100**. In other embodiments, the distal end of the panel **160** may include other means of fastening the panel **160** to the upper cuff **102**, such as zippers, buttons, magnets, snaps, buckles, and the like.

The coupling means (e.g., hook and loop fastener **162**) couple the panel **160** to the upper cuff **102** independent of the tension member **112** and thus, tensioning of the tension member **112** does not affect the coupling of the panel **160**. In this manner, the panel **160** remains tightened around the user's leg regardless of if the footwear **100** is flexed and/or if the lower leg is pressed outward against the tongue. The reel based closure device **110** may be positioned on the panel

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160 as illustrated or may be positioned elsewhere on the footwear **100** as desired. The tension member **112** is routed from the reel based closure device **110** to the lace guides **118** that are positioned near the forefoot **104** of the footwear **100**.

Referring now to FIGS. **7A-B**, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear **100**. In particular, the footwear **100** includes a fastener, or fasteners, that are positioned on the exterior surface of the footwear's tongue **170** and that engage with the footwear's upper to keep the tongue **170** from moving outward as the user's lower leg presses against the tongue **170**. In the illustrated embodiment, a first edge **171** and a second edge **172** of the tongue **170** include hook and loop type fasteners (e.g., Velcro®) that couple with corresponding hook and loop type fasteners positioned on corresponding inner edges of the footwear's upper. In other embodiments, the first edge **171** and the second edge **172** of the tongue **170** may include other means of fastening the tongue **170** to the footwear's upper, such as zippers, buttons, magnets, snaps, buckles, and the like.

The coupling means (e.g., hook and loop fasteners) secure the tongue **170** to the footwear's upper independent of the tension member **112** and thus, tensioning of the tension member **112** does not affect the coupling of the tongue **170** and upper. In this manner, the tongue **170** remains secure around the user's leg regardless of if the footwear **100** is flexed and/or if the lower leg is pressed outward against the tongue **170**. The reel based closure device **110** may be positioned on a side of the footwear **100** as illustrated or may be positioned elsewhere as desired. The tension member **112** is routed from the reel based closure device **110**, across the upper cuff **102**, and to the lace guides **118** that are positioned near the forefoot **104** of the footwear **100**.

Referring now to FIGS. **8A-C**, illustrated is another embodiment of a configuration that may be employed to reduce dynamic lace movement in the footwear **100**. In particular, the footwear **100** includes a lace guide **182** that is positioned around a heel or collar **180** of the footwear **100**. The lace guide **182** directs, routes, or guides a portion of the tension member **112** from the reel based closure device **110** and around the heel or collar **180** to an upper lace guide **184** that is positioned in the upper cuff **102** of the footwear **102**. The positioning of the lace guide **182** around the heel or collar **180** minimizes dynamic movement of the tension member **112** by increasing the frictional engagement of the tension member **112** with the footwear **100** and/or by providing a force that cinches the heel or collar **180** about the lower leg. An increase in tension of the tension member **112** may result in some cinching or tightening of the heel/collar **180** about the lower leg, which may offset some of the widening or opening of the footwear **100** in response to flexing of the footwear **100** and/or movement of the lower leg within the footwear.

The tension member **112** is routed from the upper lace guide **184** to the lower lace guides **118** that are positioned near the forefoot **104**. In some embodiments, the lace guide **182** may be formed or, or include, a segment of tubing that is disposed around a portion of the heel or collar **180**. The tubing segment may be formed of various fabric or plastic based materials. In other embodiments, a non-tubing lace guide may be employed to route or direct the tension member around the heel/collar **180** of the footwear **100**.

In any of the embodiments described herein, the footwear may include a component that is positioned on or near the tongue that aids in locking or securing the tension member **112** to the footwear's tongue, or any other part of the

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footwear. The dynamic movement or shifting of the lace may be greatly minimized or prevented with the use of a component that locks or secures the tension member **112** in this manner since the tension member **112** in the forefoot **104** and in the upper cuff **102** is essentially locked or fixed in place.

Referring now to FIGS. 9A-E, illustrated is a reel based closure device **110** that is configured so that a tension member **112** is routed axially below the interior region of the reel based closure device's housing component. The tension member **112** is routed in this manner so that the lace frictionally engages with itself and/or with the housing component to minimize dynamic movement or shifting of the tension member as described herein. As illustrated in FIGS. 9C-E, the reel based closure device **110** includes a housing component **110a** that includes an interior region within which one or more components of the reel based closure device **110** are positioned. The components that may be positioned within the interior region of the housing include a spool (not shown), a pawl disc or component (not shown), and/or any of the other components that are described herein or that are described in the applications that are incorporated by reference herein. In regards to the spool, the spool is rotatably positioned within the interior region of the housing component **110a** so that the spool may be rotated via operation of a knob or tightening component **110b** and thereby wind the tension member **112** about the spool.

In most embodiments, the housing component **110a** includes a bottom member and a wall that extends upward from the bottom member. The wall is typically cylindrical and the bottom member and wall define the interior region of the housing component **110a**. The tightening component or knob **110b** (hereinafter knob **110b**) is rotatably coupled with the housing component **110a** and operably coupled with the spool so that an operation of the knob **110b** causes the spool to rotate within the interior region of the housing component to wind the tension member about the spool. The housing component **110a** is configured so that the tension member **112** is routed axially below the housing component's interior region and to a lace path that is positioned along an opening of an article, such as a shoe as illustrated in FIG. 9A. The tension member **112** may be routed axially below the housing component interior region via tubing that is positioned within or axially below the bottom member or via a channel that is formed within or below the bottom member.

As illustrated in FIG. 9E, a channel **197** may be formed or defined in a bottom surface of the bottom member. The channel **197** extends axially upward from a bottom surface of the bottom member so that a channel is formed between the bottom member and an upper surface of an article (e.g., a shoe's upper or tongue) when the reel based closure device **110** is attached to the article. The channel **197** is typically about 10-50 mm wide and more commonly about 20-40 mm wide. In a specific embodiment, the channel **197** has a width of approximately 30 mm. The channel **197** also typically extends from one side of the housing component **110a** to the opposite side of the housing component **110a** so that when the reel based closure device **110** is attached to the article, the channel **197** extends entirely through, and typically underneath, the housing component **110a**. Stated differently, when the reel based closure device **110** is attached to the article, the channel **197** typically extends between opposing sides of the housing component **110a**, which allows the tension member **112** to be inserted entirely through the reel based device **110** below the housing component's interior region.

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The channel **197** typically also has a depth of between 1 and 4 mm, and more commonly between 2 and 3 mm. In a specific embodiment, the depth of the channel **197** is approximately 2 mm. The depth of the channel **197** is measured from the bottom surface of the bottom member and axially upward from the bottom surface. The depth is sufficient to enable the tension member **112** to be inserted through the channel **197**, but is shallow or small enough so that the tension member **112** frictionally engages with itself within the channel **197**, or frictionally engages with the channel itself. In some embodiments, the channel **197** or the housing component **110a** is made of a material that increases the frictional engagement of the tension member **112** with the channel **197**.

As illustrated in FIG. 9C, the tension member **112** is inserted through the channel **197** so that the tension member **112** crosses itself at a point **198** within the channel **197**. Specifically, a first portion **112a** of the tension member **112** is routed from the housing component **110a** and around a first guide **192**. The first tension member portion **112a** is routed from the first guide **192** and through the channel **197**. The first tension member portion **112a** is routed from the channel **197** and toward a lace path via a second guide **193**. Similarly, a second portion **112b** of the tension member **112** is routed from the housing component **110a** and around a first guide **192**. The second tension member portion **112b** is routed from the first guide **192** and through the channel **197**. The second tension member portion **112b** is routed from the channel **197** and toward a lace path via a second guide **193**. The first tension member portion **112a** crosses the second tension member portion **112b** at point **198** within the channel **197**. The first tension member portion **112a** frictionally engages with the second tension member portion **112b** at point **198**, which minimizes movement of the first tension member portion **112a** relative to the second tension member portion **112b** and thereby minimizes dynamic movement of the tension member **112** as described herein.

The depth of the channel **197** may be configured so that at the tension member crossing point **198**, the combined thickness of the first tension member portion **112a** and the second tension member portion **112b** is approximately equal to or thicker than the depth of the channel **197**. In such embodiments, the channel **197** may effectively pinch or squeeze the first tension member portion **112a** and the second tension member portion **112b** at point **198** between the bottom surface of the bottom member and an outer material of the article that the reel based closure device **110** is attached to. The pinching or squeezing of the first tension member portion **112a** and the second tension member portion **112b** at point **198** may further reduce relative movement of the tension member portions and thereby further reduce dynamic movement of the tension member **112** within the system.

As illustrated in FIG. 9E, the channel **197** is typically U-shaped and opposing sides or walls **195** of the U-shaped channel typically have an arcuate or curved shape. The arcuate or curved walls **195** allow the tension member **112** to slide within the channel **197** without overly abraising or wearing on the walls **195**. In this manner, the tension member **112** does not prematurely fall due to sliding of the tension member through the channel **197**. The U-shaped channel typically extends between opposing sides of the housing component **110a**.

The reel based closure device **110** may also include a base component **196** or bayonet that is configured to releasably couple with the housing component **110a**. The base component **196** includes a relatively large flange or surface that

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is configured to allow the base component **196** to be easily attached to the article via stitching, adhesive bonding, RF or other welding, mechanical fastening, and the like. After the base component **196** is attached to the article, the housing component **196** may be coupled with the base component **196** to attach the reel based closure device **110** to the article. The reel based closure device **110** may be subsequently detached or removed from the base component **196** for inspection, replacement, etc. In some embodiments, the base component **196** may be integrally formed or attached to the housing component **110a** as desired.

The base component **196** includes an aperture **194** that aligns with the channel **197** of the housing component **110a** when the housing component **110a** is coupled with the base component **196**. The aperture **194** allows the tension member **112** to be routed from the housing component **110a**, through the base component's aperture **194**, and through the channel **197** of the housing component **110a**. The aperture **194** may have a width that corresponds to the width of the channel **197** or is typically larger than the channel **197**. This arrangement helps ensure that the tension member **112** does not abrade or wear on the aperture **194**, which may cause premature failure of the tension member **112**. Coupling of the base component **196** and the housing component **110a** may be achieved via the configuration described in U.S. patent application Ser. No. 15/836,475, filed Dec. 8, 2017 and titled "Reel Based Closure System," the entire disclosure of which is incorporated by reference herein.

Alternative Closure Systems, Mechanisms, and/or Configurations for Footwear

Referring now to FIG. **10A**, illustrated is an embodiment of a shoe **200** that is configured to allow easy entry/exit for a user's foot. This is achieved, in part, by positioning the alternative closure system about the shoe **200** so that it tensions and pulls a flap closed about an ankle portion of the shoe **200**. In addition to the alternative closure system, the shoe **200** may also include traditional laces or a reel based closure system (see FIG. **11B**) on the front of the shoe **200** in a traditional location for laces (i.e., about a tongue of the shoe). The use of traditional laces, a reel based closure system, and/or other shoe closure device (e.g., Velcro®) may be employed on any of the embodiments described and illustrated herein. The shoe **200** includes a reel assembly **202** that is positioned on a side of the shoe **200**. In some embodiments, the reel assembly **202** include walls or members **203** that function to shield or protect the reel assembly's knob from contact with external objects by blocking or preventing the external objects from contacting the knob. The reel assembly **202** is operably coupled with a lace **204** so that rotation of the knob causes the lace to be tensioned. As illustrated in FIG. **10A**, one end of the lace **204** is attached to the reel assembly **202** while the opposite end of the lace **204** is attached to a lower portion of the shoe **200**. As such, operation of the reel assembly **202** tensions only a single end of the lace **204**.

The reel assembly **202** is positioned on the back or heel of the shoe **200**, however it may also be positioned elsewhere on the shoe **200** as desired—e.g., on the medial side, lateral side, or on the tongue of the shoe **200**. The reel assembly **202** may be sewn onto the shoe **200** or may be heat welded, adhered, or otherwise affixed to the sole of the shoe **200**. If the reel assembly **202** is located at or near the top of the shoe **200**, then the non-sewing alternatives may be preferable to avoid compressing or altering the shape of the collar of the shoe **200**.

The lace **204**, typically traverses the opening or discontinuity in the ankle portion of the shoe **200** and when

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tensioned, will draw a flap **208** over the opening or discontinuity to tighten or secure the shoe **200** about a user's ankle. The flap **208** may begin at the top of the shoe **200** and continue either partially or fully to the sole **206** of the shoe **200**. The flap **208** may be substantially vertical, however the flap **208** may curve or become partially horizontal towards the bottom of the shoe **200** to follow the natural contours of the shoe **200** and/or foot. Tightening of the shoe **200** about the ankle is accomplished by tensioning the lace **204** via the reel assembly **202**. Since the lace **204** is routed along opposing sides of the opening via several guides **210** traversing between the flap **208** and the remainder of the shoe, tensioning of the lace **204** causes the shoe **200** to constrict about the user's ankle. The guides **210** may include channels or lumens about which the lace **204** slides. In some embodiments the guides **210** may be tubing through which the lace **204** is inserted. The guides **210** may be sewn onto, adhered, or otherwise affixed in the various locations along the opening of the shoe **200**. Alternatively, the guides may be mounted within the outsole or midsole of the shoe **200**. There may be any number of guides **210** between the reel assembly **202** and the attachment point at the sole **206**, as well as many different lace configurations or paths. In the embodiment of FIG. **10A**, the lace **204** begins at the reel assembly **202** at the back of the shoe and traverses along the opening via multiple guides **210** that are attached on opposing sides of the flap **208**. The lace **204** terminates and is anchored, affixed, or otherwise attached, either permanently or removably, to the body of the shoe **200** near the sole **206**.

FIG. **10B** illustrates a shoe **220** having an alternative closure system that is similar to that illustrated in FIG. **10A**. The main difference between FIG. **10A** and FIG. **10B** is that instead one of the guides **210** is replaced with a strap **222** having a guide portion **224** through which the lace **214** is routed. The strap **222** is attached to the shoe **220** on the lateral side of the shoe **220** and extends across the front of the shoe **220** to a location near an edge of the flap. The strap **222** may be made of a textile, plastic, or other such durable material suitable for shoe construction. The use of the strap **222** changes how the shoe **212** tightens in response to tensioning of the lace **214**. In some embodiments any and/or all of the guides **221** may be replaced with a similar strap. The guide portion **224** may be affixed to the strap **222** in a manner similar to the guides **221**; namely via sewing, adhesive, heat welding, or other such attachment means. Alternatively, the guide portion **224** may be integrally formed with, or internal to, the strap **222**, for example by folding a distal end of the strap **222** and affixing the distal end to create a loop for the guide portion **224**.

FIG. **11A** is similar to the embodiment of FIG. **10B** except that the reel assembly **232** is centrally located on the side of the shoe **230**, and both ends of the lace **234** are operationally attached or coupled to the reel assembly **232**. The reel assembly **232** is located midway up the extended ankle of the shoe **230** so that the lace path has a lace configuration that promotes a more even or uniform tightening of the lace **234** along the entire lace path. For example, a typical lace experiences frictional losses due to routing of the lace through the various guides positioned along the lace path. The frictional losses often result in a lower lace tension as the lace gets farther or more remote from the reel assembly. The central location of the reel assembly **232** of FIG. **11A** helps promote more even tightening of the lace **234** by compensating or accounting for frictional losses due to routing of the lace **234** through the various guides **231**. In particular, the central positioning of the reel assembly **232** helps to ensure that the lace **234** remains relatively close to

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the reel assembly **232** along the entire length of the lace path. For example, the lace **234** is immediately routed to the top and bottom of the lace path, which ensure that the lace tension at the top and bottom of the lace path is the greatest. By ensuring that the portions of the lace **234** that experience the greatest tension are immediately routed to the top and bottom of the lace path, the lace **234** will tighten the flap **238** against the upper ankle portion more evenly or uniformly. The guides **231** shown in FIG. **11A** may be textile or fabric guides, such as a loop sewn into a small piece of strapping or webbing that is sewn onto the shoe. Alternatively, the guides may be a loop sewn into one end of a longer strap that is positioned within or atop the shoe **230**. For example, the guide **231** may be attached to the medial side of the shoe **230** and routed across the front of the shoe to the lateral side of the shoe **230**, either over the top of the shoe or between one or more layers of the shoe upper assembly. These straps may function similarly to the strap **222** of FIG. **10B**.

The lace path of FIG. **11A** is shown having an overlapping figure-eight style, in which the lace **234** exits the reel assembly **232** and traverses between various guides **231** positioned on opposing sides of the flap **238**. Though the embodiment shown in FIG. **11A** has five total guides **231**—i.e., three guides located on one side of the flap **238** and a pair of guides directly above and below the reel assembly **232**—other embodiments may have additional or fewer guides while maintaining the criss-crossing figure-eight lace path and centrally located reel assembly **232** illustrated in FIG. **11A**.

FIG. **11B** is similar to the embodiment of FIG. **11A**, in that FIG. **11B** employs a centrally located reel assembly **242** that provides even or uniform tightening of the shoe despite frictional losses associated with routing the lace **244** through multiple guides **241**. FIG. **11B** displays a lace path in which the lace **244** crosses itself as it is routed along an opening of the shoe **240**. Specifically, the lace **244** is routed from the reel assembly **242** to a pair of guides **241** that are positioned near an edge of the flap **248**. The lace is then routed back across the opening and underneath the lace **244** near the reel assembly **242**. FIG. **11B** additionally displays a traditional shoelace **246** disposed on the front of the shoe over the tongue or gusset of the shoe **240**. Though not pictured here, as described above, other traditional closure mechanisms (e.g., Velcro straps, an additional reel assembly, etc.) may be used in this and any embodiment of the present disclosure. In some embodiments, the shoelace **246** or other closure mechanism may be used for aesthetic purposes so that the shoe **240** has a more traditional appearance. In other embodiments, the shoelace **246** or closure mechanism may be functional and employed to additionally tighten the shoe **240**. In such embodiment, the shoelace **246** may be the primary means or mode of closing and tightening the shoe **240** and the reel assembly **242** may be used as a secondary closure system.

FIG. **12A** illustrates a lace path having a square or D-shaped configuration. The reel assembly **252** is illustrated as being positioned near a top or upper end of the shoe **250**, although the reel assembly **252** may be positioned elsewhere on the shoe **250**, such as lower on the shoe **250** or toward a front or back of the shoe **250**. The illustrated embodiment employs a single guide **256** that is used to route the lace **254**, although in other embodiments multiple guides may be used. Alternatively, the guide **251** may not be present in some embodiments of the invention so that the lace **254** defines a triangular shape between the reel assembly **252** and the guide **256**. The reel assembly **252** may be attached or

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connected to the shoe through any method described herein or otherwise known in the art.

The guide **256** of FIG. **12A** is pictured as a partially enclosed u-shaped tubular design through which the lace **254** travels. The guide **256** may have one or more openings **255** that allow debris or other material to escape the guide **256** and prevent additional friction on the lace **254**. The guide **256** may be positioned so that it primarily aligns with the edge of the flap **258** or is offset from the edge of the flap **258**. In this configuration, the reel assembly **252** becomes difficult to tension when the guide **256** is in a vertical position with the flap **258** fully tightened. This results in the flap **258** bottoming out, or returning to an identical position each time the shoe **250** is worn providing an identical fit and tightness for the user. To further achieve a desired “bottoming out” of the system, the tensioning power or ability of the reel assembly **252** may be selected so that the reel assembly **252** is not further operable to tension the lace after the desired lace tension is achieved.

FIG. **12B** illustrates an embodiment that is similar to FIG. **12A**, except that the guide **266** is open on one side to allow the lace **264** to be removed completely from the guide **266**. FIG. **12B** further illustrates the lace **264** being removed from the guide **266**. Specifically, FIG. **12B**, step A, shows the shoe **260** with the lace **264** positioned within the guide **266**. In this position, the reel assembly **262** may be operated to tension the lace **264** and thereby close or pull the flap **268** secure about a user's ankle. FIG. **12B**, step B, shows the shoe **260** with the tension in the lace **264** loosened such that the lace **264** is able to be partially removed from the guide **266**. To remove the lace **264**, a rear surface of portion of the guide **266** is open so that the lace **264** may be withdrawn from a channel of the guide **266** through the opening. FIG. **12B**, step C, illustrates the reel assembly **262** with the lace **264** fully loosened so that the lace **264** may be easily removed and uncoupled from the guide **266**. Removing the lace **264** from the guide **266** allows the flap **268** to open or to be pulled towards the front or toe of the shoe to allow easy donning/doffing of the shoe. The other lace guides described herein may have open rear surfaces that allow the lace to be removed as illustrated in FIG. **12B**.

FIG. **13** illustrates an embodiment of a shoe **270** that includes multiple alternative closure systems. Specifically, the shoe **270** features a first reel assembly **272** that is positioned on the shoe **270** in a manner similar to the reel assembly **232** of FIG. **11A**. The lace **274** passes through guides **271** in a figure-eight pattern about a path that extends only along a lower portion of the flap **278**. The lace path of the lace **274** may differ from the figure-eight pattern illustrated in FIG. **13** while extending only along the lower portion of the flap **278**. A second reel assembly **275** is positioned on an upper portion of the flap **278**. The second reel assembly **275** is operable with a lace **277** and a single guide **276** that are positioned on the upper portion of the flap **278**. The second reel assembly **275** is employed to tighten the upper portion of the flap **278** independently of the first reel assembly **272**. This embodiment allows for independent and differential tensioning or tightening of the upper and lower portions of the shoe **270**, which allows the user to customize the fit of the shoe **270** to their preference. The differential or zonal tightening of the shoe **270** in FIG. **13** can be accomplished through use of either multiple reel assemblies, such as reel assemblies **272** and **275**, or through use of a single reel assembly with multiple laces routed through different zones of the shoe and traversing differing

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numbers of guides. The guides, **271** and **276**, illustrated in FIG. **13** may be any of the guides and/or straps described herein.

FIG. **14A** illustrates an embodiment in which the path of the lace **284** is designed to ensure a more even or uniform tensioning and tightening of the shoe **280** along the entire edge of the flap **288**. The reel assembly **282** is positioned near the top of the shoe **280**, although the reel assembly **282** may be positioned elsewhere on the shoe, such as near the middle or bottom of the ankle portion of the shoe **280**. The guides **281** of the embodiment of FIG. **14A** are tubes which have a u-shape configuration, though other tube configurations may be employed. The guides **281** may either be attached to the surface of the shoe **280**, using any method described above, or may be positioned under one or more layers of the shoe **280** so that the guide **281** is essentially not visible from the exterior surface of the shoe **280**.

The lace path of FIG. **14A** is designed to provide an even or uniform tensioning through the use of a sequenced or spiral configuration. An upper portion of the lace **284** is routed directly from the reel assembly **282** at the top of the shoe into a first guide **281** that immediately routes or directs the lace **284** to the bottom of the lace path. The lace **284** then traverses across the opening to a guide **281** that is positioned near the back of the shoe **280**, which guide **281** routes or directs the lace **284** to a mid-point of the lace path below the reel assembly **282**.

A lower portion of the lace **284** is routed directly from the reel assembly **282** to a second guide **281** that also immediately routes or directs the lace **284** to the bottom of the lace path. The lace **284** then traverses across the opening to a guide **281** that routes or directs the lace **284** to the mid-point of the lace path below the reel assembly **282**. The upper and lower portions of the lace **284** intersect or meet at the mid-point of the lace path. This lace path configuration provides a more even tensioning and tightening of the shoe **280** by minimizing the amount of frictional loss that the lace **284** experiences as it is routed from the top of the lace path to the bottom of the lace path. Specifically, by directing or routing each lace from the reel assembly **282** to the bottom of the lace path, the frictional loss in the lace is minimized and the lace tension is more uniformly distributed along the opening.

FIG. **14B** illustrates an embodiment similar to FIG. **14A**, except that the sequential or spiral lace path is used as described to close or tighten the front of the shoe **290**. In this embodiment of the invention, rather than the guides **291** being located on the flap and rear portion of the shoe **290**, the guides are positioned on the medial and lateral sides of the tongue **298** of the shoe to guide the lace **294** about the opposing sides of the tongue. The reel assembly **292** is likewise located on the side of the shoe **290**, near the front of the shoe **290** rather than the rear as illustrated in FIG. **14A**. The lace path provides a more even and uniform tensioning or tightening of the shoe **290** along the shoe's tongue. Though a particular arrangement for the sequential lacing has been described herein, further modifications of the lace path may be employed that incorporate additional guides or fewer guides while maintaining the principle of equalizing the tension across an opening of the shoe **290** between the top and bottom end of the lace path.

FIG. **15A** illustrates another embodiment of a sequential or spiral lace path. The lace path configuration of FIG. **15A** may provide quicker tensioning and tightening around the throat or collar **307** of the shoe **230**. In this embodiment, the reel assembly **232** is positioned on the side, or at the back of the shoe **230**, optionally at the top or the bottom of the shoe

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230. The reel assembly **232** may be attached to the shoe **230** using any of the methods described above, and may be used in concert with one or more other reel assemblies on the shoe **230**. The path of the lace configuration begins at the reel assembly **232** and proceeds in an diagonally downward direction towards the front of the shoe **230** onto the flap **238**. The path then traverses back from the flap **238** toward the back of the shoe **230** before continuing in an upward diagonal direction towards the front of the shoe **230**. The diagonally upward and diagonally downward directed laces criss-cross as illustrated. The laces are directed via guide members **306**, which in some embodiments may be formed of sections of tubing. As illustrated, an upper portion of the lace **234** is routed along an outer perimeter of the lace path while a lower portion of the lace is routed along an inner perimeter of the lace path. The lace **234** may be routed via tubular guides **306**, or via any other type of guide described herein.

FIG. **15B** is substantially the same as FIG. **15A**, with the exception that instead of placing the closure system on a side of the shoe, the reel assembly **312**, lace **314**, and guides **311** are all located on the front portion of the shoe surrounding the tongue **318**. Specifically, the reel assembly **312** is located on the lateral side of the shoe near the top and near the tongue **318**. The path that the lace **314** follows is substantially the same as FIG. **15A**, with an upper portion of the lace **314** routed along the perimeter or outside of the lace path and an inner portion of the lace **314** routed along an inner portion of the lace path. The embodiment shown in FIG. **15B** may be used in conjunction with another closure system such as a reel system, traditional laces, or a zipper on the side of the boot. Additionally, there may be a second reel assembly and lace which serve to tighten a different region or portion of the shoe **231** than the system shown in FIG. **15B**.

Although the embodiments described herein have been illustrated employing manual reel based devices, it should be realized that various other tensioning mechanisms or devices may be used. For example, a pull cord device and/or motorized device may be employed to tension the lace as desired. In a specific embodiment, a motorized lace tensioning device may be concealed within or adjacent the sole of the shoe and the lace may be routed to the motorized lace tensioning device under one or more layers of the upper.

Footwear Component Adjustment and Variability

Referring now to FIG. **16**, illustrated is a shoe or footwear **400** that includes an upper with a pair of eyestays **401** that are positioned on opposite sides of an opening of the shoe **400** (e.g., the shoe's tongue). The shoe is configured so that the tension or fit of the shoe about the user's foot may be quickly and easily adjusted. The adjustment of the shoe **400** is achieved via a strap **402** that is positioned across the opening and that extends between the opposing eyestays **401**. A proximal end of the strap is attached or fixed to one side of the shoe **400** (e.g., the medial side) while a distal end of the strap **402** is freely moveable and positionable about the shoe. The distal end of the strap **402** may be coupled or removably attached to an opposite side the shoe **400** (e.g., the lateral side) in one of multiple positions or orientations, which affects how the shoe **400** will close and/or tighten about the user's foot.

In the illustrated embodiment, the distal end of the strap **402** may be removably attached to the opposite of the shoe **400** in one of three distinct positions or zones, which are labeled "a", "b", and "c" in FIG. **16**. Coupling of the distal end of the strap **402** to one of the respective zones (e.g., positions "a", "b", or "c") is achieved via a fastener **406**,

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which in the illustrated embodiment is a hook and loop type fastener, such as those commonly sold under the brand name Velcro®. Although the fastener 406 is illustrated as being positioned in zone “c”, it should be realized that similar fasteners 406 are positioned in each of the other zones (i.e., positions “a” and “b”) to enable the distal end of the strap 402 to be coupled with those respective zones of the shoe 400.

In FIG. 16, the distal end of the strap 402 is illustrated as being attached or coupled with a reel based closure device 404 via a length or segment of lace 405 that extends between the reel based closure device 404 and a distal tip or end of the strap 402. As described above, the reel based closure device 404 is configured so that operation of the device causes the lace 405 to be tensioned, which in turn tensions the strap 402 and tightens the shoe 400 about the user's foot. To attach the distal end of the strap 402 to the shoe 400, a rear or bottom surface of the reel based closure device 404 includes a corresponding hook and loop fastener (not shown) that enables the reel based closure device 404 to be coupled with the appropriate fastener 406 in the desired zone.

FIG. 16 illustrates that the distal end of the strap 402, and the reel based closure device 404, is attached to zone “a” of the shoe 400. With the distal end of the strap 402 attached to zone “a”, the strap 402 extends across the opening or tongue of the shoe 400 near the shoe's collar. Thus, tensioning of the strap 402, via the reel based closure device 404 and lace 405, causes the shoe 400 to tighten or constrict to a greater degree about the shoe's collar in comparison with other portions of the shoe 400 (e.g., the toe box). Tensioning of the strap 402 in this manner causes the shoe 400 to constrict and tighten about the user's ankle to a greater degree than other portions of the shoe 400. If a user desires to change the fit of the shoe 400, and more specifically the manner in which the shoe constricts about the foot, the user may simply change the orientation or position of the strap 402 about the shoe 400. For example, FIG. 16 illustrates a second position of the strap 402 (i.e., in dashed lines) in which the distal end of the strap 402 is attached to zone “b” and in which the strap 402 extends across a mid-portion of the shoe 400. Tensioning of the strap 402 in this position would cause the shoe 400 to tighten or constrict to a greater degree about the mid-portion of the shoe. Similarly, the distal end of the strap 402 may be attached to zone “c” so that the strap 402 extends across a lower portion of the shoe 400 nearer to the toe box. Tensioning of the strap 402 in this position would cause the shoe 400 to tighten or constrict to a greater degree about the lower portion of the shoe near the toe box in comparison with the shoe's collar.

The user may adjust the fit of the shoe 400 based on an activity that the user is engaging in, such as a sporting event, an outdoor activity (i.e., hiking ascent vs. descent), and the like, or the user may adjust the fit of the shoe 400 to achieve a desired comfort level. In other instances, the position of the strap 402 may be changed to visually and/or physically demonstrate how a configuration of the shoe 400 may be modified in order to determine an optimal placement of one or more shoe components in designing and/or manufacturing the shoe 400. For example, it may be determined that the shoe 400 should include a single strap 402, but the final or optimal placement of the strap 402 in relation to the shoe 400 may be unknown. In order to facilitate in this decision making process, the adjustability of the strap 402 may be employed so that a team of designers and/or engineers may quickly assess how the various strap orientations affect the fit and/or functionality of the shoe 400. Once an optimal

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placement of the strap 402 has been determined, the shoe 400 may be constructed with the desired/selected strap placement. Similar rationales for the adjustable fit and/or visual/physical demonstration of the shoe configuration are applicable to the various other embodiments described and/or contemplated herein.

FIG. 17 illustrates a concept similar to that of FIG. 16 except that the distal end of the strap 410 is not coupled with a reel based closure device. Rather, the distal tip or end of the strap 410 is directly attached or coupled to the side of the shoe 400 in one of multiple positions or areas illustrated by zones “a”, “b”, and “c”. The configuration and use of the strap 410 is similar to that of FIG. 16 except that the reel based closure device (not shown), or another tensioning mechanism, is position on the opposite side of the shoe or elsewhere on the shoe 400 as desired, such as on the shoe's heel or tongue. The distal end of the strap 410 is coupled with the shoe 400 via fasteners 412, which in the illustrated embodiment are hook and loop type fasteners. The strap 410 is illustrated as being coupled with zone “a” of the shoe, but is also shown in dashed lines as being coupled with zones “b” and “c”.

In another embodiment, the concepts of FIGS. 16 and 17 could be combined so that the reel based closure device 404 and the strap 410 are both movable between one or more positions along the eyestay of the shoe 400. In such embodiments, the entire strap could be removed from the shoe 400 and both ends could be fastened to a desired fastener—i.e., the reel based closure device 404 could be attached to a desired fastener 406 and the strap 410 could be attached to a desired fastener 412. In such instances, the entire orientation and positioning of the strap about the shoe 400 could be modified as desired.

FIGS. 18A-B illustrate alternative means of coupling a strap with a side of a shoe 400. In FIG. 18A, the strap 420 includes a snap button 421 that is able to snap into engagement with a corresponding snap button 422a-c that is attached to the shoe and positioned within a respective zone. The snap button 421 may be uncoupled with one of the snap buttons, 422a-c, and re-engaged with another snap button, 422a-c, to change the position or orientation of the strap 420 about the shoe 400. FIG. 18B likewise illustrates a strap 430 that includes a component 431 that is configured to releasably engage with a corresponding component 432a-c attached to the shoe and positioned within a desired zone. The components of the strap 430 and shoe 400 may be magnetic components, adhesive based components, and the like.

FIGS. 19A-B illustrate other alternative means of coupling a strap with a side of a shoe 400. In particular, the shoe 400 is made of an outer material 444 that may be mesh, woven, knitted, or any other material that is relatively open and/or porous. The reel based closure device 404 is attached to a side of the shoe 400 (or elsewhere as desired) and the lace 405 that is coupled with the reel based closure device 404 is coupled with an attachment component 440, such as by inserting or threading the lace 405 through a looped proximal end of the attachment component 440. The attachment component 440 includes a pair of hooks or grip members 442 that are configured to grip or couple with the outer material 444 of the shoe 400. This design allows the component 440 to be pulled across an opening of the shoe 400, such as the tongue, and attached to an opposite side of the shoe 400 as illustrated in FIG. 19B. Subsequent operation of the reel based closure device 404 functions to tension

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the attachment component **440** and the opposite side of the shoe **400**, which closes and tightens the shoe about the user's foot.

Although not illustrated, in some embodiments the length of lace **405** between the reel based closure device **404** and the attachment component **440** may be replaced with a strap or other tensionable member, such as those illustrated in FIGS. **25A-26C**. The design of the outer material **444** and the attachment component **440** enables the lace **405**, or a strap, to be positioned or oriented essentially anywhere in relation to the shoe **400** rather than the pre-designated zones (i.e., zones "a", "b", and "c") of the previous figures. Thus, the fit of the shoe **400** may be customized to a greater degree with the design of FIGS. **19A-B** in comparison with the previous figures. In other embodiments, the outer material **444** may extend over only a designated portion of the shoe **400** rather than essentially the entire outer surface, which may provide great latitude in customizing the fit of the shoe within a pre-designated area or range.

FIGS. **20A-C** illustrate an embodiment in which the reel based closure device **404** is coupled with a plurality of straps so that operation of the reel based closure device **404** simultaneously tensions each of the straps. Specifically, the reel based closure device **404** is coupled with the lace **405**, which is in turn coupled with a proximal guide **423**. The proximal guide **423** is attached or coupled with a proximal end of a first strap **502** and a proximal end of a second strap **504**. Operation of the reel based closure device **404** causes the first strap **502** and the second strap **504** to be simultaneously tensioned.

The first strap **502** and the second strap **504** are each routed across an opening of the shoe **400**, such as across the shoe's tongue. As shown in FIG. **20B**, the first and second straps, **502** and **504**, are each routed through a guide or ring **505** that redirects or routes the respective strap back across the opening of the shoe **400**. A distal end of each of the straps, **502** and **504**, includes an attachment component that is able to couple with the side of the shoe. In particular, a distal end of the first strap **502** includes an attachment component **506** that includes a fastener or fastening member (e.g., hook and loop, magnet, button, pair or hooks or grips, etc.) that is able to attach to the side of the shoe **400** as described herein. The distal end of the second strap **504** likewise includes an attachment component **508** that is able to attach to the side of the shoe **400**.

In some embodiments, the shoe **400** may include a tightening area or region **510** within which the attachment components, **506** and **508**, are able to attach to the shoe **400**. The tightening region **510** may include a mesh material, Velcro® material, woven material, magnets, and the like, as described or contemplated herein to enable the attachment components, **506** and **508**, to attach or couple with the shoe **400**. In some instances, the attachment components, **506** and **508**, may not be able to attach or coupled with the shoe **400** outside of the tightening region **510**. As shown in FIG. **20C**, the distal ends of the respective straps, **502** and **504**, may be moved or positioned essentially anywhere within the tightening region **510** and attached to the side of the shoe, which affects how the shoe **400** will fit and/or tighten about the user's foot. For example, each of the straps, **502** and **504**, may tighten a different zone or portion of the shoe **400**, such as the first strap **502** tightening the shoe's collar while the second strap **504** tightens the shoe's toe box. The tightness of the shoe in these zones is affected by how the distal end of the respective strap, **502** or **504**, is positioned within the tightening region **510** and coupled with the shoe.

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In some instances, the tightening region **510** may include discrete or individual coupling members **512** such as strips or sections of material to which the attachment components, **506** and **508**, may be attached. In this manner, the position of the distal end of the straps, **502** and **504**, may be limited to pre-designated attachment areas or orientations. In some instances, the straps, **502** and **504**, could be made of a flexible material that changes its visual appearance under tension, such as those described in relation to FIGS. **25A-B**. In such embodiments a level of tension that is applied in each zone via the straps, **502** and **504**, may be visually indicated.

FIGS. **21A-B** illustrate an embodiment in which the adjustability of the shoe's fit is constrained to pre-designated areas. Specifically, the shoe **400** includes multiple dedicated or pre-designated coupling zones, **532** and **534**, to which a strap **530** may be attached. The dedicated coupling zones, **532** and **534**, may be elongate strips or sections of a hook and loop type fastener material or another fastening material, such as magnets, adhesive based or tacky materials, and the like. The strap **530** includes a corresponding fastening material that enables the strap **530** to attach to one of the dedicated coupling zones, **532** or **534**. The use of the elongate strips of fastening material may enable a greater or stronger coupling of the strap **530** to the shoe, which may be desired when substantial forces are imposed on the foot and shoe. The reel based closure device **404** is attached to the lace **405**, which is in turn attached to the proximal end of the strap **530**. FIG. **21B** illustrates the strap **530** being attached to an upper coupling zone **532** and to a lower coupling zone **534**.

An advantage of the dedicated coupling zones, **532** and **534**, is that the shoe **400** is still configured with the ability to adjust the fit or tightness of the shoe **400**, but the adjustability of the fit or tightness is simplified to a degree that may appeal to a greater number of people. Stated differently, the adjustability is not too complex so as to prevent an average person from utilizing or underutilizing the adjustability. The position or orientation of the dedicated coupling zones, **532** and **534**, about the shoe may be designed to provide or achieve a tightness or fit based on a given activity or event. For example, in hiking it may be desirable to tighten the shoe **400** nearer to the toe box during ascent and to tighten the shoe **400** nearer to the collar or ankle during descent in order to prevent the user's toes from striking the front of the shoe **400**. The dedicated zones, **532** and **534**, may be oriented about the shoe and configured to provide this desired fit and tightness and thus, the user may adjust the fit of the shoe **400** without engaging in an overly complex process.

FIGS. **22A-B** illustrate an embodiment in which a separate tensioning panel or member **600** is attached to the shoe **400**. The tensioning panel **600** includes a first member **602** and a second member **604** that are separated by a gap **603**. The reel based closure device **404** is positioned on the first member **602** and is coupled with the lace **405**, which is routed or directed across the gap **603** via one or more guides **606**. The tensioning panel **600** is configured to be removably coupled with the shoe **400**. Specifically, an inner edge of the first member **602** and the second member **604** includes a fastening material or component **612** that is able to attach or couple with an outer material **610** of the shoe **400**. The outer material **610** of the shoe **400** is positioned on both sides of the shoe **400**, typically near the sole, so that the tensioning panel **600** may be placed or positioned atop the shoe **400** with the inner edge of the first member **602** and the second member **604** contacting the outer material **610**, which causes

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the tensioning panel 600 to attach or couple with the shoe 400. This placement of the tensioning panel 600 about the shoe 400 is illustrated in FIG. 22B. Operation of the reel based closure device 404 causes the tensioning panel 600 to constrict atop and about the shoe 400, which causes the shoe 400 to constrict or tighten about the foot.

As further illustrated in FIG. 22B, the tensioning panel 600 may be moved laterally about the shoe 400 toward the heel or toe box as desired, which affects how the tensioning panel 600 causes the shoe 400 to constrict or tighten about the foot. In some embodiments, one side of the tensioning panel 600 (e.g., the second side 604) may be fixedly attached to a side of the shoe 400 while the other side of the tensioning panel 600 (e.g., the first side 602) is detached from, but coupleable to, the opposite side of the shoe 400. An advantage of the separate tensioning panel 600 is that the panel may be removed for various reasons, such as replacing the panel with a different panel that is designed to provide a different fit and/or feel when tensioned. The tensioning panel 600 may also be replaced to provide a different aesthetic such as replacing hard and rigid plastic guides with softer more flexible fabric or woven guides.

FIGS. 23A-C illustrate a similar embodiment except that the tensioning panel 630 includes additional panel members. Specifically, the tensioning panel 630 includes a first member 632, a second member 634, a third member 636, a fourth member 638. The first member 632 and the third member 636 are positioned on one side of the gap or opening 603 while the second member 634 and the fourth member 638 are positioned on an opposite side of the gap 603. As illustrated in FIG. 23B, when the tensioning panel 630 is positioned atop the shoe 400, the first member 632 and the third member 636 are positioned on one side of the shoe 400 (e.g., lateral side) while the second member 634 and the fourth member 638 are positioned on the opposite side of the shoe 400 (e.g., medial side).

In one embodiment, each of the members (i.e., 632, 634, 636, and 638) include the fastening material or component 612 that is able to attach or couple with the outer material 610 of the shoe 400. In such an embodiment, each of the members (i.e., 632, 634, 636, and 638) may be moved independently along the shoe 400 between the heel and toe box and coupled with the outer material 610 as desired. For example, FIG. 23C illustrates the third member 636 being repositioned about the outer material and shoe 400 while the first member 632, the second member 634, and the fourth member 638 remain attached to the outer material 610. The tensioning panel 630 of FIGS. 23A-C provides a greater latitude or freedom in positioning and attaching the tensioning panel 630 to the shoe 400 in comparison with the tensioning panel 300 of FIGS. 22A-B.

In other embodiments, one or more of the members (i.e., 632, 634, 636, or 638) may be fixedly attached to the shoe 400 while the other members are able to be moved and repositioned about the outer material 610 and shoe 400. For example, the first member 632 and/or the second member 634 may be fixed or anchored to the shoe 400 while the third member 636 and/or the fourth member 638 are removably attachable to the shoe 400, or vice versa.

FIGS. 24A-B illustrate a similar tensioning panel 650 except that the tensioning panel 650 includes six separate and individual panel members. Specifically, the tensioning panel 650 includes a first member 651, a third member 653, and a fifth member 655 that are positioned on one side of the gap 603 and the shoe 400 and also include a second member 652, a fourth member 654, and a sixth member 656 that are positioned on an opposite side of the gap 603 and the shoe

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400. In one embodiment, each of the members (651-656) include the fastening material or component 612 that is able to attach or couple with the outer material 610 of the shoe 400 and as such, each of the members (651-656) may be moved independently from one another along the shoe 400 between the heel and toe and coupled with the outer material 610. Thus, the tensioning panel 650 of FIGS. 24A-B provides an even greater latitude or freedom in attaching the tensioning panel 650 to the shoe 400 in comparison with the tensioning panels of FIGS. 22A-23C. In other embodiments, one or more of the members (651-656) may be fixedly attached to the shoe 400 while the other members are able to be moved and repositioned about the outer material 610 and shoe 400.

FIGS. 24A-B also illustrate that the position of the reel based closure device 404 may be varied. Specifically, in FIG. 24A the reel based closure device 404 is positioned on the first member 651 whereas in FIG. 24B the reel based closure device 404 is attached to the shoe's tongue. The reel based closure device 404 may likewise be positioned elsewhere on the shoe 400, such as on the heel of the shoe 400, as desired. The differing placement of the reel based closure device 404 (e.g., tongue or heel placement) may similarly be employed in the tensioning panels of FIGS. 22A-23C as desired.

FIGS. 25A-B illustrate embodiments of straps that may be employed to visually indicate a level of tension that is imposed on the strap due to operation of the reel based closure device 404, or some other lace tensioning mechanism. The straps of FIGS. 25A-B may be used with any of the embodiments described or contemplated herein, or may be used with any other products or articles that require closure and/or tightening.

In FIG. 25A, the strap 700 is illustrated in an un-tensioned or relaxed state (i.e., image "a") and in a tensioned state (i.e., image "b"). The strap 700 includes indicia 702, such as a plurality of relatively evenly spaced dots or markings. In the un-tensioned or relaxed state, the indicia 702 is broadly spaced apart. In the tensioned state, the indicia 702 is more tightly or densely packed or spaced, which may make the strap 700 appear slightly darker in color or provide some other visual signal. The visual signal that is provided by the strap 700, such as due to the tightly or densely packed indicia 702, correlates or corresponds to the tension that is imposed on the strap 700. The visual signal may vary in accordance with the tension that is imposed on the strap 700, such as the strap 700 becoming visually darker as the tension increases. A user may learn to recognize the visual signal and may correlate that signal with a given tension in the strap 700 and/or a given fit or tightness of the shoe 400 about the foot. In this manner, the user may tension the strap 700 until a given visual signal is produced, which would indicate that a desired fit or tightness of the shoe 400 has been achieved.

In FIG. 25B, the strap 710 is also shown in an un-tensioned or relaxed state (i.e., image "a") and in a tensioned state (i.e., image "b"). The strap 710 likewise includes indicia 712, but the strap 710 is configured so that the indicia 712 is not visually apparent until the strap 710 is tensioned. For example, in the relaxed state (i.e., image "a"), the strap 710 may appear to be devoid of or otherwise lacking the indicia 712. Stated differently, the indicia 712 may be hidden from view in the relaxed state. As the strap 710 is tensioned, however, the indicia 712 may become visually apparent as illustrated in image "b". The indicia 712 may be a word, a phrase, a symbol, or any other desired object that becomes visible as the strap 710 is tensioned. The indicia 712 may be

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marked or etched on the strap **710** so that as the material of the strap **710** is relaxed and more tightly packed/spaced, the indicia **712** is not visible. As the material stretches and becomes less dense, the markings or etching may align and appear visible to the user. As with the strap **700** of FIG. **25A**, the user may correlate the indicia with a given tension in the strap **710** and/or tightness or fit of the shoe **400**.

FIGS. **26A-C** illustrate another embodiment of a strap that may be used with any of the embodiments described or contemplated herein, or may be used with any other products or articles that require closure and/or tightening. The strap of FIGS. **26A-C** is designed to distribute a force that is imposed on the shoe **400** or other article due to tensioning the lace **405**. In particular, when the lace **405** or a strap is positioned atop a shoe **400**, the lace **405** or strap may impose a downward force onto the shoe, which a user may feel pressing on the top of their foot. The downward force may be relatively uncomfortable and/or irritating, especially when the lace is tensioned to a significant degree. The strap of FIGS. **26A-C** helps alleviate these potential issues by distributing the force across a greater area of the shoe **400**. The straps distribute the force across a greater area by flexing laterally outward as the lace **405** is tensioned.

For example, FIG. **26A** illustrate the strap in which the lace **405** is in a relaxed or un-tensioned state. The strap includes a proximal end **711** and a distal end **713** with a woven material **714** that is disposed there between. The lace **405** is disposed within the woven material **714**, which forms a channel or tunnel through which the lace **405** is inserted. The lace **405** is operationally coupled with the reel based closure device **404** and with the distal end **713** of the strap so that tensioning of the lace **405** causes the distal end **713** of the strap to move toward the proximal end **711**. The woven material **714** is woven so that as the distal end **713** of the strap moves toward the proximal end **711**, the woven material **714** flexes or moves laterally outward. FIG. **26B** illustrates the strap in which the lace **405** is in a tensioned state and in which the distal end **713** of the strap has moved toward the proximal end **711**. As a result of this movement, the woven material **714** is flexed or moved laterally outward so that the strap has a greater lateral width in comparison with that of FIG. **26A**. In some instances, the woven material **714** may have a bulging appearance toward the center of the strap. The woven material **714** may have a helically wound braid or any other weave that results in the material flexing or moving laterally outward as the lace **405** is tensioned. The material that is used in the woven portion of the strap may be relatively rigid to resist buckling from the compressive forces that are imparted on the woven material **714** as the distal end **713** of the strap moves toward the proximal end **711**.

The greater lateral width of the strap that is due to the flexing or widening of the woven material **714** results in woven material **714** having a greater surface area atop the shoe **400**. As a result, any downward forces that are imposed on the shoe **400** from the strap due to tensioning of the lace **405** are distributed over a greater surface area of the shoe **400**, which results in the forces being less concentrated on any given area or portion of the shoe and the underlying foot. FIG. **26C** illustrates a positioning of the strap atop and across the shoe **400**.

It should be noted that any of the designs described herein may be employed with other means of tightening a shoe, such as conventional shoe laces, pull cords, reel based closure systems, and the like. In such embodiments, the tension adjustment components and systems described and contemplated herein may be employed to essentially adjust

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or vary the fit of the shoe after a primary means (e.g., shoe lace, pull cord, reel based closure systems, etc.) of closing and tightening the shoe has been employed.

FIGS. **27A-29C** illustrate shoes that are constructed so that the means of closing and tightening the shoe via a reel based closure device are built or manufactured into the shoe. The embodiments of FIGS. **27A-29C** are different than conventional shoes in which separate components are attached to the shoe after manufacture in order to enable the shoe to be closed and tightened via a reel based closure device. In FIGS. **27A-C** the shoe **800** includes a panel **804** that is constructed from the shoe's upper **802**. Stated differently, the upper **802** is designed and formed so that the panel **804** is an integral part of the upper **802**. In conventional shoes, the panel **804** would be a separate component that is stitched, adhered, welded, mechanically fastened, or otherwise attached to the upper **802** after manufacture of the shoe **800**. With the panel **804** integrally formed with the upper **802**, the fit of the shoe **800** about the foot may be improved and/or the process of the manufacturing the shoe **800** may be improved.

The panel **804** is operationally coupled with the reel based closure device **404** via lace **405** that extends from the reel based closure device **404** to a distal end of the panel **804**. The panel **804** may include a pair of apertures **806**, a channel, or any other feature that allows the distal end of the panel **804** to be attached to the lace **405**. As the lace **405** is tensioned via the reel based closure device **404**, the panel **804** is also tensioned, which closes and/or tightens the shoe **800** about the user's foot. FIG. **27C** illustrates an exemplary pattern of the upper **802** that may be used to form the upper **802** with the integrated panel **804**. The pattern includes a first material pattern **810**, a second material pattern **812**, and a third material pattern **814**. The first material pattern **810** includes the panel **804** and may be positioned on the medial side of the foot when the pattern **810** is attached to the other patterns to form the shoe **800**. The second material pattern **812** may be positioned on the lateral side of the foot when attached to the first material pattern **810** in forming the shoe **800**. The third material pattern **814** attaches to the first material pattern **810** and the second material pattern **812** to form the heel of the shoe **800**.

FIGS. **28A-B** illustrate another shoe **820** that includes a panel **824** that is built or integrated into the upper **822**. The panel **824** is positioned about the shoe **820** similar to the panel **804** of FIGS. **27A-C** and similar material patterns may be used in constructing the shoe **820**. The main difference in the shoe **820** of FIGS. **28A-B** is that the panel **824** includes multiple guides **826** that allow the lace **405** to traverse multiple times between the shoe's sole and the distal end of the panel **824**. The guides **826** may be made of soft materials, such as loops of fabric, in order to eliminate or reduce the rigid components that are positioned on the shoe, which may be uncomfortable to the user.

FIGS. **29A-C** illustrate another shoe **830** that includes a pair of panels that are built or integrated into the upper **832**. Specifically, the shoe **830** includes a first panel **834** that is integrated into the upper **832** in a manner similar to that illustrated in FIGS. **27A-C**. The first panel **834** is positioned across the top of the shoe **830** and is used to close and tighten the shoe about the top of the foot. The upper **832** also includes a second panel **838** that is wrapped around the heel of the shoe **830**. The second panel **838** is used to constrict the shoe **830** about the ankle. The reel based closure device **404** is illustrated as being positioned on the distal end of the first panel **834**, although in other embodiments the reel based closure device **404** may be positioned elsewhere on the shoe

830. The upper **832** includes guides **836** that route or direct the lace **405** from the reel based closure device **404** to the second panel **838** and to other parts of the upper **832**. Operation of the reel based closure device **404** causes the first panel **834** and the second panel **838** to constrict about the foot, thereby tightening the shoe about the foot.

FIG. **29C** illustrates an exemplary pattern of the upper **832** that may be used to form the upper **832** with the integrated first and second panels, **834** and **838**. The pattern includes a first material pattern **840** and a second material pattern **842**. The second material pattern **842** includes the first panel **834** and the second panel **838**. When coupled with the first material pattern **840**, the first panel **834** is positioned atop the shoe **830** and the second panel **838** is disposed around the heel as illustrated in FIGS. **29A-B**. When coupled together, the first material pattern **840** may be positioned on the medial side of the shoe **830** while the second material pattern **842** is positioned on the lateral side of the shoe **830**.

FIGS. **30A-B** illustrate another means of forming panels on a shoe **850**. In particular, the embodiments illustrate how portions of the shoe **850** may be rendered stiffer in comparison with other portions of the shoe to enable the stiffer portions to function more as a panel or strap that closes and/or tightens the shoe **850** about the foot. In particular, the shoe **850** may be made of a relatively soft and flexible material. The soft and flexible material may be more difficult to properly tension than a more rigid material and thus, in order to facilitate in tensioning the shoe **850**, one or more portions of the shoe may be rendered more stiff or rigid.

In the illustrated embodiments, the shoe **850** has a configuration similar to that of FIGS. **29A-C** in which a first panel **834** and a second panel **838** are integrated into the upper **832**. Since the material is soft and flexible, however, tensioning of the first and second panels may be more difficult without increasing the stiffness of one or both of these panels. To increase the stiffness, a more rigid or less flexible material is overlaid and bonded to the upper **832**. For example, to increase the stiffness of the second panel **838**, a more rigid material **856** is overlaid and bonded to the second panel **838** via heat welding, adhesive bonding, and the like. The stiffer material overlay **856** allows the second panel **838** to be more easily tensioned via the lace and reel based closure device **404** than an unaltered second panel **838**. The upper **832** may include additional material overlays as desired. For example, a stiffer material overlay **852** may be positioned near and/or over the toe box to increase the stiffness of this portion of the shoe. The stiffer material overlay **852** may extend to the guides **836** that route or direct the lace **405** to reinforce the coupling of the guides **836** with the shoe. An additional material overlay **854** may be attached to a portion of the first panel **834** to increase the stiffness and/or reinforce the first panel **834** as desired. More or fewer portions of the upper **832** may be stiffened via material overlays as desired.

Referring now to FIG. **31**, illustrated is a method **900** of assembling a reel based closure device. At block **902**, a housing component is provided. As described herein, the reel based closure device includes a bottom member and an exterior wall that extends upward from the bottom member to define an interior region of the housing component. The bottom member typically includes a channel that is defined or formed on or within a bottom surface of the bottom member. The channel is configured so that a tension member is positionable within the channel to route the tension member axially below the interior region of the housing component. The channel may be a U-shaped channel that is defined or formed on the bottom surface of the housing

component. The U-shaped channel may extend from a first side of the housing component to a second side of the housing component. Opposing sides of the U-shaped channel may have an arcuate or curved shape.

At block **904**, a spool is inserted within the interior region of the housing component. The spool is configured so that the tension member is windable about the spool. At block **906**, a tightening component is coupled with the housing component. The tightening component is configured so that an operation of the tightening component causes the spool to rotate within the interior region of the housing component to wind the tension member about the spool.

In some embodiments, the method also includes attaching the reel based closure device to a shoe. The method may further include coupling the housing component with a base component that is attached to a shoe to releasably coupleable the housing component with the shoe. The base component may include an aperture that is aligned with the channel of the housing component when the housing component is coupled with the base component. In such embodiments, the method may additionally include inserting the tension member through the aperture of the base component and through the channel of the housing component.

Panel Configuration for Footwear

In some embodiments, a panel configuration may be employed to enhance the fit or tightness of an article of footwear about a user's foot. Specifically, the panel configuration may be able to enhance contact between the footwear and the user's foot so that the footwear feels more comfortable and/or is more supportive during a physical activity. The panel configuration described herein may be useful in many instances or physical activity, but may be particularly useful in cycling or running related activities. The panel configuration may further be specifically useful in cooperation with a reel based tensioning device or mechanism, such as those incorporated by reference herein. It should be recognized that the panel configuration could be used in tightening other articles, such as bags, boots, packs, helmets, jackets, articles of clothing or apparel, and the like.

Referring to FIG. **32**, illustrated is an article of footwear **1000**, such as a cycling shoe (hereinafter footwear **1000**). The footwear **1000** includes a reel based closure device **1006** that is operably coupled with a tension member **1008** so that an operation of the reel based closure device **1006** (e.g., rotation of a knob) causes the tension member **1008** to be tensioned, such as by winding the tension member **1008** about an internal spool (not shown) of the reel based closure device **1006**. The tension member **1008** is routed or directed about a path along the footwear **1000** via guides **1010** that are directly or indirectly attached to the footwear **1000**. In the illustrated embodiment, the guides **1010** are positioned on a lateral side of the footwear **1000** between a sole and upper portion of the footwear **1000**. The guides **1010** are illustrated as strips of webbing that are folded to form loops or channels within which the tension member **1008** is inserted, although the guides **1010** may be formed of plastic components or other materials.

The footwear **1000** also includes one or more panels or straps, **1004**, **1005**, & **1007**, that are coupled with the medial side of the footwear **1000** and that extend over an upper portion of the footwear **1000** so that a distal end of the one or more panels or straps, **1004**, **1005**, & **1007**, (hereinafter one or more panels **1004**, **1005**, & **1007**) is positioned on the lateral side of the footwear **1000**. The one or more panels, **1004**, **1005**, & **1007**, may be directly or indirectly coupled with the medial side of the footwear **1000**. For example, a proximal end of the one or more panels, **1004**, **1005**, & **1007**,

may be stitched to an upper of the footwear **1000** on the medial side or the proximal end may be attached to a material or component that is attached to the upper. In other embodiments, the one or more panels, **1004**, **1005**, & **1007**, may be attached to a sole or other portion of the footwear **1000** as desired. The distal end of the one or more panels, **1004**, **1005**, & **1007**, is not directly attached to the footwear **1000**. Rather, the distal end of the one or more panels is free or unattached from the footwear **1000** so that the panels, **1004**, **1005**, & **1007**, are able to move and slide about the footwear **1000**. A body portions between the proximal and distal ends of the one or more panels, **1004**, **1005**, & **1007**, is also typically not attached to the footwear, or is loosely attached to the footwear **1000**, which allows the one or more panels, **1004**, **1005**, & **1007**, to move in a relatively free manner.

In the illustrated embodiment, the footwear **1000** includes three panels, although more or fewer panels may be employed as desired. The reel based closure device **1006** is directly attached to an upper panel **1004** that is positioned immediately adjacent a collar of the footwear **1000**. In other embodiments, the reel based closure device **1006** may be attached to the upper, to the heel, to another panel, or elsewhere about the footwear **1000**.

The tension member **1008** is routed, via the guides **1010**, about a path on the lateral side of the footwear **1000** between the sole and the distal end of the one or more panels **1004**, **1005**, & **1007**. In the illustrated embodiment, a central panel **1005** and a distal panel **1007** that is positioned adjacent the toe box each include a guide that is configured to route the tension member **1008**. The central panel **1005** includes a pair of guides, **1012** and **1014**, that route the tension member **1008** in a strategic manner that results in the central panel **1005** being pulled more securely over the footwear so that the footwear is pulled firmly against an instep of the user's foot. Specifically, the central panel **1005** includes a lower guide **1012** and an upper guide **1014**, which are formed of folded fabric or webbing materials that define a channel or loop for the tension member **1008**. The lower guide **1012** may be positioned on the distal end of the central panel **1005** so that the lower guide **1012** extends distally outward from the distal end of the central panel **1005**. The upper guide **1014** is positioned proximally of the lower guide **1012** and may be partially covered or concealed by an outer surface or material of the central panel **1005** so that only the opposing ends of the upper guide **1014** are visible. In other embodiments, the upper guide **1014** may be positioned under the central panel **1004** so that the upper guide **1014** is not visible, or the upper guide **1014** may be fully exposed and visible on a top surface of the central panel **1005**.

The tension member **1008** is routed through both the lower guide **1012** and the upper guide **1014**. The lower guide **1012** typically routes the tension member **1008** between a pair of guides that are positioned immediately below the central panel **1005** while the upper guide **1014** typically routes the tension member **1008** between guides that are positioned adjacent the collar and toe box respectively. As such, the portion of the tension member **1008** that exits from the opposing sides of the lower guide **1012** forms an angle θ_1 and the portion of the tension member **1008** that exits from the opposing sides of the upper guide **1014** forms an angle θ_2 . The angle, θ_2 , is typically greater than the angle, θ_1 , since the upper guide **1014** routes the tension member **1008** between a pair of guides that are spaced relatively far apart. In some embodiments, the routing of the tension member may be reversed so that the portion of the tension member **1008** that exits from the upper guide **1014** forms the

angle θ_1 and the portion of the tension member **1008** that exits from the lower guide **1012** forms the angle θ_2 . In such embodiments, the tension member **1008** may crisscross after exiting the upper and lower guides, **1012** and **1014**.

The tension member routing configuration, and the use of the upper and lower guides **1012** and **1014**, results in an increased closure force being applied to the central panel **1005**. Specifically, the closure force is significantly greater in the central panel in comparison with other panels because the tension member **1008** is routed through the central panel twice, which results in an additional amount of tension being imparted on the central panel **1005** from the tension member **1008**. As such, the central panel **1005** constricts about the outer surface of the footwear **1000** to a greater degree than conventional panel(s), which results in an increased amount of contact and tightening of the footwear **1000** about the user's foot.

The routing configuration and the use of the upper and lower guides, **1012** and **1014**, may be employed on footwear that includes a single panel, two panels, or four or more panels as desired. The upper and lower guides, **1012** and **1014**, may be positioned on the single panel or any panel in which additional closure force is desired. In some embodiments, multiple separate panels may include an upper and lower guide, **1012** and **1014**, having the same configuration or a similar configuration to the upper and lower guides illustrated. The panels, **1004**, **1005**, & **1007**, may be connected via a flexible webbing or other material, or may be essentially separate and isolated from one another as desired. Additionally, the proximal ends of separate panels may be connected or completely separate and/or a portion of the main body or of the distal ends may be connected or separate as desired.

One or more guides **1010** may have an angled orientation relative to the sole of the footwear **1000** to properly guide the tension member **1008** about the footwear **1000**. Alternatively, or in addition, the tension member **1008** may be routed around the heel or may be routed within a guide that is positioned on, or operably coupled with, the heel.

Minimizing Dynamic Lace Movement in Toe Region

In some embodiments it may be desirable to minimize movement of the tension member in the toe region of the footwear. Movement of the tension member in the toe region as described herein means that the tension member shifts or slides through, and relative to, guides that are positioned within or near the toe region of the footwear. Minimizing movement of the tension member in the toe region may be preferred in order to maintain a tight fit of the footwear about a wearer's toes. For example, in hiking shoes, it may be preferable to minimize dynamic movement of the tension member in the toe region in order to prevent a wearer's foot from shifting or sliding forward in the footwear. This may be particularly desirable during the descent portion of a hike when the hiker is hiking downhill. During the descent portion of a hike, the foot commonly slides forward within the shoe, which can cause the toes to contact or strike the front of the shoe. Repeated shifting of the foot within the footwear and repeated striking of the toes against the front of the shoe can irritate the wearer and cause pain or discomfort to the foot and toes.

Footwear that is sufficiently tensioned can greatly minimize this problem by securing the footwear about the foot. When the footwear is sufficiently tensioned about the foot, the foot is not able to slide or shift within the footwear, or such movement is greatly minimized. Therefore, the problem of the toes striking the front of the shoe is greatly reduced.

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In footwear that employs a reel based closure device as described herein, the tension member that is used with the reel based closure device may be prone to such movement and shifting. For example, the guides that are used in such systems commonly include low friction material(s) in order to minimize wear between the guide and the tension member and/or to allow tension in the tension member to be transferred to the front of the footwear to tighten the footwear about the front of the foot and the toes. In addition, the tension member that is used in such systems is commonly a thin lace, wire, or cable, which promotes sliding or shifting of the tension member within the guides. The use of these tension members and guides may result in increased movement or shifting of the tension member within the guides. This shifting or movement may be particularly prominent in the toe region of the shoe during the descent portion of a hike. For example, as the user hikes downhill, the toe region may bend and flex, which may increase the tension in the tension member in the toe region of the shoe. The increased tension may cause the tension member to shift within the guides, which may widen the toe portion of the footwear and thereby allow the foot to shift or slide forward within the footwear.

The embodiments described herein below provide footwear, and more particularly lace paths and/or guide configurations, that are designed to minimize movement or shifting of the tension member within the guides. Referring now to FIG. 33, illustrated is a shoe 1100 that includes a guide configuration that minimizes shifting of a tension member 1104 within the guides 1108 that are positioned in or near the toe region of the shoe 1100. As illustrated, the shoe 1100 includes a reel based closure device 1102 as described or incorporated herein. The shoe 1100 also includes a plurality of different sized and/or spaced guides 1106 and 1108. Specifically, the shoe 1100 includes upper guides 1106 that are positioned about the shoe 1100 so that the upper guides 1106 are positioned closer to a collar of the shoe 1106. The shoe 1100 also includes toe guide 1108 that are positioned near or within the toe region of the shoe 1100. The upper guides 1106 are typically longitudinally wider than the toe guides 1108, although in some embodiments, the upper guides 1106 and the toe guides 1108 may have essentially the same longitudinal width. In yet other embodiments, the size of the toe guides 1108 may be varied as desired, such as becoming increasingly narrower longitudinally toward the toe of the shoe 1100. The use of narrower toe guides 1108 may result in an increased number of tension member crossings in the toe region of the shoe 1100. Stated differently, the narrower toe guides 1108 may result in the tension member crossing an opening or closure zone of the shoe 1100 (e.g., the shoe's tongue) more times than would occur with longitudinally wider guides, such as the upper guides 1106. The increased crossings of the tension member 1104 results in increased friction between the tension member 1104 and the shoe 1100 and between the tension member 1104 itself. The increased friction resists shifting or sliding of the tension member 1104 through the toe guides 1108.

In addition, a spacing between the guides typically narrows or decreases toward the toe region of the shoe 1100. Specifically, as illustrated in FIG. 33, a spacing 1110 between the upper guides 1106 is wider than a spacing 1112 between the toe guides 1108 in the toe region of the shoe 1100. The spacing at a transition between the upper guides 1106 and the toe guides 1108 is also typically narrower as illustrated. The narrower spacing 1112 between the toe guides 1108 results in additional crossings in the toe region and results in a more horizontal orientation of the tension

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member 1104 across the shoe 1100. The increased horizontal orientation may increase the normal force of the tension member 1104 on the outer portions of the toe guides 1108, thereby increasing the frictional engagement of the tension member 1104 and the toe guides 1108. The increased frictional engagement minimizes movement of the tension members 1104 through the toe guides 1108.

The upper guides 1106 and/or toe guides 1108 may be attached to the shoe 1100 in various ways. For example, the upper guides 1106 and/or toe guides 1108 may be directly attached to the upper of the shoe 1100, attached underneath the upper of the shoe 1100, attached to one or more straps (not shown) that are attached to the shoe, attached to a canopy or shell that is attached to the shoe 1100, and the like. The guide configuration of FIG. 33 may be particularly designed to minimize or prevent instantaneous or quick shifting of the tension member 1104 through the toe guides 1108, which may greatly reduce the problems associated with the descent portion of a hike in which tension is quickly induced in the toe region of the shoe 1100.

Referring now to FIG. 34, illustrated is an embodiment of a shoe 1120 that includes a guide configuration that balances movement of the tension member. The guide configuration does not necessarily prevent or minimize movement of the tension member through the guides, but may balance the movement of the tension member so that the toe region of the shoe 1100 stays firmly secured about the foot and toes of the wearer. The guide configuration is achieved by shifting the guides diagonally about the shoe 1120. For example, a line A-A that is drawn through centers of a pair of guides is oriented at an angle θ with respect to a horizontal line 1123 of the shoe 1120. The pair of guides includes a left guide 1126 and a right guide 1128. The angle θ that is formed from the line A-A and a horizontal line 1123 may be between 15 and 45 degrees, although an angle θ of between 20 and 35 degrees is more common.

The angle θ may be selected to achieve a plurality of diagonal tension member crossings 1124 and a plurality of horizontal tension member crossings 1122. To achieve the diagonal crossings 1124 and horizontal crossings 1122, the angle θ may be based on the size and spacing of the guides. Specifically, the angle θ may be selected so that an upper portion or opening of the left guide 1126 is aligned with a lower portion or opening of the right guide 1128. The alignment of these portions of the left and right guides, 1126 & 1128, results in the tension member crossing the shoe 1120 roughly horizontally 1122 between these portions of the left and right guides, 1126 & 1128. The lower portion or opening of the left guide 1126 is not aligned with the upper portion or opening of the right guide 1128, which results in the diagonal crossing 1124 of the tension member between these portions of the left and right guides, 1126 & 1128. The alternation of diagonal and horizontal crossings, 1124 & 1122, balances the lace movement, especially during lateral torsion of the shoe 1120, which maintains a secure fit of the shoe 1120 about the foot and toes. The shoe 1120 may include a variety of differently shaped and/or sized guides in order to route the tension member from the reel based closure device into the diagonal and horizontal lace path illustrated in FIG. 34.

Referring to FIG. 35, illustrated is another embodiment of a shoe 1140 that includes toe guides that are configured to minimize shifting of the tension member within the toe region of the shoe 1140. The shoe 1140 includes one or more upper guides 1144 that route or direct the tension member from the reel based closure device toward the toe region of the shoe 1140. The shoe 1140 also includes a plurality of toe

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guides in the toe region that function cooperatively to increase a frictional engagement of the tension member and thereby limit shifting or movement of the tension member within the toe guides. Specifically, the toe guides include a first guide 1146 that functions with a pair of guides, 1148a & 1148b, that are positioned on an opposite side of a closure zone of the shoe 1140. The first guide 1146 routes the tension member to both guides of the guide pair, 1148a & 1148b. More specifically, a lower portion 1142b of the tension member is routed between the first guide 1146 and a lower guide 1148b of the guide pair while an upper portion 1142a of the tension member is routed between the first guide 1146 and an upper guide 1148a of the guide pair. The lower portion 1142b and the upper portion 1142a are each routed through the upper guide 1148a while only the lower portion 1142b is routed through the lower guide 1148b. The lower portion 1142b and the upper portion 1142a may be routed through the upper guide 1148a so that they contact one another and contact the upper guide 1148a. In this manner, frictional engagement of the tension member may be increased in the upper guide 1148a and in the toe region of the shoe 1140. The increased frictional engagement of the tension member minimizes dynamic movement and/or shifting of the tension member in and/or near the toe region of the shoe 1140.

In addition, the lower portion 1142b and the upper portion 1142a of the tension member may each be routed across the closure zone of the shoe 1140 so that the lower portion 1142b and the upper portion 1142a overlap, engage, or substantially contact one another across the closure zone. An example of such overlapping, engagement, and/or contact is illustrated by reference number 1149 in FIG. 35. The overlapping, engagement, and/or contact of the lower portion 1142b and the upper portion 1142a of the tension member increases frictional engagement of the tension member near the toe region of the shoe, which further minimizes dynamic movement of the tension member in and/or near the toe region of the shoe 1140. The shoe 1140 may include additional single guide and guide pair arrangements similar to the guide 1146 and guide pair, 1148a and 1148b, illustrated in FIG. 35. The additional single guide and guide pair arrangement(s) may be located near the toe region of the shoe 1140 or elsewhere about the shoe 1140 to provide strategic lace movement management in the toe regions or various other regions of the shoe 1140 as desired. The guide configuration of FIG. 35 may be particularly designed to minimize or prevent instantaneous or quick shifting of the tension member through the guide 1146 and guide pair, 1148a and 1148b, which may greatly reduce the problems associated with the descent portion of a hike in which tension is quickly induced in the toe region of the shoe 1140.

Referring now to FIG. 36, illustrated is an embodiment of a shoe 1160 that is designed to compensate for dynamic movement of the tension member 1162 in the toe region of the shoe 1160. Unlike the embodiments of FIGS. 33 and 35, which are designed to minimize movement of the lace in the toe region of the shoe 1160, the embodiment of FIG. 36 is designed to allow for such movement, but to compensate for such movement to keep the toe region of the shoe 1160 wrapped and secured tightly about the wearer's foot. To compensate for dynamic movement of the tension member 1162, the shoe 1160 includes a strap 1166 or material strip (hereinafter strap 1166) that is positioned in the shoe's toe region at or near a distal end of a lace path of the tension member 1162. A proximal end of the strap 1166 is attached to one side of a closure zone and the strap 1166 extends across the closure zone to an opposite side of the closure

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zone. The proximal end of the strap 1166 is typically attached to the shoe 1160 at or near the shoe's eyestay, although the proximal end of the strap 1166 may extend to the sole or to a position that is adjacent to the sole.

A distal end of the strap 1166 is positioned through a slit or opening 1167 (hereinafter opening 1167) in an upper of the shoe 1160. The strap 1166 is typically positioned through the opening 1167 so that the strap 1166 is able to slide within and relative to the opening 1167. The distal end of the strap 1166 is attached to a guide 1167 or is folded over to form a loop that functions as a guide for the tension member 1162. The tension member 1162 is routed through the guide 1167 so that tensioning of the tension member 1162, via the reel based closure device, causes the strap 1166 to be tensioned. Tensioning of the strap 1166 causes the strap 1166 to slide within the opening 1167 and to close or narrow the closure zone, which causes the toe region of the shoe 1160 to constrict about the wearer's foot.

As the tension member 1162 shifts about the toe region, such as during the descent portion of a hike, the strap 1166 frictionally engages with the opposing sides of the closure zone due to the coupling of the proximal portion of the strap 1166 with one side of the closure zone and the insertion of the strap 1166 through the opening 1167 on the opposite side of the closure zone. The frictional engagement of the strap 1166 with the opposing sides of the closure zone minimizes opening of the toe region of the shoe 1160, and thereby keeps the shoe secured about the wearer's toes. For example, the frictional engagement of the strap 1166 with the opposing sides of the closure zone helps prevent the toe region of the shoe from widening as the toe region is bent and flexed during the descent portion of a hike. In this manner, the strap 1166 compensates for movement of the tension member 1162 in the toe region of the shoe 1160.

In some embodiments, the strap 1166 may be made of a relatively elastic or flexible material that is able to stretch and flex with the shoe 1160. In other embodiments, the strap 1166 may be made of a more inflexible material that resists stretching and flexing of the strap 1166 as the shoe 1160 moves. In yet other embodiments, the strap 1166 may be made of a material that includes a relatively high coefficient of friction in order to minimize shifting or sliding of the tension member 1162 through the opening 1167 as the tension member 1162 shifts and adjusts. In such embodiments, the wearer may pull on the strap 1166 during an initial tightening of the shoe 1160 to cause the strap 1166 to move within the opening 1167 or the strap 1166 may slowly slide through the opening 1167 over a period of time as the shoe is worn. The frictional engagement of the strap 1166 and opening 1167 may be sufficient to minimize or prevent sudden movement or sliding of the strap 1166 through the opening 1167, such as due to a sudden increase in tension on the tension member. In some embodiments, the strap 1166 may include means for resisting such movement and sliding, such as employing hook and loop fastening means, button, clamp, or mechanical fastening means, and the like. In yet other embodiments, the tension member 1162 may be made of material having a low coefficient of friction in order to promote or facilitate sliding of the strap 1166 through the opening 1167.

While several embodiments and arrangements of various components are described herein, it should be understood that the various components and/or combination of components described in the various embodiments may be modified, rearranged, changed, adjusted, and the like. For example, the arrangement of components in any of the described embodiments may be adjusted or rearranged and/

or the various described components may be employed in any of the embodiments in which they are not currently described or employed. As such, it should be realized that the various embodiments are not limited to the specific arrangement and/or component structures described herein.

In addition, it is to be understood that any workable combination of the features and elements disclosed herein is also considered to be disclosed. Additionally, any time a feature is not discussed with regard to an embodiment in this disclosure, a person of skill in the art is hereby put on notice that some embodiments of the invention may implicitly and specifically exclude such features, thereby providing support for negative claim limitations.

Having described several embodiments, it will be recognized by those of skill in the art that various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the invention. Additionally, a number of well-known processes and elements have not been described in order to avoid unnecessarily obscuring the present invention. Accordingly, the above description should not be taken as limiting the scope of the invention.

Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Each smaller range between any stated value or intervening value in a stated range and any other stated or intervening value in that stated range is encompassed. The upper and lower limits of these smaller ranges may independently be included or excluded in the range, and each range where either, neither or both limits are included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included.

As used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a process” includes a plurality of such processes and reference to “the device” includes reference to one or more devices and equivalents thereof known to those skilled in the art, and so forth.

Also, the words “comprise,” “comprising,” “include,” “including,” and “includes” when used in this specification and in the following claims are intended to specify the presence of stated features, integers, components, or steps, but they do not preclude the presence or addition of one or more other features, integers, components, steps, acts, or groups.

What is claimed is:

1. A shoe comprising:

a sole;

an upper that is attached to the sole and configured to fit around a foot of a wearer, the upper having a medial side and a lateral side that fit around the medial and lateral side of the foot respectively;

a strap or panel having a proximal end that is attached to the shoe on the medial side or the lateral side of the shoe, the strap having a distal end that extends toward an opposite side of the shoe;

a tension member that is guided or directed about a path along the shoe; and

a tightening mechanism that is operable with the tension member to effect tensioning of the tension member upon operation of the tightening mechanism;

wherein the strap or panel includes a first guide and a second guide and wherein the tension member is routed through both the first guide and the second guide of the strap or panel;

wherein the first guide is attached to the strap or panel so that a first portion of the tension member exits from opposing sides of the first guide and forms a first angle; wherein the second guide is attached to the strap or panel so that a second portion of the tension member is routed through the second guide and exits from opposing sides of the second guide and forms a second angle;

wherein an entire length of the second portion of the tension member that is routed through the second guide is positioned between the first portion of the tension member exiting from opposing sides of the first guide and within the first angle formed by the first portion of the tension member; and

wherein the first angle is greater than the second angle.

2. The shoe of claim 1, wherein the path of the tension member is positioned on either a medial or lateral side of the shoe so that the tension member is positioned along the medial or lateral side of the shoe without being positioned atop the shoe.

3. The shoe of claim 2, wherein the tightening mechanism is positioned on the medial or lateral side of the shoe adjacent to the path of the tension member.

4. The shoe of claim 1, wherein the first guide is attached to the strap or panel so that the first guide is positioned proximally of the second guide on the strap or panel.

5. The shoe of claim 4, wherein the second guide is attached to the strap or panel so that the second guide extends distally of a distal end of the strap or panel.

6. The shoe of claim 1, wherein the strap or panel is a first strap and wherein the shoe includes a plurality of straps.

7. The shoe of claim 6, wherein the shoe includes three straps and wherein the first strap is a middle strap.

8. A shoe comprising:

a sole;

an upper that is attached to the sole, the upper defining a first side of the shoe and a second side of the shoe;

a first strap or panel that is attached to the first side of the shoe so that a body of the first strap or panel extends in a longitudinal direction over the upper toward the second side of the shoe, the longitudinal direction being defined as a direction from the first side of the shoe toward a second side of the shoe;

a second strap or panel that is attached to the first side of the shoe so that a body of the second strap or panel extends in a longitudinal direction over the upper toward the second side of the shoe;

a tension member that is guided or directed about a path along the shoe and that is coupled with the first strap or panel and the second strap or panel; and

a tightening mechanism that is operable with the tension member to effect tensioning of the tension member upon operation of the tightening mechanism;

wherein the first strap or panel includes a first guide and a second guide and wherein the tension member is routed through both the first guide and the second guide of the first strap or panel;

wherein the first guide and the second guide are aligned in the longitudinal direction along the body of the first strap or panel such that the first guide is positioned proximally of the second guide in the longitudinal direction along the body of the first strap or panel;

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wherein the first guide is attached to the first strap or panel so that the tension member exits from opposing sides of the first guide and forms a first angle;
 wherein the second guide is attached to the first strap or panel so that the tension member exits from opposing sides of the second guide and forms a second angle;
 wherein an entire length of the tension member that is routed through the second guide is positioned between the first angle formed by the tension member exiting from opposing sides of the first guide; and
 wherein the first angle is greater than the second angle.

9. The shoe of claim 8, wherein the second guide is attached to the first strap or panel so that the second guide extends distally of a distal end of the first strap or panel.

10. The shoe of claim 8, wherein the shoe includes a third strap or panel.

11. A strap or panel for use in tightening a shoe, the strap or panel comprising:

a main body having:

a proximal end that is attachable to a first side of the shoe; and

a distal end that is configured to extend in a longitudinal direction over the shoe toward an opposite side of the shoe when the strap or panel is attached to the shoe, the longitudinal direction being defined as a direction from the proximal end to the distal end of the main body;

a first guide through which a tension member is insertable, the first guide being configured to route or direct the tension member along a first path about the shoe; and

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a second guide through which the tension member is insertable, the second guide being configured to route or direct the tension member along a second path about the shoe;

wherein the first path is different than the second path;
 wherein the first guide and second guide are aligned along the main body such that the first guide is positioned proximally of the second guide along the main body in the longitudinal direction;

wherein the first guide and the second guide are the only guides coupled with the main body;

wherein the first guide is configured so that when attached to the shoe, the tension member exits from opposing sides of the first guide and forms a first angle;

wherein the second guide is configured so that when attached to the shoe, the tension member exits from opposing sides of the second guide and forms a second angle; and

wherein the first angle is greater than the second angle.

12. The strap or panel of claim 11, wherein the first guide is formed from a strip of material that is folded to form a loop through which the tension member is insertable.

13. The strap or panel of claim 12, wherein the second guide is formed from a strip of material that is folded to form a loop through which the tension member is insertable.

14. The strap or panel of claim 11, wherein the second guide is attached to the main body so that the second guide extends distally of a distal end of the main body.

15. The strap or panel of claim 11, wherein a middle portion of the first guide is covered by material of the strap or panel.

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